

Acknowledgements

I thank my supervisor, Professor Giuseppe Savaré, for supervising this thesis and for his support at decisive stages of my path.

My sincere gratitude goes to Professor Alessandro Pigati, the professor from whom I have learned the most over these years. I thank him for his guidance, his generosity, and for a kindness and availability that I have rarely found elsewhere.

I am profoundly grateful to Dr. Massimiliano Esposito, who has always believed in me and has done everything in his power to help me achieve my goals. I also warmly thank my high-school teacher, Professor Maria Antonietta Venturelli, who believed in me from the beginning and whose support continued well beyond my high-school years.

My deepest gratitude goes to my family. To my mother, my father, and my brother Daniel: I owe my projects, my ambitions, and who I am to your love and your sacrifices. I am also deeply grateful to my aunt Anna, my uncle Gianluca, and my grandparents, whose presence and support have always been a sure foundation.

My thoughts then go to my closest friends and loved ones, who have accompanied my growth and shared this journey with me: 민경, Leonardo, Anna, Kabir, Filippo, Carlo, and everyone who, though not named here, knows what they have meant to me.

Finally, I thank the institutions that shaped me and my trajectory: Bocconi University for the opportunities and growth it provided; the Fondazione Paolo e Carolina Zani for its support and the trust it placed in me; and the Unione Matematica Italiana, which indelibly shaped me through the Mathematical Olympiads.

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Notation and Conventions

We collect here the main notation used throughout the thesis.

$\langle \cdot, \cdot \rangle$	Euclidean or Frobenius inner product
$\mathbb{R}_+$	the set $\{x \in \mathbb{R} : x \geq 0\}$
$[I], [J], [L]$	the sets $\{1, \dots, I\}, \{1, \dots, J\}, \{1, \dots, L\}$
$N := IJ$	dimension of the ambient Euclidean space for transport plans
$K := I + J - 1$	number of linearly independent marginal constraints
$\bar{\varepsilon} : \mathbb{R}^N \rightarrow \mathbb{R}^K$	affine constraint map encoding marginal constraints
$d := (I - 1)(J - 1)$	affine, and Hausdorff, dimension of the transport polytope
μ, ν, ρ	discrete probability measures
$\mathcal{G}(\mu, \nu)$	transport polytope, i.e. the set of admissible couplings between μ and ν
$C \in \mathbb{R}_+^{I \times J}$	cost matrix, with entries $C_{ij} = C(x_i, y_j)$
$\mathcal{U}(G)$	transport cost, or internal energy, of a coupling $G \in \mathcal{G}(\mu, \nu)$
$\text{OT}_C(\mu, \nu)$	optimal transport cost with cost matrix C
$u \in \mathbb{R}^I, v \in \mathbb{R}^J$	Kantorovich dual potentials, or finite-temperature dual variables
$\theta \in \mathbb{R}^I, \eta \in \mathbb{R}^{J-1}$	Fourier dual potentials associated with the marginal constraints,
$(p, q) \in \mathbb{C}^I \times \mathbb{C}^{J-1}$	complexified and rescaled Fourier variables, with $(\theta, \eta) = \beta(p, q)$ on the original real cycle
$\mathcal{L}^d$	d -dimensional Lebesgue measure
$\mathcal{H}_{\mu\nu}^d$	d -dimensional Hausdorff measure restricted to $\mathcal{G}(\mu, \nu)$
$\overline{\mathcal{H}}_{\mu\nu}^d$	normalized d -dimensional Hausdorff measure on $\mathcal{G}(\mu, \nu)$
$\beta := 1/T$	inverse temperature
$\mathbf{P}_\beta(\mu, \nu)$	regularized primal finite-temperature optimal value
$\mathbf{D}_\beta(\mu, \nu)$	Lagrangian dual optimal value, equivalently $\sup_{(u,v)} F_\beta(u, v)$
$Z_\beta(\mu, \nu)$	partition function on the transport polytope
$Z_\beta^n(\mu, \nu)$	Gaussian-regularized partition function
$\mathcal{F}_\beta(\mu, \nu)$	finite-temperature free energy
$E_\beta(\mu, \nu)$	average internal energy
$S_\beta(\mu, \nu)$	Gibbs entropy
$F_\beta(u, v)$	effective free energy
$F_\beta(u^\beta, v^\beta)$	mean-field free energy
$\mathcal{U}_\beta$	mean-field internal energy
G^β	mean-field transport plan

Introduction

1. Statistical-physics methods beyond traditional physics

Statistical physics was originally developed to relate macroscopic observables to probability distributions over a very large number of microscopic degrees of freedom. In its classical formulation, this relation is expressed through ensembles, entropy, energy, partition functions, and variational principles [1, 2, 3].

More recently, statistical physics has become a general language for high-dimensional systems with many interacting components, especially when collective behaviour or concentration phenomena make local descriptions ineffective, suggesting instead probabilistic, variational, or mean-field approaches [4, 5, 6]. This modern perspective is closely related to the notion of a *complex system*, namely a system composed of many interacting parts whose collective behaviour does not follow trivially from that of its isolated constituents [6].

Of particular relevance for this thesis is the interface between statistical physics, optimization, theoretical computer science, and machine learning [4, 5, 7, 8]. While the use of temperature-based algorithms for difficult optimization tasks was sparked by the seminal work on simulated annealing [9], this interest has grown even further with recent advances in artificial intelligence and deep learning [10, 7, 8].

Despite their remarkable effectiveness, these physics-inspired methods often rely on heuristic manipulations. While these techniques are deeply rooted in physical intuition and widely accepted in practice [4, 5, 11], finding an explicit and self-contained mathematical justification adapted to the specific model under consideration is often difficult. This thesis studies one representative instance of this gap.

Specifically, we analyze the finite-temperature approach to discrete optimal transport introduced by Koehl, Delarue, and Orland [12], with the aim of proving the correctness of the resulting algorithm through the language of convex optimization and of identifying the mathematical framework and the precise hypotheses, under which the physical derivation could be made rigorous. Beyond this specific setting, our analysis serves as a tractable case study of the mathematical structure behind the saddle-point manipulations that are standard in statistical physics and mean-field methods [13, 5, 11].

2. Historical background of optimal transport

The theory of optimal transport (OT) dates back to the 18th century, when Gaspard Monge posed the problem of finding the most efficient way to move a pile of soil into an excavation of equal volume, minimizing the total transportation effort. His seminal memoir [14] laid the foundations of what is now called the Monge formulation of optimal transport.

More than a century later, Leonid Kantorovich relaxed Monge’s problem by introducing probability measures and linear programming techniques. Motivated by applications in logistics and production planning, this reformulation revealed the broader applicability of OT [15].

Analytic advances in the late 20th century revived interest in OT, notably through the work of Brenier and Benamou–Brenier [16, 17]. These developments were systematically presented in the monographs by Villani [18] and Santambrogio [19], which established optimal transport as a central topic at the intersection of analysis, geometry, probability, and optimization.

Furthermore, OT plays an increasingly important role in statistics and data analysis. It provides not only a metric to compare probability distributions over a metric space, but also a way to construct optimal transport maps, endowing the space of probability measures with a rich geometric structure.

On the applied side, recent algorithmic advances have established OT as a versatile tool in data science and related fields. Applications range from machine learning and computer vision to image analysis and graphics, as well as economics, computational biology, and signal processing.

The wide applicability of optimal transport is tied to our ability to solve it efficiently [20]. Classical linear-programming methods scale poorly with the number of support points and become impractical in large-scale applications. A major breakthrough in this direction was Cuturi’s entropic regularization of OT, which adds an entropy term to the transport plan and yields a strictly convex problem solvable by Sinkhorn iterations [21]. This approach has had enormous impact, both because of its computational efficiency and because it created a bridge between OT, information theory, and matrix scaling.

The finite-temperature OT method of Koehl, Delarue, and Orland [12] can be viewed as a competitive alternative to Cuturi’s approach. By treating each transport plan as a Gibbs ensemble at inverse temperature β and the cost function as an energy, they define

a free-energy functional whose unique maximiser determines a temperature-dependent transport plan. In the zero-temperature limit, this plan is expected to recover an optimal Kantorovich coupling. This mean-field approach leads to a Newton–Raphson solver with $\mathcal{O}(n^2)$ complexity per iteration, which remains stable even for extremely large values of β .

3. Aim and structure of the thesis

The aim of this thesis is to give a mathematically explicit analysis of the paper *Optimal Transport at Finite Temperature* by Koehl, Delarue, and Orland [12].

After introducing in Chapter 1 the minimal optimal transport setting needed in the rest of the thesis, Chapters 2 and 3 study the finite-temperature algorithm independently of its physical origin. Using standard mathematical formalisms from convex optimization and duality theory, we prove the convergence of the algorithm and study quantities that monotonically converge to the optimal transport plan at zero temperature. We then provide a variational interpretation of this finite-temperature setting as a convex regularized transport problem, directly comparable with entropic regularization of optimal transport [21].

The second part of the thesis studies the analytical structure behind the formal physical derivation of the method. Here, the optimal transport cost associated with a coupling is reinterpreted as the minimum energy of a system described by a Gibbs distribution. This minimum energy is recovered as the zero-temperature limit of the free energy, whose value is obtained by integrating over the polytope of admissible couplings. This integral is simplified by replacing the marginal constraints with products of Dirac delta functions, which are then expanded through a Fourier representation. This allows one to perform the integrations explicitly and to arrive at an unconstrained dual expression involving only dual variables. For this reason, Chapters 4 and 5 are entirely devoted to a reformulation based on geometric measure theory, Fourier analysis, and complex analysis.

The final analytic step concerns the saddle-point approximation [5, 11] of the previous expression, eventually leading to the optimization algorithm. Here, we formulate the required multidimensional steepest-descent structure explicitly, and we state under which hypotheses the claimed approximation can be made mathematically rigorous under the language of complex analysis.

Finally, in Chapter 6 we present the statistical-physics derivation in terms of the mean-

field effective free energy, showing that it leads to the same zero-temperature limit as the convex-optimization approach of Chapter 2, and Chapter 7 discusses further properties arising in the framework of [12].

4. Main contributions of the thesis

The main contributions of the thesis can be summarized as follows:

- First, we provide an original convex-optimization proof of the validity of the finite-temperature algorithm, independent of its formal statistical-physics derivation.
- Second, we formulate a convex regularized transport problem whose optimality system is precisely the finite-temperature nonlinear system that defines the algorithm.
- Third, we define the appropriate measure-theoretic framework for integration over the transport polytope. Using the intrinsic Hausdorff measure and the coarea formula, we replace the use of Dirac delta functions by Gaussian approximate identities.
- Fourth, we derive the corresponding regularized Fourier representation and isolate the hypotheses under which a temperature-dependent multidimensional steepest-descent analysis would be sufficient to justify the claimed logarithmic asymptotics. We do not claim to verify these global hypotheses in full generality.
- Finally, we critically reassess the mathematical claims made in [12]: some are proved rigorously, some are true only after adding missing hypotheses, some are based on unjustified formal arguments, and at least one is demonstrably false, as shown in Section 7.3.3.

1 Kantorovich Problem and Dual Potentials

In this chapter we introduce the finite-dimensional optimal-transport framework used throughout the thesis. Although optimal transport admits a far more general formulation on arbitrary Polish spaces [18, 19], the present work is intrinsically discrete and finite-dimensional.

1.1 The discrete Kantorovich problem

We consider the Kantorovich formulation of optimal transport between finitely supported probability measures. Let

$$\mu = \sum_{i=1}^I \mu_i \delta_{x_i}, \quad \nu = \sum_{j=1}^J \nu_j \delta_{y_j},$$

where $\mathcal{X} = \{x_1, \dots, x_I\}$, $\mathcal{Y} = \{y_1, \dots, y_J\}$, $\mu_i > 0$, $\nu_j > 0$, and

$$\sum_{i=1}^I \mu_i = \sum_{j=1}^J \nu_j = 1. \quad (1.1)$$

A transport plan, or coupling, between μ and ν is a probability measure on $\mathcal{X} \times \mathcal{Y}$ of the form

$$G = \sum_{i=1}^I \sum_{j=1}^J G_{ij} \delta_{(x_i, y_j)}$$

whose marginals are μ and ν . Equivalently, its coefficients satisfy

$$G_{ij} \geq 0, \quad \sum_{j=1}^J G_{ij} = \mu_i, \quad \sum_{i=1}^I G_{ij} = \nu_j.$$

We denote the set of all such couplings by $\mathcal{G}(\mu, \nu)$.

Let $C : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$ be a cost function, and write

$$C_{ij} := C(x_i, y_j).$$

The transport cost, i.e., the cost of a transport plan G , is

$$\mathcal{U}(G) := \int_{\mathcal{X} \times \mathcal{Y}} C(x, y) dG(x, y) = \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij}.$$

The discrete Kantorovich problem is

$$\text{OT}_C(\mu, \nu) := \min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G). \quad (1.2)$$

The optimal transport cost $\text{OT}_C(\mu, \nu)$ is called the (discrete) *optimal transport cost* (or distance) between μ and ν . When the cost C comes from a metric, it defines a metric between probability measures, called discrete Wasserstein distance, as shown in Theorem 7.6.

1.2 Matrix representation and the transport polytope

In the discrete setting, the optimal transport problem can be reformulated entirely in terms of real-valued vectors and matrices, without further reference to measures. We adopt the measure-theoretic description only to fix the optimal-transport interpretation, while using the matrix framework as the primary working basis throughout the thesis.

Thus, given vectors $\mu \in \mathbb{R}_+^I$ and $\nu \in \mathbb{R}_+^J$, a coupling is identified with the matrix

$$G = \begin{pmatrix} G_{11} & G_{12} & \dots & G_{1J} \\ G_{21} & G_{22} & \dots & G_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ G_{I1} & G_{I2} & \dots & G_{IJ} \end{pmatrix} \in \mathbb{R}_+^{I \times J}$$

whose row sums and column sums are prescribed by μ and ν , respectively. With this notation, the transport cost is

$$\mathcal{U}(G) = \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij} = \langle C, G \rangle,$$

where $\langle C, G \rangle := \sum_{i,j} C_{ij} G_{ij}$ denotes the Frobenius inner product.

The set $\mathcal{G}(\mu, \nu)$ is called the *transport polytope*. The terminology is justified by the following elementary observation:

Proposition 1.1. *The set $\mathcal{G}(\mu, \nu)$ is a convex and compact polytope.*

Proof. By definition, $\mathcal{G}(\mu, \nu)$ is determined by finitely many linear equalities and inequalities in $\mathbb{R}^{I \times J}$, hence it is a polyhedron. Moreover, for every $(i, j) \in [I] \times [J]$,

$$0 \leq G_{ij} \leq \sum_{i=1}^I G_{ij} = \nu_j \leq 1 \quad \text{and} \quad 0 \leq G_{ij} \leq \sum_{j=1}^J G_{ij} = \mu_i \leq 1$$

so that $\mathcal{G}(\mu, \nu) \subseteq [0, 1]^{I \times J}$. Therefore $\mathcal{G}(\mu, \nu)$ is bounded. Since it is also closed, being defined by finitely many linear equalities and inequalities, it follows from the Heine–Borel theorem that $\mathcal{G}(\mu, \nu)$ is compact. In particular, as a bounded polyhedron, it is a polytope.

It remains to prove convexity. Let $G, G' \in \mathcal{G}(\mu, \nu)$ and let $t \in [0, 1]$. Then, for every $(i, j) \in [I] \times [J]$, $tG_{ij} + (1 - t)G'_{ij} \geq 0$,

$$\sum_{j=1}^J [tG_{ij} + (1 - t)G'_{ij}] = t\mu_i + (1 - t)\mu_i = \mu_i,$$

and

$$\sum_{i=1}^I [tG_{ij} + (1 - t)G'_{ij}] = t\nu_j + (1 - t)\nu_j = \nu_j,$$

concluding that $tG + (1 - t)G' \in \mathcal{G}(\mu, \nu)$, and $\mathcal{G}(\mu, \nu)$ is convex. $\square$

Lemma 1.2. *The transport polytope $\mathcal{G}(\mu, \nu)$ has nonempty relative interior.*

Proof. Observe that the transport plan defined by

$$G_{ij} = \mu_i \nu_j > 0$$

satisfies the defining marginal constraints of $\mathcal{G}(\mu, \nu)$, hence it is a feasible transport plan for which the inequalities in $\mathcal{G}(\mu, \nu)$ are inactive. $\square$

Corollary 1.3. *The discrete optimal transport problem admits an optimal solution.*

Proof. The functional $\mathcal{U}(G) = \langle C, G \rangle$ is affine, hence continuous, on the compact set $\mathcal{G}(\mu, \nu)$. Moreover, by Lemma 1.2 this set is nonempty. Therefore, by the Weierstrass theorem, it attains its minimum on $\mathcal{G}(\mu, \nu)$. $\square$

1.3 Duality and Kantorovich potentials

In this section we recall the Kantorovich dual formulation in the finite-dimensional setting. It introduces dual variables that transform the constrained minimization problem (1.2) into a corresponding maximization problem, thereby revealing a complementary variational structure that leads to the finite-temperature algorithm formulated in this thesis. From the viewpoint of convex analysis, the same dual problem may be derived through Lagrange multipliers or, equivalently, through Fenchel duality [22, 23]. More broadly, this fits into the general minimax philosophy underlying many duality principles in analysis and optimization [24]. In full generality, establishing the equivalence between primal and dual formulations is part of the broader theory of Kantorovich duality [18, Thm. 5.10], [19, Thm. 1.46]. In the finite-dimensional discrete setting considered here, however, the dual formulation is a particular instance of linear programming duality (see, e.g., [25, 26]).

We therefore begin by recalling the basic facts from linear programming that will be used in the sequel.

Let

$$\begin{aligned}
 (\mathbf{P}) \quad & \min \quad c^T x \\
 & \text{s.t.} \quad Ax = b, \\
 & \quad \quad x \geq 0,
 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$, be a primal linear program. Its dual is

$$\begin{aligned}
 (\mathbf{D}) \quad & \max \quad b^T y \\
 & \text{s.t.} \quad A^T y \leq c, \\
 & \quad \quad y \in \mathbb{R}^m.
 \end{aligned}$$

Weak duality prescribes that

$$b^T y \leq c^T x.$$

This follows immediately: if x and y are feasible, then $Ax = b$, $x \geq 0$ and $A^T y \leq c$, so

$$b^T y = (Ax)^T y = x^T A^T y \leq x^T c = c^T x.$$

A more precise dual relation between the two problems is given by the strong duality principle [26, Section 6.2], which states that, if either $(\mathbf{P})$ or $(\mathbf{D})$ admit optimum, then there exist optimal feasible solutions x^* of $(\mathbf{P})$ and y^* of $(\mathbf{D})$ such that

$$c^T x^* = b^T y^*.$$

In other words, knowing that optimum is attained for either $(\mathbf{P})$ or $(\mathbf{D})$ guarantees that both optima are attained, and they coincide. As a corollary, if both primal and dual are feasible, then both admit optimal solutions.

Combining strong and weak duality, we see that feasible vectors x and y are optimal if and only if $0 = c^T x - b^T y$, which is equivalent to

$$0 = c^T x - b^T y = c^T x - (Ax)^T y = c^T x - x^T A^T y = x^T (c - A^T y).$$

This observation yields the *complementary slackness condition*: feasible x and y are optimal if and only if

$$x_\ell (c_\ell - a_\ell^T y) = 0, \quad \ell = 1, \dots, n. \tag{1.3}$$

We now recast the discrete Kantorovich problem as a linear program.

Definition 1.4 (Kantorovich primal). Let $\mu \in \mathbb{R}_+^I$, $\nu \in \mathbb{R}_+^J$ be positive vectors such that $\langle \mu, 1 \rangle = \langle \nu, 1 \rangle = 1$, and let $C \in \mathbb{R}_+^{I \times J}$ be a nonnegative matrix. Then the discrete Kantorovich problem (1.2) is

$$\begin{aligned} \min \quad & \mathcal{U}(G) = \sum_{i=1}^I \sum_{j=1}^J G_{ij} C_{ij} \\ \text{s.t.} \quad & \sum_{j=1}^J G_{ij} = \mu_i \quad \forall i \in [I], \\ & \sum_{i=1}^I G_{ij} = \nu_j \quad \forall j \in [J], \\ & G_{ij} \geq 0 \quad \forall (i, j) \in [I] \times [J]. \end{aligned} \tag{1.4}$$

Proposition 1.5 (Kantorovich dual). Let $\mu \in \mathbb{R}_+^I$, $\nu \in \mathbb{R}_+^J$ be positive vectors such that $\langle \mu, 1 \rangle = \langle \nu, 1 \rangle = 1$, and let $C \in \mathbb{R}_+^{I \times J}$ a nonnegative matrix. Introduce dual variables $u \in \mathbb{R}^I$ and $v \in \mathbb{R}^J$, associated with the row and column constraints in (1.4). Then the dual problem is

$$\begin{aligned} \max \quad & \mathcal{D}(u, v) = - \sum_{i=1}^I u_i \mu_i - \sum_{j=1}^J v_j \nu_j \\ \text{s.t.} \quad & C_{ij} + u_i + v_j \geq 0 \quad \forall (i, j) \in [I] \times [J]. \end{aligned} \tag{1.5}$$

Proof. Set

$$x = (G_{11}, \dots, G_{1J}, G_{21}, \dots, G_{2J}, \dots, G_{IJ})^T \in \mathbb{R}_+^{IJ}$$

and

$$c = (C_{11}, \dots, C_{1J}, C_{21}, \dots, C_{2J}, \dots, C_{IJ})^T \in \mathbb{R}_+^{IJ}.$$

Then (1.4) can be seen as a standard linear program $Ax = b$, with $x \geq 0$. To illustrate, we show an example with $I = 2$ and $J = 3$:

$$A = \left(\begin{array}{cccccc} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} \leftarrow u_1 \\ \leftarrow u_2 \\ \leftarrow v_1 \\ \leftarrow v_2 \\ \leftarrow v_3 \end{array} \implies Ax = b \iff \left\{ \begin{array}{ll} G_{11} + G_{12} + G_{13} & = \mu_1 \\ G_{21} + G_{22} + G_{23} & = \mu_2 \\ G_{11} + G_{21} & = \nu_1 \\ G_{12} + G_{22} & = \nu_2 \\ G_{13} + G_{23} & = \nu_3 \end{array} \right.$$

Here, we also defined the dual variables $\xi \in \mathbb{R}^I$ and $\zeta \in \mathbb{R}^J$ for each constraint. Writing $y = (\xi, \zeta)^T$, the dual objective is

$$b^T y = b_I^T \xi + b_J^T \zeta = \sum_{i=1}^I \mu_i \xi_i + \sum_{j=1}^J \nu_j \zeta_j,$$

and the dual feasibility $A^T y \leq c$ reads

$$\begin{pmatrix} A_I^T & A_J^T \end{pmatrix} \begin{pmatrix} \xi \\ \zeta \end{pmatrix} \leq c \iff A_I^T \xi + A_J^T \zeta \leq c \iff \xi_i + \zeta_j \leq C_{ij}, \forall i \in [I], j \in [J].$$

Setting $(u, v) := -(\xi, \zeta)$, the objective becomes $-\langle u, \mu \rangle - \langle v, \nu \rangle$ and the dual feasibility is $C_{ij} + u_i + v_j \geq 0$, for $(i, j) \in [I] \times [J]$. Hence we recover exactly (1.5). $\square$

The dual variables introduced above are called *Kantorovich potentials*. We denote by $\mathcal{K}(C)$ the set of admissible dual potentials, namely

$$\mathcal{K}(C) := \{(u, v) \in \mathbb{R}^I \times \mathbb{R}^J : C_{ij} + u_i + v_j \geq 0 \text{ for all } i \in [I], j \in [J]\}.$$

We can now state the main result of this section:

Theorem 1.6 (Kantorovich duality). *The discrete optimal transport problem satisfies*

$$\text{OT}_C(\mu, \nu) = \min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G) = \max_{(u, v) \in \mathcal{K}(C)} \mathcal{D}(u, v).$$

Moreover, both the primal and the dual problem admit optimal solutions.

Proof. The primal problem admits an optimal solution by Corollary 1.3. By the strong duality principle, the dual problem also admits an optimal solution, and the primal and dual optimal values coincide. $\square$

The same duality can also be expressed through a Lagrangian formulation. Here, we follow [23, Chapter 5], [22, Section 28] and [27, Section 3.1]. A more detailed description of this framework is provided in the Appendix A. We assume that the non-negativity constraint is kept as part of the primal domain, while only the affine marginal constraints are dualized. Thus, for a minimization problem of the form

$$\min_{x \in X} f(x) \quad \text{subject to } h_\alpha(x) = 0,$$

where X is a convex domain, its Lagrangian is the function

$$\mathcal{L}(x; \lambda) := f(x) - \sum_{\alpha} \lambda_{\alpha} h_{\alpha}(x), \quad x \in X, \quad (1.6)$$

where λ is a *Lagrange multiplier* associated with the constraint. In this thesis, we will use interchangeably the terms Lagrange multipliers, potentials and fields, to emphasize that they are structurally identical.

The corresponding *Lagrangian dual* problem is a maximization in λ of the dual objective:

$$\sup_{\lambda} \inf_{x \in X} \mathcal{L}(x; \lambda) \quad (1.7)$$

In the Kantorovich linear program, we consider the convex set $X := \{G \in \mathbb{R}^{IJ} : G_{ij} \geq 0\}$. Starting from (1.4), we introduce Lagrange multipliers u_i and v_j for the marginal constraints. With the sign convention used throughout the thesis, we set

$$\begin{aligned} \mathcal{L}(G; u, v) &\stackrel{P}{=} \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij} - \sum_{i=1}^I u_i \left(\mu_i - \sum_{j=1}^J G_{ij} \right) - \sum_{j=1}^J v_j \left(\nu_j - \sum_{i=1}^I G_{ij} \right) \\ &\stackrel{D}{=} -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i=1}^I \sum_{j=1}^J (C_{ij} + u_i + v_j) G_{ij}. \end{aligned} \quad (1.8)$$

From P we see that

$$\inf_{G \geq 0} \sup_{(u,v) \in \mathbb{R}^{I+J}} \mathcal{L}(G; u, v) = \inf_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G).$$

Indeed, if $G \geq 0$ satisfies the marginal constraints, then the multiplier terms vanish and the supremum over (u, v) equals $\mathcal{U}(G)$. If one of the marginal constraints is violated, the supremum over the corresponding multiplier is $+\infty$.

On the other hand, from D we see that

$$\inf_{G \geq 0} \mathcal{L}(G; u, v) = -\langle u, \mu \rangle - \langle v, \nu \rangle + \inf_{G \geq 0} \sum_{i,j} (C_{ij} + u_i + v_j) G_{ij}.$$

The last infimum is finite if and only if

$$C_{ij} + u_i + v_j \geq 0 \quad \forall (i, j) \in [I] \times [J].$$

In that case it equals 0, attained at $G = 0$. Otherwise it equals $-\infty$. Therefore

$$\sup_{(u,v) \in \mathbb{R}^{I+J}} \inf_{G \geq 0} \mathcal{L}(G; u, v) = \sup_{(u,v) \in \mathcal{K}(C)} \mathcal{D}(u, v).$$

Thus the Kantorovich primal and dual problems can be written in Lagrangian form as

$$\inf_{G \geq 0} \sup_{(u,v) \in \mathbb{R}^{I+J}} \mathcal{L}(G; u, v) \geq \sup_{(u,v) \in \mathbb{R}^{I+J}} \inf_{G \geq 0} \mathcal{L}(G; u, v),$$

where the inequality is the general min-max inequality [27, Thm. 3.3]. Moreover, at optimum,

$$\text{OT}_C(\mu, \nu) = \inf_{G \geq 0} \sup_{(u,v) \in \mathbb{R}^{I+J}} \mathcal{L}(G; u, v) = \sup_{(u,v) \in \mathbb{R}^{I+J}} \inf_{G \geq 0} \mathcal{L}(G; u, v).$$

In the discrete setting, this equality is certified by the KKT conditions, which are necessary and sufficient for primal-dual optimality; see Appendix A.

We conclude this section deriving the complementary slackness condition for the discrete Kantorovich problem:

Corollary 1.7 (Complementary slackness). *Let $G \in \mathcal{G}(\mu, \nu)$ and $(u, v) \in \mathcal{K}(C)$. Then G and (u, v) are primal-dual optimal if and only if*

$$G_{ij}(C_{ij} + u_i + v_j) = 0 \quad \text{for every } (i, j) \in [I] \times [J].$$

Proof. By Kantorovich duality, feasible G and (u, v) are optimal if and only if

$$\mathcal{U}(G) = \mathcal{D}(u, v).$$

Equivalently,

$$\begin{aligned} 0 &= \mathcal{U}(G) - \mathcal{D}(u, v) \\ &= \sum_{i,j} C_{ij} G_{ij} + \sum_i u_i \mu_i + \sum_j v_j \nu_j \\ &= \sum_{i,j} C_{ij} G_{ij} + \sum_i u_i \sum_j G_{ij} + \sum_j v_j \sum_i G_{ij} \\ &= \sum_{i,j} G_{ij} (C_{ij} + u_i + v_j). \end{aligned}$$

Since $G_{ij} \geq 0$ and $C_{ij} + u_i + v_j \geq 0$, the sum is zero if and only if every term is zero. Hence

$$G_{ij}(C_{ij} + u_i + v_j) = 0 \quad \forall i, j.$$

□

Remark 1.8 (Non-uniqueness of Kantorovich potentials). The potentials are translational invariant in the following sense: For every $a \in \mathbb{R}$,

$$(u, v) \mapsto (u + a\mathbf{1}_I, v - a\mathbf{1}_J)$$

leaves both the constraints $C_{ij} + u_i + v_j \geq 0$ and the optimal dual value $\mathcal{D}(u, v)$ unchanged. Thus the effective number of dual degrees of freedom is $I + J - 1$, not $I + J$.

In the sequel we often remove this degeneracy by fixing $v_J = 0$. After this convention, one may either regard v as an element of $\mathbb{R}^{J-1}$, containing only the first $J - 1$ components, or as an element of $\mathbb{R}^J$ with last component equal to zero. We shall use both viewpoints without further comment. In particular,

$$\langle v, \nu \rangle = \sum_{j=1}^J v_j \nu_j = \sum_{j=1}^{J-1} v_j \nu_j$$

whenever $v_J = 0$.

2 The Finite-Temperature Algorithm

In this chapter we introduce the finite-temperature algorithm studied in this thesis. We first define the nonlinear dual system, prove that it is well posed, and show that the induced transport plans converge to Kantorovich optimizers in the zero-temperature limit. In later chapters, after the algorithmic and variational structure has been clarified, we return to the physical derivation that motivates this system. This separation shows that the correctness of the algorithm can be proved independently of the physical arguments, using convex optimization and Kantorovich duality.

2.1 Definition of the Algorithm

Fix an inverse temperature $\beta > 0$. The finite-temperature algorithm is defined through a nonlinear system in dual variables. We impose the invariance condition $v_J = 0$ (cf. Remark 1.8), and therefore write $u^\beta \in \mathbb{R}^I$, $v^\beta \in \mathbb{R}^{J-1}$. The unknowns u^β and v^β are the solutions of the system

$$\begin{cases} \sum_{j=1}^J \varphi(\beta(C_{ij} + u_i + v_j)) = \mu_i, & i = 1, \dots, I, \\ \sum_{i=1}^I \varphi(\beta(C_{ij} + u_i + v_j)) = \nu_j, & j = 1, \dots, J-1, \end{cases} \quad (2.1)$$

where

$$\varphi(y) = \frac{e^{-y}}{e^{-y} - 1} + \frac{1}{y} \quad \varphi(0) = \frac{1}{2}.$$

Once the dual variables have been found, the associated finite-temperature transport plan is defined componentwise by

$$G_{ij}^\beta = \varphi(\beta(C_{ij} + u_i^\beta + v_j^\beta)). \quad (2.2)$$

Theorem 2.1. *G^β is a feasible coupling for every $\beta > 0$. Moreover, if G^∞ is a limit of G^{β_k} along a subsequence $\beta_k \rightarrow +\infty$, then G^∞ is an optimal solution for the Kantorovich problem 1.4. In particular,*

$$\mathcal{U}_\infty := \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij}^\infty = \text{OT}_C(\mu, \nu).$$

Theorem 2.1 is the main result of this chapter. It justifies the following annealing algorithm, analogous to that proposed in [12]:

i) **Input:** Marginals $\mu \in \mathbb{R}_+^I$, $\nu \in \mathbb{R}_+^J$, cost matrix $C \in \mathbb{R}_+^{I \times J}$, initial dual variables $(u^{(0)}, v^{(0)}) = (0, 0)$, initial inverse temperature $\beta_0 > 0$, temperature-increase factor $\gamma > 1$, and tolerance $\epsilon > 0$.

ii) **Temperature update:** For $i = 0, 1, 2, \dots$, set

$$\beta_{i+1} = \gamma \beta_i, \quad \beta_i \rightarrow +\infty.$$

iii) **Dual solve:** At each β_{i+1} , solve the optimality system (2.1), for instance by a Newton method, using $(u^{(i)}, v^{(i)})$ as the initial guess. Denote the solution by $(u^{(i+1)}, v^{(i+1)})$.

iv) **Compute coupling and energy:** Define

$$G_{k\ell}^{\beta_{i+1}} := \varphi(\beta_{i+1}(C_{k\ell} + u_k^{(i+1)} + v_\ell^{(i+1)})),$$

and

$$\mathcal{U}_{\beta_{i+1}} := \sum_{k=1}^I \sum_{\ell=1}^J C_{k\ell} G_{k\ell}^{\beta_{i+1}}. \quad (2.3)$$

v) **Stopping criterion:** Terminate if

$$\frac{|\mathcal{U}_{\beta_{i+1}} - \mathcal{U}_{\beta_i}|}{\mathcal{U}_{\beta_i}} < \epsilon.$$

2.2 Well-Posedness of the Algorithm and Effective Free Energy

In this section we prove Theorem 2.1. In particular, we show that the nonlinear system (2.1) is the first-order optimality system of a strictly concave dual functional, *called effective free energy*.

We start by collecting the elementary analytic properties of the following scalar functions:

$$h(y) := \frac{1 - e^{-y}}{y} = \int_0^1 e^{-ty} dt, \quad h(0) := 1, \quad (2.4)$$

$$\psi(y) := \log \left(\frac{1 - e^{-y}}{y} \right) = \log h(y), \quad (2.5)$$

$$f_\beta(x) := -\frac{1}{\beta} \log \left(\frac{1 - e^{-\beta x}}{\beta x} \right) = -\frac{1}{\beta} \psi(\beta x). \quad (2.6)$$

The function h is strictly positive on $\mathbb{R}$, so ψ and f_β are well defined on the real line, with the values at zero understood by continuity.

Lemma 2.2. $\varphi(y)$ is the expectation of $t \in [0, 1]$ under the probability density proportional to e^{-ty} :

$$\varphi(y) = \frac{\int_0^1 te^{-ty} dt}{\int_0^1 e^{-ty} dt}. \quad (2.7)$$

In particular, φ is smooth, with derivative given by

$$\varphi'(y) = \left(\frac{\int_0^1 te^{-ty} dt}{\int_0^1 e^{-ty} dt} \right)^2 - \frac{\int_0^1 t^2 e^{-ty} dt}{\int_0^1 e^{-ty} dt}.$$

Moreover, it is a strictly decreasing homeomorphism between $\mathbb{R}$ and $(0, 1)$.

Proof. Observe that, for every $y \in \mathbb{R} \setminus \{0\}$,

$$\begin{aligned} \frac{\int_0^1 te^{-yt} dt}{\int_0^1 e^{-yt} dt} &= \frac{1}{[-e^{-yt}/y]_0^1} \int_0^1 te^{-yt} dt = \frac{y}{1 - e^{-y}} \left([-te^{-yt}/y]_0^1 + \frac{1}{y} \int_0^1 e^{-yt} dt \right) \\ &= \frac{y}{1 - e^{-y}} \left(-\frac{e^{-y}}{y} + \frac{1 - e^{-y}}{y^2} \right) = \frac{e^{-y}}{e^{-y} - 1} + \frac{1}{y} = \varphi(y). \end{aligned}$$

Furthermore,

$$\frac{\int_0^1 te^{-0 \cdot t} dt}{\int_0^1 e^{-0 \cdot t} dt} = \int_0^1 t dt = \frac{1}{2} = \varphi(0).$$

This shows that (2.7) holds pointwise. In particular, φ is continuous. Let $N(y) := \int_0^1 te^{-ty} dt$ and $D(y) = \int_0^1 e^{-ty} dt$. Since $[0, 1]$ is compact and te^{-ty} and e^{-ty} are C^∞ , by the Leibniz rule we conclude that

$$N'(y) = \int_0^1 \partial_y (te^{-ty}) dt = - \int_0^1 t^2 e^{-ty} dt,$$

and

$$D'(y) = \int_0^1 \partial_y (e^{-ty}) dt = - \int_0^1 te^{-ty} dt = -N(y).$$

Hence φ is C^∞ and

$$\varphi'(y) = \frac{N'(y)D(y) - N(y)D'(y)}{D(y)^2} = \frac{N(y)^2}{D(y)^2} + \frac{N'(y)}{D(y)} = \left(\frac{\int_0^1 te^{-ty} dt}{\int_0^1 e^{-ty} dt} \right)^2 - \frac{\int_0^1 t^2 e^{-ty} dt}{\int_0^1 e^{-ty} dt}.$$

This is the opposite of the variance of the random variable t under the law with pdf proportional to e^{-ty} , hence it is strictly negative.

Alternatively, the result holds by Cauchy-Schwarz inequality, taking $f_y(t) = te^{-ty/2}$ and $g_y(t) = e^{-ty/2}$:

$$\varphi'(y) < 0 \iff \left(\int_0^1 f_y(t)g_y(t) dt \right)^2 < \left(\int_0^1 f_y(t)^2 dt \right) \left(\int_0^1 g_y(t)^2 dt \right),$$

with equality occurring if and only if $f_y(t) \propto g_y(t)$ for every t , contradiction. Finally, by strict monotonicity and continuity,

$$\text{Im}(\varphi) = \left(\lim_{x \rightarrow +\infty} \varphi(x), \lim_{x \rightarrow -\infty} \varphi(x) \right) = \left(\lim_{x \rightarrow +\infty} \frac{1}{1 - e^x} + \frac{1}{x}, \lim_{x \rightarrow -\infty} \frac{1}{1 - e^x} + \frac{1}{x} \right) = (0, 1).$$

In particular, φ is a homeomorphism between $\mathbb{R}$ and $(0, 1)$. $\square$

Lemma 2.3. *The following relationships hold:*

$$\psi'(y) = -\varphi(y) \tag{2.8}$$

$$f'_\beta(x) = \varphi(\beta x) \tag{2.9}$$

$$f''_\beta(x) = \beta \varphi'(\beta x) \tag{2.10}$$

Proof. Since $\psi(y) = \log h(y)$, it holds

$$\psi'(y) = \frac{d}{dy} \log h(y) = \frac{h'(y)}{h(y)} = \frac{\int_0^1 -te^{-ty} dt}{\int_0^1 e^{-ty} dt} = -\varphi(y),$$

where differentiation under the integral operator is valid by continuity of the integrand function and compactness of $[0, 1]$. Next, we see that

$$f'_\beta(x) = \frac{d}{dx} \left(-\frac{1}{\beta} \psi(\beta x) \right) = -\psi'(\beta x) = \varphi(\beta x).$$

Finally,

$$f''_\beta(x) = \frac{d}{dx} (\varphi(\beta x)) = \beta \varphi'(\beta x).$$

$\square$

Lemma 2.4. *$f_\beta(x)$ is a strictly concave function of x . Furthermore, it is a decreasing function of β .*

Proof. Combining Lemma 2.2 and Lemma 2.3, we deduce that $f_\beta(x)$ is strictly concave:

$$f''_\beta(x) \stackrel{2.3}{=} \beta \varphi'(\beta x) \stackrel{2.2}{<} 0.$$

Moreover,

$$\partial_\beta f_\beta(x) = \partial_\beta \left(-\frac{1}{\beta} \psi(\beta x) \right) = \frac{1}{\beta^2} [\psi(\beta x) - \beta x \psi'(\beta x)].$$

Setting $y := \beta x$, we want to show that $\psi(y) - y\psi'(y) \leq 0$. Differentiating this expression,

$$\psi'(y) - (\psi'(y) + y\psi''(y)) = -y\psi''(y) \stackrel{2.3}{=} y\varphi'(y).$$

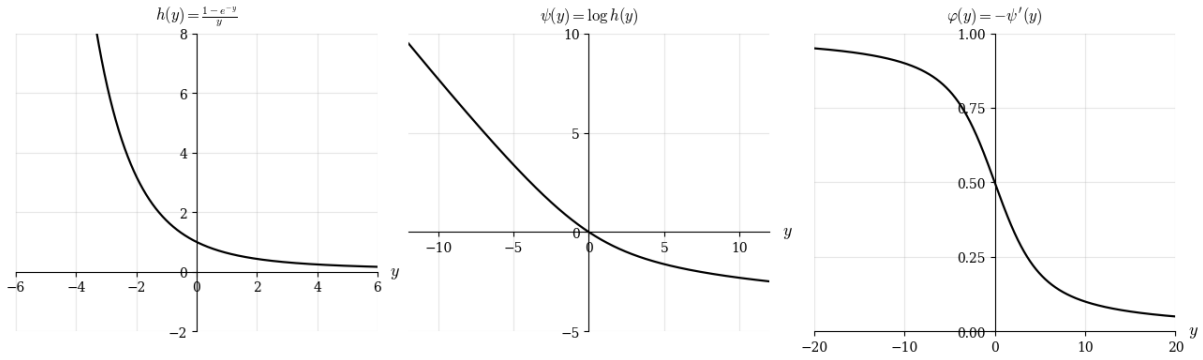


Figure 1: Profiles of the scalar functions h , $\psi = \log h$, and $\varphi = -\psi'$ appearing in Lemma 2.3.

Since $\varphi'(y) < 0$ by Lemma 2.2, the derivative $y\varphi'(y)$ has the opposite sign of y : it is positive for negative values of y , and negative for positive values of y . Since φ is smooth, we conclude that $y = 0$ is a global maximizer for $\psi(y) - y\psi'(y)$. Hence, we compute its value at $y = 0$, obtaining

$$\psi(0) - 0 \cdot \psi'(0) = \psi(0) = \log h(0) = \log 1 = 0.$$

Therefore, $\psi(y) - y\psi'(y) \leq 0$ for all $y \in \mathbb{R}$, which implies $\partial_\beta f_\beta(x) \leq 0$. $\square$

The previous scalar identities allow us to introduce the dual functional whose critical points solve the finite-temperature system. We define the *effective free energy* $F_\beta : \mathbb{R}^I \times \mathbb{R}^J \rightarrow \mathbb{R}$ by

$$F_\beta(u, v) = -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i=1}^I \sum_{j=1}^J f_\beta(x_{ij}), \quad (2.11)$$

where

$$x_{ij} := C_{ij} + u_i + v_j.$$

The terminology "free energy" will acquire its statistical-physics interpretation later. In this chapter, F_β is simply a function of the Kantorovich potentials $(u, v) \in \mathbb{R}^{I+J}$ whose structure is, in a sense, reminiscent of the Lagrangian formulation in (1.8): the dual term

$$-\langle u, \mu \rangle - \langle v, \nu \rangle$$

is the same, while the primal contribution $G_{ij}x_{ij}$ is replaced by the effective term $f_\beta(x_{ij})$. Thus F_β may be viewed as an effective dual functional $\mathcal{L}$ obtained after replacing the primal variables G_{ij} by their finite-temperature response. The next lemma is a consequence of Remark 1.8: because of it, we will assume that F_β is defined on the restricted domain $\mathcal{P} := \{(u, v) \in \mathbb{R}^{I+J} : v_J = 0\}$.

Lemma 2.5. *The effective free energy is translation invariant in the following sense: for every $c \in \mathbb{R}$,*

$$F_\beta(u + \mathbf{1}_I \cdot c, v - \mathbf{1}_J \cdot c) = F_\beta(u, v),$$

where $\mathbf{1}_I$ and $\mathbf{1}_J$ denote the vectors $(1, \dots, 1)^T \in \mathbb{R}^I$ and $(1, \dots, 1)^T \in \mathbb{R}^J$, respectively.

Proof. Direct computation shows that

$$\begin{aligned} F_\beta(u + \mathbf{1}_I c, v - \mathbf{1}_J c) &= -\langle u + c, \mu \rangle - \langle v - c, \nu \rangle + \sum_{i=1}^I \sum_{j=1}^J f_\beta(C_{ij} + u_i + c + v_j - c) \\ &= -\langle u, \mu \rangle - \langle v, \nu \rangle - c \left(\sum_{i=1}^I \mu_i - \sum_{j=1}^J \nu_j \right) + \sum_{i=1}^I \sum_{j=1}^J f_\beta(C_{ij} + u_i + v_j) \\ &= F_\beta(u, v) + c(1 - 1) = F_\beta(u, v). \end{aligned}$$

□

Lemma 2.6. *The Hessian of the effective free energy $F_\beta(u, v)$ is negative semidefinite with $(I + J - 1)$ negative eigenvalues and one zero eigenvalue. Furthermore, the eigenvector corresponding to the zero eigenvalue is*

$$\underbrace{(1, \dots, 1)}_I, \underbrace{(-1, \dots, -1)}_J$$

and thus corresponds to the constant translation invariance in Lemma 2.5. Setting $v_J := 0$, the effective free energy on $\mathcal{P}$ is strictly concave.

Proof. We compute, for every $i, i' \in [I]$ and for every $j, j' \in [J]$, the following quantities:

$$\frac{\partial^2 F_\beta}{\partial u_i \partial u_{i'}}, \quad \frac{\partial^2 F_\beta}{\partial v_j \partial v_{j'}}, \quad \text{and} \quad \frac{\partial^2 F_\beta}{\partial u_i \partial v_j}.$$

Notice that for any of these quantities, the linear terms in $F_\beta(u, v)$ vanish. Hence:

$$\begin{aligned} \frac{\partial^2 F_\beta}{\partial u_i \partial u_{i'}} &= \sum_{k=1}^I \sum_{\ell=1}^J \frac{d^2 f_\beta(x_{k\ell})}{dx_{k\ell}^2} \frac{\partial x_{k\ell}}{\partial u_i} \frac{\partial x_{k\ell}}{\partial u_{i'}} = \sum_{k,\ell} f''_\beta(x_{k\ell}) \delta_{k,i} \delta_{k,i'} = \delta_{i,i'} \sum_{j=1}^J f''_\beta(x_{ij}), \\ \frac{\partial^2 F_\beta}{\partial v_j \partial v_{j'}} &= \sum_{k=1}^I \sum_{\ell=1}^J \frac{d^2 f_\beta(x_{k\ell})}{dx_{k\ell}^2} \frac{\partial x_{k\ell}}{\partial v_j} \frac{\partial x_{k\ell}}{\partial v_{j'}} = \sum_{k,\ell} f''_\beta(x_{k\ell}) \delta_{\ell,j} \delta_{\ell,j'} = \delta_{j,j'} \sum_{i=1}^I f''_\beta(x_{ij}), \\ \frac{\partial^2 F_\beta}{\partial u_i \partial v_j} &= \sum_{k=1}^I \sum_{\ell=1}^J \frac{d^2 f_\beta(x_{k\ell})}{dx_{k\ell}^2} \frac{\partial x_{k\ell}}{\partial u_i} \frac{\partial x_{k\ell}}{\partial v_j} = \sum_{k,\ell} f''_\beta(x_{k\ell}) \delta_{k,i} \delta_{\ell,j} = f''_\beta(x_{ij}). \end{aligned}$$

Since $f''_\beta(x_{ij}) < 0$ by Lemma 2.4, the entries of the Hessian are either zero or negative. It can be written in the following form:

$$\nabla^2 F_\beta = \begin{bmatrix} D_u & H \\ H^T & D_v \end{bmatrix},$$

where D_u is a diagonal matrix with entry $H_{ii} = \sum_{j=1}^J f''_\beta(x_{ij})$, D_v is a diagonal matrix with entries $H_{jj} = \sum_{i=1}^I f''_\beta(x_{ij})$, while $H \in \mathbb{R}^{I \times J}$ and $H^T \in \mathbb{R}^{J \times I}$ are matrices with entries $H_{ij} = f''_\beta(x_{ij})$. Let $(\alpha, \beta) \in \mathbb{R}^I \times \mathbb{R}^J$ be arbitrary. Then

$$\begin{aligned} (\alpha, \beta)^T \nabla^2 F_\beta (\alpha, \beta) &= \sum_{i,i'} \alpha_i [D_u]_{ii'} \alpha_{i'} + 2 \sum_{i,j} \alpha_i [H]_{ij} \beta_j + \sum_{j,j'} \beta_j [D_v]_{jj'} \beta_{j'} \\ &= \sum_{i=1}^I \sum_{j=1}^J f''_\beta(x_{ij}) \cdot (\alpha_i + \beta_j)^2 \leq 0, \end{aligned}$$

since $f''_\beta(x_{ij}) < 0$ for all i, j . Therefore $\nabla^2 F_\beta$ is negative semidefinite and F_β is weakly concave. Moreover, the equality can only occur if $\alpha_i + \beta_j = 0$ for every $(i, j) \in [I] \times [J]$. Setting $\alpha_1 := c$, for every $j \in [J]$ we have $\beta_j = -\alpha_1 = -c$. On the other hand, for every $i \in [I]$, we have $\alpha_i = -\beta_1 = -(-c) = c$. Hence 0 is an eigenvalue of $\nabla^2 F_\beta$, with eigenvector

$$\underbrace{(c, \dots, c)}_I, \underbrace{(-c, \dots, -c)}_J \quad \text{or} \quad \underbrace{(1, \dots, 1)}_I, \underbrace{(-1, \dots, -1)}_J.$$

This eigenvector corresponds to the translation invariance 2.5 for F_β . It can be removed by setting $v_J := 0$. On this restricted parameter space, F_β is strictly concave. $\square$

Next, we establish the existence and uniqueness of a global maximizer for F_β .

Theorem 2.7. *F_β admits a unique global maximizer on $\mathcal{P}$.*

Proof. It is easy to see that $h(x) \in [0, 1]$ for $x \geq 0$. Since $C_{ij} \geq 0$ for every $(i, j) \in [I] \times [J]$, then

$$F_\beta(0, 0) = \sum_{i=1}^I \sum_{j=1}^J f_\beta(C_{ij}) = \sum_{i=1}^I \sum_{j=1}^J -\frac{1}{\beta} \log \left(\frac{1 - e^{-\beta C_{ij}}}{\beta C_{ij}} \right) \geq 0.$$

Let $F_\beta^\geq := \{(u, v) \in \mathbb{R}^{I+J} : v_J = 0, F_\beta(u, v) \geq 0\}$. Since F_β is strictly concave by Lemma 2.6, then $F_\beta^\geq$ is convex, closed, and contains $(0, 0)$. It is therefore enough to show that no ray from the origin is entirely contained in $F_\beta^\geq$, for that would imply $F_\beta^\geq$ to be bounded [22, Thm. 8.4], hence compact. A strictly concave function on a finite dimensional vector

space is continuous. If it admits a compact nonempty level set, then there exists a unique maximizer in it, by the Weierstrass theorem. Equivalently, it is enough to prove that

$$F_\beta(\lambda a, \lambda b) \rightarrow -\infty \quad \text{as } \lambda \rightarrow +\infty$$

for every unit vector $(a, b) \in \mathbb{R}^{I+J}$ with $b_J = 0$.

Observe that, by Equation (1.1),

$$\begin{aligned} F_\beta(u, v) &= -\sum_{i=1}^I u_i \mu_i - \sum_{j=1}^J v_j \nu_j + \sum_{i=1}^I \sum_{j=1}^J f_\beta(C_{ij} + u_i + v_j) \\ &= -\sum_{i=1}^I \sum_{j=1}^J (u_i \mu_i \nu_j + v_j \mu_i \nu_j) + \sum_{i=1}^I \sum_{j=1}^J f_\beta(C_{ij} + u_i + v_j) \\ &= \sum_{i=1}^I \sum_{j=1}^J f_\beta(C_{ij} + u_i + v_j) - \mu_i \nu_j (u_i + v_j) \\ &= \sum_{i=1}^I \sum_{j=1}^J f_\beta(C_{ij} + \lambda(a_i + b_j)) - \lambda \mu_i \nu_j (a_i + b_j). \end{aligned}$$

We set $a_i + b_j := y_{ij}$ and we study the asymptotics as $\lambda \rightarrow +\infty$ componentwise:

- If $y_{ij} > 0$, then

$$\begin{aligned} f_\beta(C_{ij} + \lambda y_{ij}) - \lambda \mu_i \nu_j y_{ij} &= -\frac{1}{\beta} \log \left(\frac{1 - e^{-\beta(C_{ij} + \lambda y_{ij})}}{\beta(C_{ij} + \lambda y_{ij})} \right) - \lambda \mu_i \nu_j y_{ij} \\ &= \frac{\log(\beta(C_{ij} + \lambda y_{ij}))}{\beta} - \frac{\log(1 - e^{-\beta(C_{ij} + \lambda y_{ij})})}{\beta} - \lambda \mu_i \nu_j y_{ij} \\ &\sim \frac{1}{\beta} \log \lambda - \lambda \mu_i \nu_j y_{ij} \sim -\lambda \mu_i \nu_j y_{ij} \rightarrow -\infty, \end{aligned}$$

since $\mu_i \nu_j y_{ij} > 0$;

- If $y_{ij} < 0$, then

$$\begin{aligned} f_\beta(C_{ij} + \lambda y_{ij}) - \lambda \mu_i \nu_j y_{ij} &= \frac{\log(-\beta(C_{ij} + \lambda y_{ij}))}{\beta} - \frac{\log(e^{-\beta(C_{ij} + \lambda y_{ij})} - 1)}{\beta} - \lambda \mu_i \nu_j y_{ij} \\ &\sim \frac{1}{\beta} \log(\lambda) + (C_{ij} + \lambda y_{ij}) - \lambda \mu_i \nu_j y_{ij} \\ &\sim \lambda y_{ij} (1 - \mu_i \nu_j) \rightarrow -\infty, \end{aligned}$$

since $1 - \mu_i \nu_j > 0$;

- If $y_{ij} = 0$, then

$$f_\beta(C_{ij} + \lambda y_{ij}) - \lambda \mu_i \nu_j y_{ij} = f_\beta(C_{ij}).$$

We conclude that if there exist $(i, j) \in [I] \times [J]$ such that $y_{ij} \neq 0$, then F_β diverges to $-\infty$ as $\lambda \rightarrow +\infty$. Such a pair (i, j) exists since, if not, then $a_i + b_j = 0$ for every $i \in [I]$, but because $b_J = 0$, then $a_i = 0$ for every $i \in [I]$, thereby implying also $b_j = 0$ for every $j \in [J]$. This observation concludes the proof. $\square$

Theorem 2.7 guarantees that, provided we assume $v_J = 0$, for every $\beta > 0$ there exists a unique maximizer (u^β, v^β) for F_β . We now verify that this maximizer is exactly the solution of the nonlinear system defining the algorithm.

Theorem 2.8. *The system (2.1) is equivalent to the condition*

$$\nabla F_\beta(u, v) = 0,$$

provided $v_J = 0$. As a consequence, it admits a unique solution (u^β, v^β) for every $\beta > 0$, coinciding with the global maximizer of F_β .

Proof. Differentiating F_β with respect to u_i in (2.11) and setting it equal to zero gives

$$0 = \frac{\partial F_\beta}{\partial u_i} = -\mu_i + \sum_{j=1}^J f'_\beta(C_{ij} + u_i + v_j) \stackrel{2.3}{=} -\mu_i + \sum_{j=1}^J \varphi(\beta(C_{ij} + u_i + v_j)),$$

giving the first I equations of the system (2.1). The same computation on $j \in [J-1]$ gives the other $J-1$ equations of the system (2.1). Since F_β is twice differentiable (because f_β is) and strictly concave, there must be a unique stationary point $\nabla F_\beta(u, v) = 0$ coinciding with the global maximizer (u^β, v^β) . As we just showed, this stationary point coincides with the solution to (2.1), concluding the proof. $\square$

Theorem 2.8 guarantees the well-posedness of step 3 of the algorithm, which prescribes solving the system (2.1). We next study the zero-temperature behaviour of the corresponding $(u^\beta, v^\beta)_{\beta \in \mathbb{N}}$ maximizers and of the finite-temperature plans G^β .

Lemma 2.9. *Let (u^β, v^β) be the unique maximizer for F_β along $v_J = 0$. Then there exists a positive constant β_0 and a radius $R > 0$ such that, for every $\beta \geq \beta_0$,*

$$\|(u^\beta, v^\beta)\| < R.$$

Proof. Let $x_{ij} := C_{ij} + u_i + v_j$. We first derive a uniform lower bound for x_{ij} . For every $(i, j) \in [I] \times [J]$,

$$\mu_i = \sum_{k=1}^J \varphi(\beta x_{ik}) \geq \varphi(\beta x_{ij}).$$

Since φ is strictly decreasing by Lemma 2.2, we deduce

$$x_{ij} \geq \frac{1}{\beta} \varphi^{-1}(\mu_i).$$

Similarly, one obtains

$$x_{ij} \geq \frac{1}{\beta} \varphi^{-1}(\nu_j).$$

Hence,

$$C_{ij} + u_i + v_j \geq \frac{1}{\beta} \max\{\varphi^{-1}(\mu_i), \varphi^{-1}(\nu_j)\}.$$

For $\beta \geq 1$, this implies

$$u_i + v_j \geq -C_{ij} + \min\{\max\{\varphi^{-1}(\mu_i), \varphi^{-1}(\nu_j)\}, 0\}.$$

Therefore, taking the minimum over (i, j) , we obtain

$$u_i + v_j \geq -\max_{(i,j) \in [I] \times [J]} C_{ij} + \min_{(i,j) \in [I] \times [J]} \min\{\max\{\varphi^{-1}(\mu_i), \varphi^{-1}(\nu_j)\}, 0\}.$$

The right-hand side is uniform in β and independent of i and j . Calling it A , we deduce

$$u_i + v_j \geq A \quad \forall i \in [I], j \in [J]. \quad (2.12)$$

We apply a similar strategy for finding an upper bound.

Fix $i \in [I]$, and let j_i be an index such that

$$\varphi(\beta x_{ij_i}) \geq \varphi(\beta x_{ij}), \quad \forall j \in [J].$$

Then

$$\mu_i = \sum_{j=1}^J \varphi(\beta x_{ij}) \leq J \varphi(\beta x_{ij_i}).$$

Since φ is strictly decreasing by Lemma 2.2, we deduce

$$x_{ij_i} \leq \frac{1}{\beta} \varphi^{-1}\left(\frac{\mu_i}{J}\right).$$

For $\beta \geq 1$, this implies

$$C_{ij_i} + u_i + v_{j_i} \leq \max\left\{\varphi^{-1}\left(\frac{\mu_i}{J}\right), 0\right\}.$$

Similarly, fixing $j \in [J]$ and choosing i_j such that

$$\varphi(\beta x_{i_j j}) \geq \varphi(\beta x_{ij}), \quad \forall i \in [I],$$

we obtain

$$C_{i,j} + u_{i,j} + v_j \leq \max \left\{ \varphi^{-1} \left(\frac{\nu_j}{I} \right), 0 \right\}.$$

Therefore, taking the maximum over columns and rows, we obtain that for every row i there exists j_i such that

$$C_{i,j_i} + u_i + v_{j_i} \leq B, \quad (2.13)$$

and for every column j there exists i_j such that

$$C_{i_j,j} + u_{i_j} + v_j \leq B, \quad (2.14)$$

for some constant B , uniform in β , i and j .

We recall that $v_J := 0$. With this fix, we propagate alternatively the lower bounds and upper bounds to every $(i, j) \in [I] \times [J]$. It holds

$$u_i \stackrel{(2.12)}{\geq} A - v_J = A, \quad \forall i \in [I], \quad (2.15)$$

and, on the other hand, there exists $i_J \in [I]$ such that

$$C_{i_J,J} + u_{i_J} + v_J \stackrel{(2.14)}{\leq} B \implies u_{i_J} \leq B. \quad (2.16)$$

Next, it holds

$$u_{i_J} + v_j \stackrel{(2.12)}{\geq} RA \implies v_j \geq A - B, \quad \forall j \in [J]. \quad (2.17)$$

Furthermore, for every $i \in [I]$ there exists $j_i \in [J]$ such that

$$C_{i,j_i} + u_i + v_{j_i} \stackrel{(2.13)}{\leq} B \implies u_i \leq B - C_{i,j_i} - v_{j_i} \leq B - (A - B) = 2B - A. \quad (2.18)$$

Finally, for every $j \in [J]$, there exists $i_j \in [I]$ such that

$$C_{i_j,j} + u_{i_j} + v_j \stackrel{(2.14)}{\leq} B \implies v_j \leq B - C_{i_j,j} - u_{i_j} \leq B - A. \quad (2.19)$$

This shows that $(u^\beta, v^\beta) \in \mathbb{R}^I \times \mathbb{R}^J$ is uniformly bounded for every $\beta > 1$. Setting $\beta_0 := 1$ and taking a radius sufficiently large, we conclude the proof. $\square$

Corollary 2.10. *For every sequence $(\beta_n)_{n \in \mathbb{N}}$ diverging to $+\infty$, there exists a subsequence such that $(u^{\beta_{n_k}}, v^{\beta_{n_k}})_{k \in \mathbb{N}} \rightarrow (u^\infty, v^\infty)$.*

Proof. The sequence $(u^{\beta_n}, v^{\beta_n})_{n \in \mathbb{N}}$ is eventually confined in a compact set $\overline{B}_R \cap \{v_J = 0\}$, where R is a proper constant found in the previous theorem. Hence it admits a limit point (u^∞, v^∞) . $\square$

Given a sequence $(u^{\beta_{k_n}}, v^{\beta_{k_n}})$ converging to a limit point (u^∞, v^∞) we define the coupling $G^{\beta_{k_n}}$ as

$$G_{ij}^{\beta_{k_n}} = \varphi(\beta_{k_n}(C_{ij} + u_i^{\beta_{k_n}} + v_j^{\beta_{k_n}})).$$

We are now ready to prove Theorem 2.1:

Proof. By construction, G_{ij}^β satisfies the marginal constraints. Indeed,

$$\sum_{i=1}^I G_{ij}^\beta \stackrel{(2.2)}{=} \sum_{i=1}^I \varphi(\beta(C_{ij} + u_i^\beta + v_j^\beta)) \stackrel{(2.1)}{=} \nu_j,$$

since (u^β, v^β) solve (2.1). Similarly,

$$\sum_{j=1}^J G_{ij}^\beta \stackrel{(2.2)}{=} \sum_{j=1}^J \varphi(\beta(C_{ij} + u_i^\beta + v_j^\beta)) \stackrel{(2.1)}{=} \mu_i.$$

Finally, $G_{ij}^\beta \in (0, 1)$ since $\text{Im}(\varphi) = (0, 1)$, by Lemma 2.2.

Since $G^\beta \in \mathcal{G}(\mu, \nu)$, and the transport polytope is compact, any limit G^∞ also belongs to $\mathcal{G}(\mu, \nu)$. We show that G^∞ is an optimal solution for the Kantorovich problem.

Fix a coordinate $(i, j) \in [I] \times [J]$. By Lemma 2.2,

$$G_{i,j}^\infty = \lim_{\beta \rightarrow \infty} \varphi(\beta(C_{ij} + u_{i;\beta}^* + v_{j;\beta}^*)) = \begin{cases} 0 & \text{if } C_{ij} + u_i^\infty + v_j^\infty > 0 \\ (0, 1) & \text{if } C_{ij} + u_i^\infty + v_j^\infty = 0 \\ 1 & \text{if } C_{ij} + u_i^\infty + v_j^\infty < 0 \end{cases} \quad (2.20)$$

The first two conditions, together, read as

$$G_{ij}^\infty(C_{ij} + u_i^\infty + v_j^\infty) = 0, \quad C_{ij} + u_i^\infty + v_j^\infty \geq 0, \quad \forall (i, j) \in [I] \times [J].$$

Since $G^\infty \in \mathcal{G}(\mu, \nu)$ and $C_{ij} + u_i^\infty + v_j^\infty \geq 0$ for every $(i, j) \in [I] \times [J]$, by complementary slackness (Corollary 1.7), the coupling G^∞ is optimal, as well as the Kantorovich potentials $(u^\infty, v^\infty) \in \mathbb{R}^{I+J}$.

On the other hand, the third condition can never occur: if $G_{i,j}^\infty = 1$, then, for every $i' \neq i$,

$$\mu_{i'} \leq 1 - \mu_i = 1 - \sum_{j=1}^J G_{ij}^\infty \leq 1 - G_{ij}^\infty = 0,$$

contradicting $\mu_i > 0$ for all $i \in [I]$.

Since G^∞ is optimal regardless the subsequence chosen, its associated transport cost $\mathcal{U}_\infty$ must also be optimal:

$$\mathcal{U}_\infty := \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij}^\infty = \text{OT}_C(\mu, \nu).$$

□

2.3 Zero-Temperature Limit and Monotonicity

We now compare the optimal value of the effective dual free energy with the transport cost of the finite-temperature plan. The goal is to prove that

$$F_\beta(u^\beta, v^\beta) \longrightarrow \text{OT}_C(\mu, \nu) \quad \text{as } \beta \rightarrow +\infty.$$

Since Theorem 2.1 already shows that the transport costs $\mathcal{U}_\beta$ converge to $\text{OT}_C(\mu, \nu)$, it is enough to study the difference $F_\beta(u^\beta, v^\beta) - \mathcal{U}_\beta$:

$$\begin{aligned} F_\beta(u^\beta, v^\beta) - \mathcal{U}_\beta &= -\langle u^\beta, \mu \rangle - \langle v^\beta, \nu \rangle + \sum_{i,j} f_\beta(C_{ij} + u_i^\beta + v_j^\beta) - \sum_{i,j} C_{ij} G_{ij}^\beta \\ &\stackrel{\dagger}{=} \sum_{i,j} \left[f_\beta(C_{ij} + u_i^\beta + v_j^\beta) - u_i^\beta G_{ij}^\beta - v_j^\beta G_{ij}^\beta - C_{ij} G_{ij}^\beta \right] \\ &= \sum_{i,j} \left[f_\beta(C_{ij} + u_i^\beta + v_j^\beta) - G_{ij}^\beta (C_{ij} + u_i^\beta + v_j^\beta) \right], \end{aligned}$$

where step $\dagger$ uses the fact that G^β , by definition, satisfies the marginal constraints $\mu_i = \sum_{j=1}^J G_{ij}^\beta$ and $\nu_j = \sum_{i=1}^I G_{ij}^\beta$. Setting

$$x_{ij}^\beta := C_{ij} + u_i^\beta + v_j^\beta$$

and observing that $G_{ij}^\beta = \varphi(\beta x_{ij}^\beta)$, we obtain

$$F_\beta(u^\beta, v^\beta) - \mathcal{U}_\beta = \sum_{i,j} \left[f_\beta(x_{ij}^\beta) - x_{ij}^\beta \varphi(\beta x_{ij}^\beta) \right] \stackrel{2.3}{=} \sum_{i,j} \left[f_\beta(x_{ij}^\beta) - x_{ij}^\beta f'_\beta(x_{ij}^\beta) \right] \quad (2.21)$$

The expression $f_\beta(x_{ij}^\beta) - x_{ij}^\beta f'_\beta(x_{ij}^\beta)$ can be further rewritten in terms of the function ψ , by means of (2.6). Setting $y := \beta x_{ij}^\beta$, we obtain

$$f_\beta(x_{ij}^\beta) - x_{ij}^\beta f'_\beta(x_{ij}^\beta) = -\frac{1}{\beta} \psi(y) + \frac{y \psi'(y)}{\beta} = \frac{y \psi'(y) - \psi(y)}{\beta}. \quad (2.22)$$

Our claim is that this expression, that can be interpreted as an error term, converges to zero in the large β limit.

Theorem 2.11 (Convergence of the effective value). *It holds*

$$\lim_{\beta \rightarrow \infty} F_\beta(u^\beta, v^\beta) - \mathcal{U}_\beta = 0.$$

As a consequence,

$$\lim_{\beta \rightarrow \infty} F_\beta(u^\beta, v^\beta) = \text{OT}_C(\mu, \nu).$$

Proof. First suppose that $y = \beta x_{ij}^\beta$ is bounded as $\beta \rightarrow \infty$. Since ψ is smooth on $\mathbb{R}$, the function $y \mapsto \psi(y) - y\psi'(y)$ is bounded on compact subsets of $\mathbb{R}$. Hence

$$\left| \frac{\psi(y_\beta) - y_\beta \psi'(y_\beta)}{\beta} \right| \leq \frac{C}{\beta} \rightarrow 0.$$

Next, suppose $y \rightarrow \pm\infty$. Therefore, we compute

$$\begin{aligned} \lim_{|y| \rightarrow +\infty} \frac{y\psi'(y) - \psi(y)}{y} &= \lim_{|y| \rightarrow +\infty} \psi'(y) - \frac{\psi(y)}{y} \\ &= \lim_{|y| \rightarrow +\infty} -\frac{1}{y} - \frac{e^{-y}}{e^{-y} - 1} - \frac{\log(|e^{-y} - 1|)}{y} + \frac{\log(|y|)}{y} \\ &= \lim_{|y| \rightarrow +\infty} -\frac{e^{-y}}{e^{-y} - 1} - \frac{\log(|e^{-y} - 1|)}{y}, \end{aligned}$$

where the derivative has been computed thanks to 2.3, that gives $\psi'(y) = -\varphi(y)$. If $y \rightarrow +\infty$, then both terms converge to zero, and the overall limit is zero. If $y \rightarrow -\infty$, then the terms converge to ± 1 , eventually canceling each other. Hence

$$\lim_{|y| \rightarrow +\infty} \frac{y\psi'(y) - \psi(y)}{y} = 0.$$

By Lemma 2.9, $|x_{ij}^\beta|$ is bounded by a positive constant K_{ij} . Therefore, $|\beta x_{ij}^\beta| \leq \beta K_{ij}$. This implies that, if $x_{ij}^\beta \neq 0$,

$$\lim_{\beta \rightarrow \infty} f_\beta(x_{ij}^\beta) - x_{ij}^\beta f'_\beta(x_{ij}^\beta) = \lim_{\beta \rightarrow \infty} \frac{\beta x_{ij}^\beta \psi'(\beta x_{ij}^\beta) - \psi(\beta x_{ij}^\beta)}{\beta x_{ij}^\beta} \cdot x_{ij}^\beta.$$

Since $|x_{ij}^\beta|$ is bounded and the other limit yields 0, we conclude that the overall limit is 0. The conclusion now follows from Equation (2.21):

$$\lim_{\beta \rightarrow \infty} F_\beta(u^\beta, v^\beta) - \mathcal{U}_\beta = \lim_{\beta \rightarrow \infty} \sum_{i,j} \left[f_\beta(x_{ij}^\beta) - x_{ij}^\beta f'_\beta(x_{ij}^\beta) \right] = 0,$$

by taking the limit componentwise.

Combining Theorem 2.1,

$$\lim_{\beta \rightarrow \infty} F_\beta(u^\beta, v^\beta) = \lim_{\beta \rightarrow \infty} \mathcal{U}_\beta = \text{OT}_C(\mu, \nu).$$

□

We can strengthen Theorems 2.1 and 2.11 by showing that the convergence is monotone:

Theorem 2.12 (Monotone Convergence to Optimal Transport Cost). *The functions $F_\beta(u^\beta, v^\beta)$ and $\mathcal{U}_\beta$ are monotonically decreasing functions of β .*

Proof. In what follows, we will use the notations $\frac{dF_\beta}{d\beta}$ and $\frac{\partial F_\beta}{\partial \beta}$ to differentiate between the total derivative and partial derivative of F_β with respect to β , respectively.

By the chain rule,

$$\frac{dF_\beta}{d\beta} = \frac{\partial F_\beta(u^\beta, v^\beta)}{\partial \beta} + \sum_{i=1}^I \frac{\partial F_\beta(u^\beta, v^\beta)}{\partial u_i} \frac{\partial u_i^\beta}{\partial \beta} + \sum_{j=1}^J \frac{\partial F_\beta(u^\beta, v^\beta)}{\partial v_j} \frac{\partial v_j^\beta}{\partial \beta}.$$

Since $\nabla F_\beta(u, v) = 0$ at (u^β, v^β) , the only remaining term is

$$\frac{dF_\beta}{d\beta} = \frac{\partial F_\beta(u^\beta, v^\beta)}{\partial \beta} = \frac{\partial}{\partial \beta} \left(-\langle u^\beta, \mu \rangle - \langle v^\beta, \nu \rangle + \sum_{i=1}^I \sum_{j=1}^J f_\beta(x_{ij}^\beta) \right) = \sum_{i=1}^I \sum_{j=1}^J \frac{\partial f_\beta(x_{ij}^\beta)}{\partial \beta} \leq 0,$$

where the last inequality follows from Lemma 2.4.

Next, we compute the total derivative of $\mathcal{U}_\beta$: setting $x_{ij}^\beta := C_{ij} + u_i^\beta + v_j^\beta$,

$$\begin{aligned} \frac{d\mathcal{U}_\beta}{d\beta} &= \sum_{i,j} C_{ij} \frac{dG_{ij}^\beta}{d\beta} = \sum_{i,j} [x_{ij}^\beta - u_i^\beta - v_j^\beta] \frac{dG_{ij}^\beta}{d\beta} \\ &= \sum_{i,j} x_{ij}^\beta \frac{dG_{ij}^\beta}{d\beta} - \sum_{i,j} u_i^\beta \frac{dG_{ij}^\beta}{d\beta} - \sum_{i,j} v_j^\beta \frac{dG_{ij}^\beta}{d\beta} \\ &= \sum_{i,j} x_{ij}^\beta \frac{dG_{ij}^\beta}{d\beta} - \sum_{i=1}^I u_i^\beta \frac{d}{d\beta} \left[\sum_{j=1}^J G_{ij}^\beta \right] - \sum_{j=1}^J v_j^\beta \frac{d}{d\beta} \left[\sum_{i=1}^I G_{ij}^\beta \right] \\ &= \sum_{i,j} x_{ij}^\beta \frac{dG_{ij}^\beta}{d\beta} - \sum_{i=1}^I u_i^\beta \frac{d}{d\beta} [\mu_i] - \sum_{j=1}^J v_j^\beta \frac{d}{d\beta} [\nu_j] = \sum_{i,j} x_{ij}^\beta \frac{dG_{ij}^\beta}{d\beta} - 0 - 0 \\ &= \sum_{i,j} x_{ij}^\beta \frac{dG_{ij}^\beta}{d\beta} + \sum_{i=1}^I \beta(u_i^\beta)' \frac{d}{d\beta} [\mu_i] - \sum_{j=1}^J \beta(v_j^\beta)' \frac{d}{d\beta} [\nu_j] \\ &= \sum_{i,j} x_{ij}^\beta \frac{dG_{ij}^\beta}{d\beta} + \sum_{i,j} \beta(u_i^\beta)' \frac{dG_{ij}^\beta}{d\beta} + \sum_{i,j} \beta(v_j^\beta)' \frac{dG_{ij}^\beta}{d\beta} \\ &= \sum_{i,j} [x_{ij}^\beta + \beta(u_i^\beta)' + \beta(v_j^\beta)'] \frac{dG_{ij}^\beta}{d\beta} = \sum_{i,j} [x_{ij}^\beta + \beta(x_{ij}^\beta)'] \frac{dG_{ij}^\beta}{d\beta} \\ &= \sum_{i,j} (\beta x_{ij}^\beta)' \frac{dG_{ij}^\beta}{d\beta} = \sum_{i,j} (\beta x_{ij}^\beta)' \frac{d(\varphi(\beta x_{ij}^\beta))}{d\beta} = \sum_{i,j} (\beta x_{ij}^\beta)' \varphi'(\beta x_{ij}^\beta) (\beta x_{ij}^\beta)' \\ &= \sum_{i,j} [(\beta x_{ij}^\beta)']^2 \varphi'(\beta x_{ij}^\beta). \end{aligned}$$

The latter quantity is negative because $\varphi' < 0$ by Lemma 2.2. □

Remark 2.13. By monotonicity,

$$\text{OT}_C(\mu, \nu) \leq \mathcal{U}_{\beta_{i+1}} \leq \mathcal{U}_{\beta_i}.$$

Hence the stopping condition

$$\frac{\mathcal{U}_{\beta_i} - \mathcal{U}_{\beta_{i+1}}}{\mathcal{U}_{\beta_i}} < \epsilon$$

is a relative stabilization test along the annealing path, rather than an exact estimate of the error $|\mathcal{U}_{\beta_{i+1}} - \text{OT}_C(\mu, \nu)|$.

Remark 2.14. We remark that the proof of Theorem 2.12 follows the derivation in [12, Appendix E]. However, we note that in [12, Eq. E17], the authors compute the mixed derivative $\partial_u \partial_\beta F_\beta$ neglecting the fact that u^β depends on β . This leads to the erroneous conclusion in [12, Eq. E19] that

$$\frac{\partial \mathcal{U}}{\partial \beta} = \sum_{i,j} \beta (x_{ij}^\beta)^2 \varphi'(\beta x_{ij}^\beta),$$

an equality that would hold if and only if $(x_{ij}^\beta)' = (u_i^\beta)' + (v_j^\beta)' = 0$, which is generally false. Therefore, we do not regard this argument as rigorous in its present form. However, this mistake does not invalidate the validity of the result.

Remark 2.15 (Comparison with [12]). Theorems 2.1, 2.11, and 2.12 provide a rigorous proof of the convergence and monotonicity properties stated in [12, Prop. 6]. The route followed here is different from the one suggested by the physical derivation.

In [12], the effective free energy F_β and the quantity $\mathcal{U}_\beta$ are interpreted as mean-field approximations of the corresponding finite-temperature free energy and internal energy. Their zero-temperature behaviour is then justified heuristically by combining this mean-field interpretation with a Laplace-type argument. The latter step is not made explicit in [12], but it is relatively standard in the present finite-dimensional setting, as shown in Theorem 6.5. By contrast, a direct justification of the mean-field asymptotics is considerably more delicate; this is precisely the issue addressed in Chapter 5. For this reason, in the present chapter we followed a different approach.

2.4 Solving the Nonlinear Optimality System

The computational bottleneck in the finite-temperature algorithm is the solution of the nonlinear system (2.1). In this section we make explicit a Newton method for computing

its unique gauge-fixed solution. The structure is analogous in spirit to the dual equations arising in Cuturi's entropic regularization [21]: in both cases, one solves for dual variables so that the transport plan reconstructed from them satisfies the prescribed marginals. However, the scalar response function is different. In entropic OT the response is exponential and leads to Sinkhorn scaling [28], whereas in the finite-temperature system studied here the response is

$$x \mapsto \varphi(\beta x),$$

and the resulting nonlinear system is solved by a Newton scheme [29, Sec. 1.2].

Define the residuals

$$R_i(u, v) := \sum_{j=1}^J \varphi(\beta x_{ij}) - \mu_i, \quad i = 1, \dots, I,$$

and

$$S_j(u, v) := \sum_{i=1}^I \varphi(\beta x_{ij}) - \nu_j, \quad j = 1, \dots, J-1,$$

where

$$x_{ij} := C_{ij} + u_i + v_j.$$

Together, they form a map

$$\mathcal{N}_\beta(u, v) := \begin{pmatrix} R(u, v) \\ S(u, v) \end{pmatrix} : \mathbb{R}^{I+J-1} \rightarrow \mathbb{R}^{I+J-1},$$

omitting $v_J = 0$ from the unknowns. We compute the Jacobian matrix of $\mathcal{N}_\beta$, that is, its linearisation: it is easy to see that, by Theorem 2.8, the Jacobian of $\mathcal{N}_\beta$ is the Hessian of F_β , computed in Lemma 2.6:

$$J_{\mathcal{N}_\beta}(u, v) := \nabla \mathcal{N}_\beta(u, v) = \nabla^2 F_\beta = \begin{pmatrix} D_u & H \\ H^T & D_v \end{pmatrix},$$

where

$$(D_u)_{ii} = \sum_{j=1}^J f''_\beta(x_{ij}), \quad (D_v)_{jj} = \sum_{i=1}^I f''_\beta(x_{ij}),$$

and

$$H_{ij} = f''_\beta(x_{ij}), \quad i = 1, \dots, I, \quad j = 1, \dots, J-1.$$

Since

$$f''_\beta(x) = \beta \varphi'(\beta x) < 0,$$

the diagonal entries of D_v are strictly negative. Hence D_v is invertible.

Starting from a given point $X_n = (u, v)_n$, we define an update as

$$X_{n+1} := X_n + \eta_{\beta,n} \delta X,$$

where $\eta_{\beta,n}$ is a step size, and the Newton increment

$$\delta X = (\delta u, \delta v)$$

is obtained by solving the Newton system

$$J_{\mathcal{N}_\beta}(X_n) \delta X = -\mathcal{N}_\beta(X_n),$$

that is,

$$\begin{cases} D_u(X_n) \delta u + H(X_n) \delta v = -R(X_n), \\ H^T(X_n) \delta u + D_v(X_n) \delta v = -S(X_n). \end{cases}$$

Since D_v is invertible, one may eliminate δv and solve the Schur complement system

$$(D_u - H D_v^{-1} H^T) \delta u = -R + H D_v^{-1} S.$$

Then

$$\delta v = D_v^{-1}(-S - H^T \delta u).$$

3 Finite-Temperature Regularization and Duality

In Chapter 2 we introduced the finite-temperature algorithm directly through its nonlinear dual system. We proved that, after fixing $v_J = 0$, this system is well posed, since it is the first-order optimality system of the strictly concave effective free energy F_β . We also proved that the associated finite-temperature plans G^β converge, along subsequences, to Kantorovich optimizers, and that

$$F_\beta(u^\beta, v^\beta) - \mathcal{U}_\beta \rightarrow 0 \quad \text{as } \beta \rightarrow +\infty$$

by Theorem 2.11.

The goal of this chapter is to reinterpret the same construction from the primal side. More precisely, we show that the finite-temperature system (2.1) is the optimality system of a regularized Kantorovich problem. This places the algorithm in a standard convex optimization framework and makes its relation with entropic regularization of optimal transport [21] explicit.

3.1 Regularization as a variational principle

A constrained optimization problem can be written as

$$\min_{x \in \mathcal{C}} f(x),$$

or, equivalently, in the form

$$\min_x \{f(x) + \iota_{\mathcal{C}}(x)\},$$

where $\iota_{\mathcal{C}}$ is the indicator function of the feasible set, equal to 0 on $\mathcal{C}$ and $+\infty$ outside it [23, Sec. 3.1.4]. It is common in the literature to replace a hard constraint by a penalty or barrier term, obtaining a family of regularized problems

$$\min_x \{f(x) + \varepsilon R(x)\}, \quad \varepsilon > 0.$$

Depending on the choice of R , this procedure may serve different purposes: it may incorporate inequality constraints through a barrier term [30, Example 1.3], improve strict or strong convexity [20, Sec. 4.1], smooth the dual problem [20, Prop. 4.6], and improve the numerical stability of iterative methods such as gradient descent or Newton methods [23, Sec. 6.3]. In particular, interior-point methods replace the nonsmooth $\iota_{\mathcal{C}}$ by smooth logarithmic barriers [23, Sec. 11.2].

This same philosophy is applied in entropic optimal transport [21]. The discrete Kantorovich problem is the linear program

$$\min_{G \in \mathcal{G}(\mu, \nu)} \sum_{i,j} C_{ij} G_{ij}.$$

Its solutions need not be unique and often lie on low-dimensional faces of the transport polytope. Cuturi's entropic regularization replaces it by

$$\min_{G \in \mathcal{G}(\mu, \nu)} \left\{ \sum_{i,j} C_{ij} G_{ij} + \varepsilon \sum_{i,j} G_{ij} (\log G_{ij} - 1) \right\},$$

which is the standard regularized optimal-transport problem studied in [20, 21]. This regularization produces a unique interior solution, yields a smooth dual formulation, and leads to a computationally fast algorithm based on the Sinkhorn method [28].

As observed in [12, Sec. II], the finite-temperature algorithm studied in this thesis fits the same variational paradigm, but with a different scalar regularizer. We look for a regularized Kantorovich problem of the form

$$\min_{G \in \mathcal{G}(\mu, \nu)} \left\{ \sum_{i,j} C_{ij} G_{ij} + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) \right\}, \quad (3.1)$$

and we determine which scalar function r produces the system of optimal solutions in (2.1), which prescribes

$$G_{ij} = \varphi(\beta(C_{ij} + u_i + v_j)), \quad \forall (i, j) \in [I] \times [J]. \quad (3.2)$$

Following the same strategy that Koehl et al. use for entropic optimal transport, we write the Lagrangian of the regularized problem, using the Lagrangian viewpoint already introduced in (1.8), and compute its critical points. Given $(u, v) \in \mathbb{R}^I \times \mathbb{R}^J$ Kantorovich potentials, the regularized Lagrangian is

$$\begin{aligned} \mathcal{L}_\beta(G; u, v) &= \sum_{i,j} C_{ij} G_{ij} + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) + \sum_{i=1}^I u_i \left(\sum_{j=1}^J G_{ij} - \mu_i \right) + \sum_{j=1}^J v_j \left(\sum_{i=1}^I G_{ij} - \nu_j \right) \\ &= -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i,j} \left[G_{ij} (C_{ij} + u_i + v_j) + \frac{1}{\beta} r(G_{ij}) \right]. \end{aligned} \quad (3.3)$$

Optimizing the Lagrangian with respect to the potentials imposes the marginal constraints. Optimizing with respect to the primal variables gives, at every interior critical point,

$$0 = \frac{\partial \mathcal{L}_\beta}{\partial G_{ij}} = C_{ij} + u_i + v_j + \frac{1}{\beta} r'(G_{ij}), \quad \forall (i, j) \in [I] \times [J].$$

Equivalently,

$$r'(G_{ij}) = -\beta(C_{ij} + u_i + v_j). \quad (3.4)$$

To identify r , we impose a posteriori that (3.2) and (3.4) coincide. Since φ is a homeomorphism from $\mathbb{R}$ to $(0, 1)$ by Lemma 2.2, its inverse exists. Thus (3.2) gives

$$\varphi^{-1}(G_{ij}) = \beta(C_{ij} + u_i + v_j).$$

Combining this with (3.4), we obtain

$$r'(G_{ij}) = -\varphi^{-1}(G_{ij}). \quad (3.5)$$

Therefore

$$r'(g) = -\varphi^{-1}(g), \quad g \in (0, 1).$$

By Lemma 2.3,

$$\psi'(y) = -\varphi(y) \implies \varphi^{-1}(g) = (\psi')^{-1}(-g)$$

Hence,

$$r'(g) = -(\psi')^{-1}(-g) \quad (3.6)$$

We recognize that the right hand-side is written precisely in the form of Corollary A.7. Since $g \mapsto \psi(-g)$ is C^1 , strictly convex (by Lemma 2.3 and 2.4 combined) and a bijection between $\mathbb{R}$ and $(-1, 0)$ (by Lemma 2.2 and 2.3 combined), then, by Corollary A.7,

$$(\psi^*)'(s) = (\psi')^{-1}(s), \quad s \in (-1, 0).$$

Taking $s = -g$, with $g \in (0, 1)$, gives

$$(\psi^*)'(-g) = (\psi')^{-1}(-g). \quad (3.7)$$

Comparing (3.6) and (3.7),

$$r'(g) = \frac{d\psi^*(-g)}{dg},$$

where the derivative differentiates the map $g \mapsto \psi^*(-g)$. By the chain rule,

$$r'(g) = -(\psi^*)'(-g) \cdot (-1) = (\psi^*)'(-g),$$

Since $(0, 1)$ is connected,

$$r(g) = \psi^*(-g) + C,$$

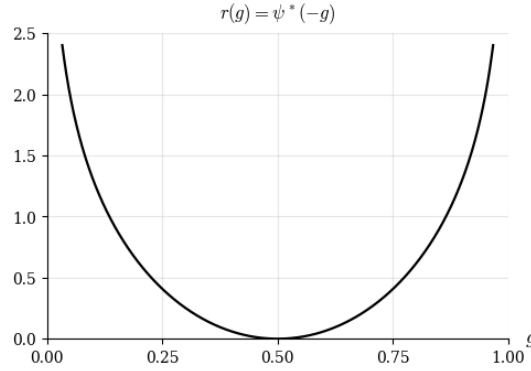


Figure 2: Profile of the scalar function $r(g) = \psi^*(-g)$, where $\psi(y) = \log \frac{1-e^{-y}}{y}$. The plot shows that r effectively acts as a penalty term, forcing $G_{ij} \in (0, 1)$ for every $(i, j) \in [I] \times [J]$.

where C is a constant of integration. We can safely take $C := 0$. The effective domain of r is $(0, 1)$, and we set

$$r(g) = +\infty \quad g \notin (0, 1).$$

Thus the regularizer acts as a barrier term [30, Example 1.3] that forces the problem to live in its natural convex domain, consistently with the partial Lagrangian convention used in (1.8). The other constraints are enforced by maximizing the dual objective in the Lagrange multipliers.

3.2 Duality with the effective free energy

We now prove that the Lagrangian dual of the regularized primal problem is precisely the effective free-energy maximization problem studied in Chapter 2.

Theorem 3.1 (Duality for the finite-temperature regularization). *Consider the regularized Kantorovich problem with parameter $\beta > 0$:*

$$P_\beta(\mu, \nu) := \inf_{G \in \mathcal{G}(\mu, \nu)} \left\{ \sum_{i,j} C_{ij} G_{ij} + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) \right\}, \quad (3.8)$$

where

$$r(g) = \psi^*(-g), \quad \psi(y) = \log \int_0^1 e^{-ty} dt.$$

Its Lagrangian dual, in the sense of (1.7), is

$$D_\beta(\mu, \nu) = \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} F_\beta(u, v),$$

where

$$F_\beta(u, v) = -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i,j} f_\beta(C_{ij} + u_i + v_j),$$

and

$$f_\beta(x) = -\frac{1}{\beta} \psi(\beta x).$$

Moreover, there is no duality gap:

$$P_\beta(\mu, \nu) = D_\beta(\mu, \nu).$$

Proof. Using the Lagrangian (3.3), we have

$$\begin{aligned} D_\beta(\mu, \nu) &= \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} \inf_{G \in \mathbb{R}^{IJ}} \mathcal{L}_\beta(G; u, v) \\ &= \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i,j} \inf_{G_{ij} \in \mathbb{R}} \left\{ G_{ij}(C_{ij} + u_i + v_j) + \frac{1}{\beta} r(G_{ij}) \right\} \\ &= \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} -\langle u, \mu \rangle - \langle v, \nu \rangle + \frac{1}{\beta} \sum_{i,j} \inf_{G_{ij} \in \mathbb{R}} \{ G_{ij} \beta (C_{ij} + u_i + v_j) + r(G_{ij}) \}. \end{aligned}$$

Clearly, the infimum is taken in the effective domain of r , which forces $G_{ij} \in (0, 1)$. For fixed (u, v) , the infimum over G can be computed componentwise as

$$\inf_{G_{ij} \in \mathbb{R}} \{ G_{ij} \beta (C_{ij} + u_i + v_j) + r(G_{ij}) \} = \inf_{g \in \mathbb{R}} \{ \beta (C_{ij} + u_i + v_j) g + \psi^*(-g) \}.$$

Setting

$$x_{ij} := C_{ij} + u_i + v_j \quad \text{and} \quad s := -g,$$

this becomes

$$\inf_{g \in \mathbb{R}} \{ \beta x_{ij} g + \psi^*(-g) \} = -\sup_{s \in \mathbb{R}} \{ \beta x_{ij} s - \psi^*(s) \}.$$

By the Fenchel–Moreau theorem A.4, since ψ is convex, smooth, and finite on $\mathbb{R}$,

$$\sup_{s \in \mathbb{R}} \{ \beta x_{ij} s - \psi^*(s) \} = \psi^{**}(\beta x_{ij}) = \psi(\beta x_{ij}).$$

Hence

$$\begin{aligned} D_\beta(\mu, \nu) &= \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} -\langle u, \mu \rangle - \langle v, \nu \rangle - \frac{1}{\beta} \sum_{i,j} \psi(\beta(C_{ij} + u_i + v_j)) \\ &= \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i,j} f_\beta(C_{ij} + u_i + v_j) \\ &= \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} F_\beta(u, v). \end{aligned}$$

It remains to certify the absence of a duality gap and the existence of optimal solutions. To this end, we apply the KKT conditions for convex optimization problems. The feasible region for the primal optimization problem is

$$X = [0, 1]^{IJ}, \quad g_i(G) = 0, \quad h_j(G) = 0,$$

where the constraints are the affine functions

$$g_i(G) = \sum_{j=1}^J G_{ij} - \mu_i, \quad i \in [I],$$

and

$$h_j(G) = \sum_{i=1}^I G_{ij} - \nu_j, \quad j \in [J].$$

Moreover, the objective is strictly convex in $(0, 1)^{IJ}$, and, by Lemma 1.2, there exists a feasible point in the effective domain of the regularizer. This is Slater's condition (A.2), which ensures strong duality for the convex problem. By the KKT conditions (A.3), there exists a feasible minimizer $G^\beta \in \mathcal{G}(\mu, \nu)$ if and only if there exist $(u, v) \in \mathbb{R}^{I+J}$ such that

$$0 \in \partial \left(\sum_{i,j} C_{ij} G_{ij}^\beta + \frac{1}{\beta} \sum_{i,j} r(G_{ij}^\beta) \right) + \sum_{i=1}^I u_i \nabla g_i(G^\beta) + \sum_{j=1}^J v_j \nabla h_j(G^\beta).$$

Since the constraints are affine and G^β lies in the effective domain of r , this is equivalent to finding critical points of the regularized Lagrangian, namely

$$C_{ij} + u_i + v_j + \frac{1}{\beta} r'(G_{ij}^\beta) = 0, \quad \forall (i, j) \in [I] \times [J].$$

Equivalently,

$$r'(G_{ij}^\beta) = -\beta(C_{ij} + u_i + v_j), \quad \forall (i, j) \in [I] \times [J].$$

By our choice of r , this is equivalent to

$$G_{ij}^\beta = \varphi(\beta(C_{ij} + u_i + v_j)), \quad \forall (i, j) \in [I] \times [J].$$

After fixing $v_J = 0$, the solution to this system exists and is unique by Theorem 2.8. Therefore, the regularized primal and the Lagrangian dual have the same optimal value, namely

$$P_\beta(\mu, \nu) = D_\beta(\mu, \nu).$$

□

The theorem shows that the effective free energy F_β is the Lagrangian dual of the regularized primal problem (3.8). However, it remains to prove that the regularized optimal value converges to the optimal transport cost:

$$\lim_{\beta \rightarrow \infty} D_\beta(\mu, \nu) = \lim_{\beta \rightarrow \infty} P_\beta(\mu, \nu) \stackrel{?}{=} \text{OT}_C(\mu, \nu).$$

This fact follows from Theorem 2.11.

3.3 Primal consequences of the finite-temperature regularization

After having established the duality between

$$D_\beta(\mu, \nu) = \sup_{(u,v) \in \mathbb{R}^I \times \mathbb{R}^J} F_\beta(u, v),$$

and

$$P_\beta(\mu, \nu) := \inf_{G \in \mathcal{G}(\mu, \nu)} \left\{ \sum_{i,j} C_{ij} G_{ij} + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) \right\}, \quad (3.9)$$

in this section we study some of the properties that this duality yields.

We recall that

$$r(g) = \psi^*(-g) = \sup_{y \in \mathbb{R}} \{-gy - \psi(y)\} = - \int_{C_0}^g \varphi^{-1}(t) dt, \quad g \in (0, 1), \quad (3.10)$$

and

$$\psi(y) = \log \int_0^1 e^{-ty} dt = \log \frac{1 - e^{-y}}{y}, \quad \varphi(y) = -\psi'(y) = \frac{e^{-y}}{e^{-y} - 1} + \frac{1}{y}.$$

The value $C_0 \in (0, 1)$ for the integration is yet to be determined. The profiles of r , ψ and φ can be visualized in Figures 1 and 2.

Theorem 3.2 (Analytic properties of the finite-temperature regularizer). *$r(g)$ is a smooth, strictly convex and symmetric function on its domain $(0, 1)$. Moreover, it is nonnegative, with value zero only at $r(1/2) = 0$, and diverges at the boundaries of its domain.*

Proof. Observe that we can write

$$\begin{aligned} r(g) &= \sup_{y \in \mathbb{R}} \{-gy - \psi(y)\} = \sup_{y \in \mathbb{R}} \left\{ -gy - \log \frac{1 - e^{-y}}{y} \right\} \\ &= \sup_{y \in \mathbb{R}} \left\{ -\log \frac{e^{gy}(1 - e^{-y})}{y} \right\} = -\log \inf_{y \in \mathbb{R}} \left\{ \frac{e^{gy} - e^{y(g-1)}}{y} \right\}. \end{aligned}$$

This expression makes it clear that r is invariant under the map $g \mapsto 1 - g$, hence r is symmetric with respect to $g = 1/2$. Furthermore, evaluating the infimum at $y = 1/g$, we obtain

$$\begin{aligned} r(g) &= -\log \inf_{y \in \mathbb{R}} \left\{ \frac{e^{gy} - e^{y(g-1)}}{y} \right\} \\ &\geq -\log \frac{e^{g \cdot 1/g} - e^{1/g(g-1)}}{1/g} = -\log \frac{e - e^{1-1/g}}{1/g} \\ &= \log g - \log e - \log(1 - e^{-1/g}) \xrightarrow{g \rightarrow 0^+} +\infty \end{aligned}$$

By symmetry,

$$\lim_{g \rightarrow 1^-} r(g) = +\infty.$$

Furthermore, r is smooth: by Lemma 2.2, the map $\varphi : \mathbb{R} \rightarrow (0, 1)$ is a smooth bijection, hence its inverse is smooth. Since

$$r'(g) = -\varphi^{-1}(g),$$

we conclude that $r \in C^\infty(0, 1)$. Differentiating $r'(g) = -\varphi^{-1}(g)$, we obtain

$$r''(g) = -\frac{1}{\varphi'(\varphi^{-1}(g))}.$$

By Lemma 2.2, φ is strictly decreasing and smooth, so $\varphi' < 0$. Thus $r''(g) > 0$ for every $g \in (0, 1)$.

By strict convexity and continuity, we deduce that r has a unique global minimizer. By symmetry, this minimizer must be $g = 1/2$. Computing its value,

$$r(1/2) = -\log \inf_{y \in \mathbb{R}} \left\{ \frac{e^{y/2} - e^{-y/2}}{y} \right\} = -\log \inf_{y \in \mathbb{R}} \left\{ \frac{\sinh(y/2)}{y/2} \right\}.$$

For every $x \in \mathbb{R}$,

$$\sinh^2(x) = \left(\sum_{i=0}^{\infty} \frac{x^{2i+1}}{(2i+1)!} \right)^2 \geq x^2,$$

with equality at $x = 0$. Hence the infimum is attained for $y = 0$, understood by continuity:

$$r(1/2) = -\log \lim_{y \rightarrow 0} \frac{e^{y/2} - e^{-y/2}}{y} = -\log 1 = 0.$$

As a byproduct, $r(g) \geq 0$ for every $g \in (0, 1)$. Moreover, for consistency, we deduce that $C_0 = 1/2$ in (3.10):

$$r(g) = -\int_{1/2}^g \varphi^{-1}(t) dt, \quad g \in (0, 1).$$

□

Theorem 3.3 (Analytic properties of the regularized primal). *The regularized primal problem*

$$P_\beta(\mu, \nu) := \inf_{G \in \mathcal{G}(\mu, \nu)} \left\{ \sum_{i,j} C_{ij} G_{ij} + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) \right\}$$

admits a unique solution for every $\beta > 0$. Moreover, it is monotonically decreasing in β .

Proof. We understand $r : [0, 1] \rightarrow \mathbb{R} \cup \{+\infty\}$ by setting $r(0) = r(1) = +\infty$. By Lemma 1.2 there exists a feasible coupling $G \in \mathcal{G}(\mu, \nu)$ with $G_{ij} \in (0, 1)$ for every $(i, j) \in [I] \times [J]$. Since $\mathcal{U}$ is affine, r is strictly convex, and $\mathcal{G}(\mu, \nu)$ is compact by Proposition 1.1, we conclude that, for every $\beta > 0$, there exists a unique minimizer

$$G^\beta = \arg \min_{G \in \mathcal{G}(\mu, \nu)} \left\{ \mathcal{U}(G) + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) \right\}.$$

Since r diverges to $+\infty$ at the boundary of its domain, $G_{ij}^\beta \in (0, 1)$ for every $(i, j) \in [I] \times [J]$.

Moreover, if $\beta_2 > \beta_1 > 0$, since $r(g) \geq 0$ for every $g \in (0, 1)$, we have

$$\mathcal{U}(G) + \frac{1}{\beta_2} \sum_{i,j} r(G_{ij}) \leq \mathcal{U}(G) + \frac{1}{\beta_1} \sum_{i,j} r(G_{ij}).$$

Taking the infimum over $G \in \mathcal{G}(\mu, \nu)$ gives

$$P_{\beta_2}(\mu, \nu) \leq P_{\beta_1}(\mu, \nu).$$

□

Remark 3.4. Combining Corollary 3.3 and the simple observation that, for every $\beta > 0$,

$$P_\beta(\mu, \nu) = \min_{G \in \mathcal{G}(\mu, \nu)} \left\{ \mathcal{U}(G) + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) \right\} \geq \min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G) = \text{OT}_C(\mu, \nu),$$

we conclude that $\beta \mapsto P_\beta(\mu, \nu)$ has a limit as $\beta \rightarrow \infty$. However, we cannot deduce easily that this limit is $\text{OT}_C(\mu, \nu)$. In fact, writing

$$P_\beta(\mu, \nu) = \mathcal{U}(G^\beta) + \frac{1}{\beta} \sum_{i,j} r(G_{ij}^\beta),$$

we see that $r(G_{ij}^\beta)$ depends on β . If it scales as $r(G_{ij}^\beta) \gg \beta$, then the convergence is not guaranteed. In fact, when we showed in Theorem 2.1, that the dual formulation converges to $\text{OT}_C(\mu, \nu)$, we allowed G_{ij}^∞ to be 0 if $C_{ij} + u_i^\infty + v_j^\infty$. Therefore, we need

a further asymptotic argument, similar to that of Theorem 2.11, to deduce the same conclusion from the primal side. This is a clear difference from entropic regularization [21]. However, we already know that $P_\beta(\mu, \nu) \rightarrow OT_C(\mu, \nu)$ as $\beta \rightarrow \infty$ by combining Theorem 2.1 and Theorem 3.1.

Finally, we can easily provide a result dual to Theorem 2.12, which showed that $F_\beta(u^\beta, v^\beta)$ and $\mathcal{U}_\beta$ are decreasing functions of β :

Theorem 3.5. *Let G^β be the unique minimizer of the regularized primal Kantorovich problem:*

$$G^\beta = \arg \min_{G \in \mathcal{G}(\mu, \nu)} \left\{ \mathcal{U}(G) + \frac{1}{\beta} \sum_{i,j} r(G_{ij}) \right\}.$$

Then $\beta \mapsto \mathcal{U}(G^\beta)$ is monotonically decreasing, while $\beta \mapsto \sum_{i,j} r(G_{ij}^\beta)$ is monotonically increasing.

Proof. If $\beta_2 > \beta_1 > 0$, then by optimality

$$\mathcal{U}(G^{\beta_1}) + \frac{1}{\beta_1} \sum_{i,j} r(G_{ij}^{\beta_1}) \leq \mathcal{U}(G^{\beta_2}) + \frac{1}{\beta_1} \sum_{i,j} r(G_{ij}^{\beta_2}),$$

and

$$\mathcal{U}(G^{\beta_2}) + \frac{1}{\beta_2} \sum_{i,j} r(G_{ij}^{\beta_2}) \leq \mathcal{U}(G^{\beta_1}) + \frac{1}{\beta_2} \sum_{i,j} r(G_{ij}^{\beta_1}).$$

Hence

$$\frac{1}{\beta_2} \left(\sum_{i,j} r(G_{ij}^{\beta_2}) - \sum_{i,j} r(G_{ij}^{\beta_1}) \right) \leq \mathcal{U}(G^{\beta_1}) - \mathcal{U}(G^{\beta_2}) \leq \frac{1}{\beta_1} \left(\sum_{i,j} r(G_{ij}^{\beta_2}) - \sum_{i,j} r(G_{ij}^{\beta_1}) \right).$$

Since $\beta_2 > \beta_1$, this implies

$$\sum_{i,j} r(G_{ij}^{\beta_2}) \geq \sum_{i,j} r(G_{ij}^{\beta_1}) \quad \text{and} \quad \mathcal{U}(G^{\beta_1}) \geq \mathcal{U}(G^{\beta_2}).$$

□

4 Geometric Measure on the Transport Polytope

The statistical-physics approach does not begin by solving the Kantorovich problem directly. Instead, it replaces the optimization of the transport cost by the study of a Gibbs ensemble on the entire set of admissible plans. This viewpoint immediately raises the problem of defining integration over the transport polytope. For this reason, the present chapter isolates the geometric measure theory of $\mathcal{G}(\mu, \nu)$. The goal is to identify its affine hull, compute its Hausdorff dimension, define its natural Hausdorff measure, and construct integration on $\mathcal{G}(\mu, \nu)$ through the area formula.

As we shall see, the geometry of $\mathcal{G}(\mu, \nu)$ is particularly simple: it is a convex subset of a d -dimensional affine space, whose natural measure is the d -dimensional Hausdorff measure, where $d = (I - 1)(J - 1)$. Once an affine identification with $\mathbb{R}^d$ is fixed, this measure coincides, up to a multiplicative constant, with Lebesgue measure on $\mathbb{R}^d$.

4.1 Affine structure and intrinsic dimension

We start the geometric analysis of $\mathcal{G}(\mu, \nu)$ from its affine structure. To avoid degenerate geometries, we assume $I, J \geq 2$, as well as $\mu_i > 0$ and $\nu_j > 0$ for every $(i, j) \in [I] \times [J]$. Here, we follow [22] and [23].

Definition 4.1 (Affine Space). A set $V \subseteq \mathbb{R}^N$ is an *affine subspace* if there exist $x_0 \in \mathbb{R}^N$ and a linear subspace W of $\mathbb{R}^N$ such that $V = x_0 + W$.

Definition 4.2 (Affine Hull). The affine hull of a subset $X \subseteq \mathbb{R}^N$ is the smallest affine subspace containing X .

Definition 4.3 (Affine Dimension). Let $X \subseteq \mathbb{R}^N$ and let $x_0 + L$ be the affine hull of X . The affine dimension of X is defined as

$$\dim_{\text{aff}}(X) := \dim(L).$$

Theorem 4.4. *The transport polytope $\mathcal{G}(\mu, \nu)$ has affine dimension*

$$\dim_{\text{aff}}(\mathcal{G}(\mu, \nu)) := (I - 1)(J - 1).$$

Proof. By Lemma 1.2, the inequalities do not affect the affine hull of $\mathcal{G}(\mu, \nu)$, since $\mathcal{G}(\mu, \nu)$ contains a point with all entries strictly positive, namely the transport plan defined by

$$G_{ij} = \mu_i \nu_j > 0.$$

Hence the affine hull is determined only by the marginal equalities

$$\sum_{j=1}^J G_{ij} = \mu_i \quad \text{for } i = 1, \dots, I, \quad \sum_{i=1}^I G_{ij} = \nu_j \quad \text{for } j = 1, \dots, J.$$

These are $I + J$ linear equations in the IJ variables $\{G_{ij}\}_{i,j}$, but one of them is redundant, since summing all row constraints and summing all column constraints both give

$$\sum_{i=1}^I \sum_{j=1}^J G_{ij} = 1.$$

It remains to show that the remaining $I + J - 1$ equations are linearly independent. For instance, consider all I row constraints together with the first $J - 1$ column constraints. Suppose that

$$\sum_{i=1}^I a_i \left(\sum_{j=1}^J G_{ij} \right) + \sum_{j=1}^{J-1} b_j \left(\sum_{i=1}^I G_{ij} \right) = 0 \quad \text{for all } G \in \mathbb{R}^{I \times J}.$$

Looking at the coefficient of G_{iJ} , we obtain $a_i = 0$ for every $i = 1, \dots, I$. Then, looking at the coefficient of G_{Ij} , we obtain $b_j = 0$ for every $j = 1, \dots, J - 1$. Hence these $I + J - 1$ equations are linearly independent.

Therefore the affine hull has codimension $I + J - 1$ in $\mathbb{R}^{I \times J}$, and

$$\dim_{\text{aff}}(\mathcal{G}(\mu, \nu)) = IJ - (I + J - 1) = (I - 1)(J - 1).$$

□

4.2 Hausdorff dimension of the polytope

Once the affine dimension is identified, we introduce a measure on $\mathcal{G}(\mu, \nu)$ by leveraging the Hausdorff measure formalism of [31].

Definition 4.5 (δ -cover). A countable collection $\{U_i\}_{i \in \mathbb{N}}$ of subsets of $\mathbb{R}^N$ is a δ -cover of a set $E \subseteq \mathbb{R}^N$ if

$$E \subseteq \bigcup_{i \in \mathbb{N}} U_i \quad \text{and} \quad \text{diam}(U_i) := \sup\{\|x - y\| : x, y \in U_i\} \leq \delta \quad \text{for all } i \in \mathbb{N}.$$

Definition 4.6 (Hausdorff outer measure). The s -dimensional Hausdorff outer measure, with $s \geq 0$, is the function $\mathcal{H}_*^s : 2^{\mathbb{R}^N} \rightarrow [0, \infty]$ defined by

$$\mathcal{H}_*^s(E) := \lim_{\delta \rightarrow 0^+} \inf \left\{ \sum_{i=1}^{\infty} \alpha(s) \left(\frac{\text{diam}(U_i)}{2} \right)^s \mid \{U_i\}_{i=1}^{\infty} \text{ is a } \delta\text{-cover of } E \right\},$$

where

$$\alpha(s) := \frac{\pi^{s/2}}{\Gamma\left(\frac{s}{2} + 1\right)}, \quad \Gamma(s) := \int_0^\infty e^{-x} x^{s-1} dx.$$

Moreover, we set $\mathcal{H}_*^s(\emptyset) := 0$.

By Carathéodory's construction, the outer measure $\mathcal{H}_*^s$ induces a measure $\mathcal{H}^s$ in the usual sense: if $\mathcal{M}^s$ denotes the σ -algebra of $\mathcal{H}_*^s$ -measurable sets, then the restriction $\mathcal{H}^s := \mathcal{H}_*^s|_{\mathcal{M}^s}$ is a Borel regular measure for $0 \leq s < \infty$ (see [31, Thm. 2.1.1]). Moreover, with this choice of the normalising constant $\alpha(s)$, the N -dimensional Hausdorff measure $\mathcal{H}^N$ coincides with the Lebesgue measure $\mathcal{L}^N$ on $\mathbb{R}^N$ (see [31, Thm. 2.2.2]).

Definition 4.7 (Hausdorff dimension). Given $E \subset \mathbb{R}^N$, the Hausdorff dimension of E is

$$\dim_{\mathcal{H}}(E) := \inf\{s \geq 0 : \mathcal{H}^s(E) = 0\}.$$

In what follows, we will use the following basic properties of Hausdorff dimension. The first two are immediate from the definition, while the third follows from [31, Thm. 2.2.2]:

- i) If $E \subseteq F$, then $\dim_{\mathcal{H}}(E) \leq \dim_{\mathcal{H}}(F)$;
- ii) If $T : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a dilation or an isometry, then

$$\dim_{\mathcal{H}}(T(E)) = \dim_{\mathcal{H}}(E); \tag{4.1}$$

- iii) for every $d \in \mathbb{N}$, $\dim_{\mathcal{H}}([0, 1]^d) = d$.

Proposition 4.8. *If $E \subseteq \mathbb{R}^d$ is open, then $\dim_{\mathcal{H}}(E) = d$. In particular, $\dim_{\mathcal{H}}(\mathbb{R}^d) = d$.*

Proof. If E is open, there exist $\varepsilon > 0$ and $x \in E$ such that $B(x, \varepsilon) \subseteq E$. Observe that $\dim_{\mathcal{H}}(B(x, \varepsilon)) = d$. Indeed, one can find cubes $[a_1, b_1]^d \subseteq B(x, \varepsilon) \subseteq [a_2, b_2]^d$, and therefore, by the previous three properties,

$$d = \dim_{\mathcal{H}}([a_1, b_1]^d) \leq \dim_{\mathcal{H}}(B(x, \varepsilon)) \leq \dim_{\mathcal{H}}([a_2, b_2]^d) = d.$$

Hence,

$$\dim_{\mathcal{H}}(E) \geq \dim_{\mathcal{H}}(B(x, \varepsilon)) = d.$$

On the other hand, every open subset of $\mathbb{R}^d$ is a countable union of open balls, say $E = \bigcup_{n \in \mathbb{N}} B_n$. Since each ball B_n has Hausdorff dimension d , we obtain

$$\dim_{\mathcal{H}}(E) \leq \dim_{\mathcal{H}}\left(\bigcup_{n \in \mathbb{N}} B_n\right) \leq \sup_{n \in \mathbb{N}} \dim_{\mathcal{H}}(B_n) = d,$$

where the second inequality is a standard consequence of the definition of Hausdorff dimension and the countable subadditivity of Hausdorff measures (see [31, Section 2.1]): if $s > \sup_n \dim_{\mathcal{H}}(B_n)$, then $\mathcal{H}^s(B_n) = 0$ for all n , hence $\mathcal{H}^s(\bigcup_n B_n) = 0$. Thus $\dim_{\mathcal{H}}(E) = d$. $\square$

Lemma 4.9. *Given $E \subseteq \mathbb{R}^d$, if $f : E \rightarrow \mathbb{R}^N$ is a bi-Lipschitz function, i.e.*

$$c_1\|x - y\| \leq \|f(x) - f(y)\| \leq c_2\|x - y\| \quad \forall x, y \in E,$$

where $0 < c_1 \leq c_2 < \infty$, then

$$\dim_{\mathcal{H}}(f(E)) = \dim_{\mathcal{H}}(E).$$

The following proof is adapted from [31, Thm. 2.4.1].

Proof. First, we show that for every $s \geq 0$,

$$\mathcal{H}^s(f(E)) \leq c_2^s \mathcal{H}^s(E). \quad (4.2)$$

Indeed, let $\{U_n\}_{n \in \mathbb{N}}$ be a δ -cover of E . Then

$$\text{diam}(f(E \cap U_n)) \leq c_2 \text{diam}(E \cap U_n) \leq c_2 \text{diam}(U_n) \leq c_2 \delta,$$

so $\{f(E \cap U_n)\}_{n \in \mathbb{N}}$ is a $c_2 \delta$ -cover of $f(E)$, and

$$\sum_{n \in \mathbb{N}} \alpha(s) \left(\frac{\text{diam}(f(E \cap U_n))}{2} \right)^s \leq c_2^s \sum_{n \in \mathbb{N}} \alpha(s) \left(\frac{\text{diam}(U_n)}{2} \right)^s.$$

Taking the infimum over all δ -covers and letting $\delta \downarrow 0$, we obtain (4.2).

Now let $s > \dim_{\mathcal{H}}(E)$. Then $\mathcal{H}^s(E) = 0$, and therefore (4.2) yields $\mathcal{H}^s(f(E)) = 0$.

Since this holds for every $s > \dim_{\mathcal{H}}(E)$, we conclude that

$$\dim_{\mathcal{H}}(f(E)) \leq \dim_{\mathcal{H}}(E).$$

For the converse inequality, we apply the same argument to $f^{-1} : f(E) \rightarrow E$. First observe that f is injective: if $f(x) = f(y)$ for some $x, y \in E$, then

$$0 = \|f(x) - f(y)\| \geq c_1\|x - y\|,$$

hence $x = y$. Thus $f : E \rightarrow f(E)$ is invertible. Moreover, for every $u, v \in f(E)$, writing $u = f(x)$ and $v = f(y)$ with $x, y \in E$, we have

$$c_1\|f^{-1}(u) - f^{-1}(v)\| \leq \|u - v\| \leq c_2\|f^{-1}(u) - f^{-1}(v)\|,$$

which is equivalent to

$$\frac{1}{c_2} \|u - v\| \leq \|f^{-1}(u) - f^{-1}(v)\| \leq \frac{1}{c_1} \|u - v\|.$$

Therefore $f^{-1} : f(E) \rightarrow E$ is bi-Lipschitz. Applying the previous argument to f^{-1} , we obtain

$$\dim_{\mathcal{H}}(E) \leq \dim_{\mathcal{H}}(f(E)).$$

Combining the two inequalities, we conclude that

$$\dim_{\mathcal{H}}(f(E)) = \dim_{\mathcal{H}}(E).$$

□

The next theorem connects the affine structure of $\mathcal{G}(\mu, \nu)$ obtained in Theorem 4.4 with the Hausdorff dimension:

Theorem 4.10. *Let $C \subseteq \mathbb{R}^N$ be a nonempty convex set and let $V = x_0 + W$ be its affine hull. Then the Hausdorff dimension of C coincides with the affine dimension of C .*

Proof. By definition, the affine dimension of C is the dimension of W . Let

$$d := \dim(W).$$

Then there exists a linear isomorphism $\varphi : W \rightarrow \mathbb{R}^d$. By Proposition 4.8, $\dim_{\mathcal{H}}(\mathbb{R}^d) = d$. If φ were bi-Lipschitz, then by Lemma 4.9,

$$\dim_{\mathcal{H}} W = \dim_{\mathcal{H}} \varphi(W) = d.$$

Let us show that φ is bi-Lipschitz. By linearity, $\|\varphi(x) - \varphi(y)\| = \|\varphi(x - y)\|$. Setting $t := x - y$, it suffices to find c_1 and c_2 such that $c_1 \|t\| \leq \|\varphi(t)\| \leq c_2 \|t\|$. Let

$$c_1 := \inf_{t: \|t\|=1} \|\varphi(t)\| \quad \text{and} \quad c_2 := \sup_{t: \|t\|=1} \|\varphi(t)\|.$$

Both quantities exist finite and positive since $t \mapsto \|\varphi(t)\|$ is continuous and $\{t : \|t\| = 1\}$ is compact. It is easy to see that c_1 and c_2 just defined satisfy the bi-Lipschitz condition, concluding φ is bi-Lipschitz.

Finally, $\dim_{\mathcal{H}} W = \dim_{\mathcal{H}} C$ by convexity: there exist $x \in C$ and $\varepsilon > 0$ such that

$$B_V(x, \varepsilon) \subseteq C,$$

where $B_V(x, \varepsilon)$ is an ball open in V (see [22, Thm. 6.2]). Hence

$$d = \dim_{\mathcal{H}} W \geq \dim_{\mathcal{H}} C \geq \dim_{\mathcal{H}} B_V(x, \varepsilon) = d,$$

by monotonicity of the Hausdorff dimension. $\square$

As a corollary of Theorem 4.10, the transport polytope $\mathcal{G}(\mu, \nu)$, being nonempty by Lemma 1.2 and convex by Proposition 1.1 has Hausdorff dimension

$$d := \dim_{\mathcal{H}}(\mathcal{G}(\mu, \nu)) = \dim_{\text{aff}}(\mathcal{G}(\mu, \nu)) \stackrel{4.4}{=} (I - 1)(J - 1).$$

4.3 Hausdorff measure and affine parametrizations

We now define the intrinsic geometric measure on $\mathcal{G}(\mu, \nu)$. Let $N := IJ$, $K = I + J - 1$, and

$$d := \dim_{\mathcal{H}}(\mathcal{G}(\mu, \nu)) = (I - 1)(J - 1) = N - K,$$

and let V be the affine hull of $\mathcal{G}(\mu, \nu)$. Since V is a d -dimensional affine subspace of $\mathbb{R}^{I \times J}$, and $\mathcal{G}(\mu, \nu) \subseteq V$, the canonical geometric measure on the transport polytope is the restricted Hausdorff measure

$$\mathcal{H}_{\mu\nu}^d := \mathcal{H}^d \llcorner \mathcal{G}(\mu, \nu).$$

To relate this measure to Lebesgue measure on $\mathbb{R}^d$, fix $X_0 \in \mathcal{G}(\mu, \nu)$ and write $V = X_0 + W$, where W is a linear subspace of dimension d . Let $L : \mathbb{R}^d \rightarrow W$ be a linear isomorphism and define the affine map $A : \mathbb{R}^d \rightarrow V$ by

$$A(y) := X_0 + L(y).$$

Set

$$D := A^{-1}(\mathcal{G}(\mu, \nu)) = \{y \in \mathbb{R}^d : A(y) \in \mathcal{G}(\mu, \nu)\}.$$

We recall the area formula [31, Thm. 3.3.2.1]:

Theorem 4.11 (Area formula). *if $f : \mathbb{R}^d \rightarrow \mathbb{R}^N$ is Lipschitz, with $d \leq N$, then for every $\mathcal{L}^d$ -measurable set $E \subseteq \mathbb{R}^d$,*

$$\int_E J_f(x) \, d\mathcal{L}^d(x) = \int_{\mathbb{R}^N} \mathcal{H}^0(E \cap f^{-1}(\{y\})) \, d\mathcal{H}^d(y). \quad (4.3)$$

In our setting, since $\mathbb{R}^d$ and W are finite dimensional vector spaces, it follows that L and A are bi-Lipschitz. Moreover, D is compact because $\mathcal{G}(\mu, \nu)$ is compact and A is an affine

homeomorphism. Furthermore, $A : \mathbb{R}^d \rightarrow V \subseteq \mathbb{R}^{IJ}$ is affine and injective, so for every Borel set $E \subseteq D$ we obtain

$$\mathcal{H}^d(A(E)) = \int_{\mathbb{R}^{IJ}} \mathcal{H}^0(E \cap A^{-1}(\{z\})) \, d\mathcal{H}^d(z) \stackrel{(4.3)}{=} \int_E J_A(y) \, d\mathcal{L}^d(y).$$

Since $A(y) = X_0 + L(y)$, its Jacobian is constant and equal to

$$J_A \equiv J_L := \sqrt{\det(L^T L)} > 0.$$

Hence, for every Borel set $E \subseteq D$,

$$\mathcal{H}^d(A(E)) = J_L \mathcal{L}^d(E).$$

Equivalently,

$$\mathcal{L}^d(E) = \frac{1}{J_L} \mathcal{H}^d(A(E)).$$

Therefore, define the pushforward measure of $\mathcal{L}^d \llcorner D$ through A by

$$A_{\#}(\mathcal{L}^d \llcorner D)(B) := \mathcal{L}^d(A^{-1}(B) \cap D),$$

for every Borel set $B \subseteq V$. Then

$$A_{\#}(\mathcal{L}^d \llcorner D) = \frac{1}{J_L} \mathcal{H}_{\mu\nu}^d.$$

Thus the parametrized Lebesgue measure and the intrinsic Hausdorff measure differ only by the constant factor J_L .

Remark 4.12 (Normalized Hausdorff measure). Since $\mathcal{G}(\mu, \nu)$ is a compact convex subset of its d -dimensional affine hull with nonempty relative interior, one has

$$0 < \mathcal{H}^d(\mathcal{G}(\mu, \nu)) < \infty.$$

Hence the normalized measure

$$\overline{\mathcal{H}}_{\mu\nu}^d := \frac{1}{\mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu))} \mathcal{H}_{\mu\nu}^d$$

is well defined and is a probability measure on $\mathcal{G}(\mu, \nu)$. At this stage it is useful to separate two notions that later play different roles. Geometrically, the natural measure on the transport polytope is the restricted Hausdorff measure $\mathcal{H}_{\mu\nu}^d$. Probabilistically, however, the statistical-physics framework is cleaner if one works with its normalization $\overline{\mathcal{H}}_{\mu\nu}^d$. The thesis will therefore distinguish these two objects from the beginning. In the geometric measure theory derivation of Chapter 5 we will work with the non-normalized measure, since it is the geometrically canonical one. Later, in Chapter 6, we will move to the normalized measure because it allows a better statistical-physics interpretation.

We conclude this section with an illustrative example:

Example 4.13 (A 2×3 transport polytope). Let $\mathcal{S} = \{x_1, x_2, x_3\}$ and consider the discrete measures

$$\mu = \frac{1}{2}(\delta_{x_1} + \delta_{x_2}), \quad \nu = \frac{1}{3}(\delta_{x_1} + \delta_{x_2} + \delta_{x_3}).$$

A coupling $G \in \mathcal{G}(\mu, \nu)$ is identified by a 2×3 matrix

$$G = \begin{pmatrix} G_{11} & G_{12} & G_{13} \\ G_{21} & G_{22} & G_{23} \end{pmatrix}, \quad G_{ij} \geq 0,$$

whose row sums are equal to $1/2$ and whose column sums are equal to $1/3$. Setting $G_{11} = s \geq 0$ and $G_{12} = t \geq 0$, the marginal constraints determine all remaining entries, and we obtain

$$\mathcal{G}(\mu, \nu) = \left\{ \begin{pmatrix} s & t & \frac{1}{2} - s - t \\ \frac{1}{3} - s & \frac{1}{3} - t & s + t - \frac{1}{6} \end{pmatrix} \mid s, t \in [0, 1/3], s + t \in [1/6, 1/2] \right\},$$

which is two dimensional, as predicted by Theorem 4.4 (which gives $\dim_{\mathcal{H}} \mathcal{G}(\mu, \nu) = (2 - 1)(3 - 1) = 2$). Thus $\mathcal{G}(\mu, \nu)$ is parametrized by the planar region

$$D := \{(s, t) \in [0, \frac{1}{3}]^2 : \frac{1}{6} \leq s + t \leq \frac{1}{2}\}.$$

Then there exists an affine bijection $A : D \rightarrow \mathcal{G}(\mu, \nu)$ mapping (s, t) to $G_{s,t}$, given by

$$A(s, t) = \begin{pmatrix} s & t & \frac{1}{2} - s - t \\ \frac{1}{3} - s & \frac{1}{3} - t & s + t - \frac{1}{6} \end{pmatrix}.$$

Its differential is constant and is represented by

$$DA_{(s,t)} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix},$$

so

$$(DA_{(s,t)})^T DA_{(s,t)} = \begin{pmatrix} 1 & 0 & -1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \\ -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}.$$

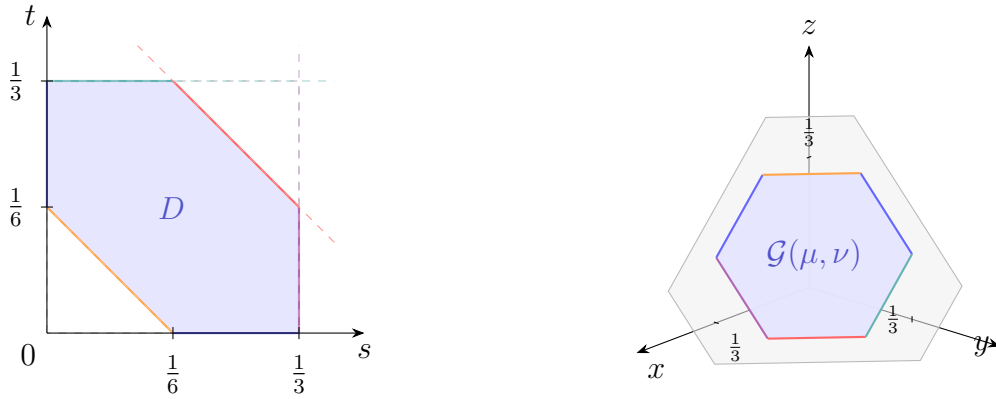


Figure 3: Left: The parameter domain $D = \{(s, t) \in [0, \frac{1}{3}]^2 : \frac{1}{6} \leq s + t \leq \frac{1}{2}\}$ associated with the transport polytope in Example 4.13. Right: the transport polytope $\mathcal{G}(\mu, \nu)$ lying on its shaded affine hull $x + y + z = \frac{1}{2}$, projected on a 3D space.

Hence

$$J_A = \sqrt{\det \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}} = \sqrt{16 - 4} = 2\sqrt{3}.$$

This bijection yields a measure $\mathcal{H}_{\mu\nu}^2$ on $\mathcal{G}(\mu, \nu)$ via pushforward. Since J_A is constant, there are no local distortions, and $\mathcal{H}_{\mu\nu}^d$ is uniform, given by

$$\mathcal{H}_{\mu\nu}^2 = J_A \cdot A_{\#}(\mathcal{L}^2|_D) = 2\sqrt{3}A_{\#}(\mathcal{L}^2|_D).$$

After normalization, the constant $2\sqrt{3}$ cancels, so the normalized Hausdorff measure corresponds exactly to the normalized Lebesgue measure on D transported by A .

5 Steepest-Descent Analysis of the Partition Function

5.1 The heuristic derivation in the physics literature

In the previous chapter, we established the geometric and measure-theoretic structure of the transport polytope $\mathcal{G}(\mu, \nu)$. We showed that it is a compact convex polytope of Hausdorff dimension $d = (I - 1)(J - 1)$ embedded in $\mathbb{R}^{IJ}$, and we endowed it with the restricted Hausdorff measure $\mathcal{H}_{\mu\nu}^d$.

Analytically integrating a continuous function f over this high-dimensional constrained domain such as $\mathcal{G}(\mu, \nu)$ is generally prohibitive. To circumvent this difficulty, the statistical physics literature ([4, 5, 11, 32, 33]) typically adopts a powerful heuristic approach based on the Fourier representation of the Dirac delta. In [12, Eqs. 16–20] and [34, Eqs. 7–11], the integration is embedded into the ambient hypercube $[0, 1]^{IJ}$, and the marginal constraints are enforced through products of Dirac delta distributions:

$$\int_{\mathcal{G}(\mu, \nu)} f(G) d\mathcal{H}_{\mu\nu}^d \approx \int_{[0, 1]^{IJ}} f(G) \prod_{i=1}^I \delta\left(\sum_{j=1}^J G_{ij} - \mu_i\right) \prod_{j=1}^J \delta\left(\sum_{i=1}^I G_{ij} - \nu_j\right) dG. \quad (5.1)$$

The standard procedure then replaces the Dirac deltas with their Fourier integral representation, $\delta(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\xi x} d\xi$. By swapping the integration order, the integrals over the primal variables G_{ij} decouple and can be evaluated explicitly. This yields a dual representation involving purely unconstrained variables, which naturally correspond to the Kantorovich potentials. The resulting dual integral is approximated through a standard saddle-point approximation [13, Sec. 1.5], which, in the present setting, is multidimensional.

While extraordinarily effective for generating analytical approximations, this derivation poses severe mathematical issues:

- i) The Dirac δ is a distribution, not a Lebesgue-integrable function, producing an ill-posed integrand within Lebesgue integration theory. Moreover, the product of delta factors in (5.1) imposes $I + J$ marginal constraints, although one is redundant (cf. Theorem 4.4).
- ii) The Fourier representation of the Dirac delta is not generally absolutely convergent, which again enforces an abuse of notation. This issue will be addressed by replacing

the Dirac deltas with Gaussian approximate identities before applying the Fourier transform.

- iii) The integral in the Fourier dual variables is then deformed from the original real cycle to a complex cycle passing through an imaginary saddle, without specifying the integration cycle and proving that the deformation preserves the value of the integral.
- iv) In our setting the steepest-descent problem is multidimensional, unbounded, and temperature-dependent, hence standard saddle-point treatments ([35, Ch. 5], [36, Sec. 3], [33, App. C], [11, Sec. 2.2.4] [13],) do not directly apply to our problem.

The goal of this chapter is to replace the formal physical manipulations with a mathematically explicit framework based on the coarea formula, Gaussian approximate identities, the Fourier transform of the Gaussian kernel, and a conditional multidimensional steepest-descent argument. We do not verify all the hypotheses for the multidimensional steepest-descent problem, but we fully describe the analytical assumptions under which the result would follow, and identify a natural candidate for the complex cycle deformation.

5.2 Gaussian Mollifiers and the Coarea Formula

To streamline the exposition, we collect the notation needed for the regularized constraint framework.

- **Dimensions and domain.** We set

$$N := IJ, \quad K := I + J - 1, \quad d := N - K = (I - 1)(J - 1).$$

The ambient space for transport plans is the hypercube $\mathcal{Q} := [0, 1]^N$, which contains the transport polytope $\mathcal{G}(\mu, \nu)$.

- **Integrand.** Let $f : \mathbb{R}^N \rightarrow \mathbb{R}_+$ be a nonnegative continuous function (in our context, the Boltzmann function $f(G) = e^{-\beta \langle C, G \rangle}$). Since $f \in C(\mathcal{Q})$ and $\mathcal{Q}$ is compact, $f \in L^1(\mathcal{Q})$. We also define $\Phi := f \cdot \mathbf{1}_{\mathcal{Q}}$.
- **Constraint map.** We define the affine map $\tilde{\pi} : \mathbb{R}^N \rightarrow \mathbb{R}^K$ encoding the indepen-

dent marginal constraints of $\mathcal{G}(\mu, \nu)$ by

$$\bar{\varepsilon}(G) := \begin{pmatrix} \sum_{j=1}^J G_{1j} - \mu_1 \\ \vdots \\ \sum_{j=1}^J G_{Ij} - \mu_I \\ \sum_{i=1}^I G_{i1} - \nu_1 \\ \vdots \\ \sum_{i=1}^I G_{i,J-1} - \nu_{J-1} \end{pmatrix}. \quad (5.2)$$

The final column constraint $\sum_{i=1}^I G_{iJ} = \nu_J$ is implicitly satisfied due to the global mass conservation $\sum_i \mu_i = \sum_j \nu_j = 1$ (cf. Theorem 4.4). Writing $\bar{\varepsilon}(G) = LG - b$, where $L : \mathbb{R}^N \rightarrow \mathbb{R}^K$ is the linear part of $\bar{\varepsilon}$; we have:

- i) $\bar{\varepsilon}$ is affine and surjective, hence Lipschitz;
- ii) the Jacobian of $\bar{\varepsilon}$ is independent of G and equals

$$J_{\bar{\varepsilon}} := \sqrt{\det(LL^T)} > 0;$$

- iii) the transport polytope is the zero level set of $\bar{\varepsilon}$ inside the hypercube:

$$\mathcal{G}(\mu, \nu) = \{G \in \mathcal{Q} : \bar{\varepsilon}(G) = 0\} = \bar{\varepsilon}^{-1}(0) \cap \mathcal{Q}. \quad (5.3)$$

- **Fiber integral.** We define the map $A_{\Phi} : \mathbb{R}^K \rightarrow \mathbb{R}$ representing the integral over the fibers of $\bar{\varepsilon}$ by

$$A_{\Phi}(t) := \int_{\bar{\varepsilon}^{-1}(t)} \Phi(G) d\mathcal{H}^d(G).$$

With these conventions, we can state the main result of this section.

Theorem 5.1. *With the notation above,*

$$\int_{\mathcal{G}(\mu, \nu)} f(G) d\mathcal{H}_{\mu\nu}^d(G) = \lim_{n \rightarrow \infty} J_{\bar{\varepsilon}} \int_{\mathcal{Q}} f(G) \rho_n(\bar{\varepsilon}(G)) dG, \quad (5.4)$$

where

$$\rho_n(t) := \left(\frac{n}{\sqrt{2\pi}} \right)^K \exp\left(-\frac{\|nt\|^2}{2} \right), \quad t \in \mathbb{R}^K. \quad (5.5)$$

The proof relies on the coarea formula in the following form [31, Theorem 3.4.2.1].

Theorem 5.2 (Coarea formula). *Let $\bar{\varepsilon} : \mathbb{R}^N \rightarrow \mathbb{R}^K$ be Lipschitz, with $N \geq K$. Then, for every $g \in L^1(\mathbb{R}^N)$, the restriction $g|_{\bar{\varepsilon}^{-1}(t)}$ is $\mathcal{H}^d$ -integrable for $\mathcal{L}^K$ -a.e. $t \in \mathbb{R}^K$, and*

$$\int_{\mathbb{R}^N} g(G) J_{\bar{\varepsilon}}(G) dG = \int_{\mathbb{R}^K} \left(\int_{\bar{\varepsilon}^{-1}(t)} g(G) d\mathcal{H}^d(G) \right) dt.$$

Before proving Theorem 5.1, let us explain how the coarea formula relates to the heuristic manipulations appearing in the statistical-physics literature, in particular in [12] and [34]. By (5.3), the integral over the transport polytope can be written as

$$\int_{\mathcal{G}(\mu, \nu)} f(G) d\mathcal{H}_{\mu\nu}^d(G) = \int_{\bar{\varepsilon}^{-1}(0)} f(G) \mathbf{1}_{\mathcal{Q}}(G) d\mathcal{H}^d(G) = \int_{\bar{\varepsilon}^{-1}(0)} \Phi(G) d\mathcal{H}^d(G) = A_{\Phi}(0). \quad (5.6)$$

On the other hand, applying the coarea formula to Φ gives

$$\int_{\mathbb{R}^K} A_{\Phi}(t) dt = J_{\bar{\varepsilon}} \int_{\mathbb{R}^N} \Phi(G) dG = J_{\bar{\varepsilon}} \int_{\mathcal{Q}} f(G) dG.$$

To isolate the 0 fiber rather than integrating over $\mathbb{R}^K$, the statistical physics literature weights the integration by the Dirac delta measure $\delta_0(t)$ (the so-called *Feynman trick*, see, e.g., [34, Eq. (7)], [12, Eq. (16)], and [37, Sec. 2.1], [13, Sec. 1.5]), yielding the heuristic identity:

$$J_{\bar{\varepsilon}} \int_{\mathcal{Q}} f(G) \delta_0(\bar{\varepsilon}(G)) dG = \int_{\mathbb{R}^K} A_{\Phi}(t) \delta_0(t) dt = A_{\Phi}(0).$$

Substituting this operation back into the ambient integral leads to the widespread expression involving $\delta_0(\bar{\varepsilon}(G))$ inside the integral over G . Mathematically, evaluating a distribution on an affine map and integrating it over a bounded domain is ill-posed within Lebesgue integration theory.

This formulation can only be rigorously justified through a limit procedure. One can frame this either as the narrow convergence of a sequence of Gaussian probability measures (see, for instance, [38, Thm. 7.1.2] or [39, Problem 1.5.7]), or by approximating the Dirac mass at the origin by a family of Gaussian kernels ([40, Section 8.2] and [41, Chapter 2, Section 4]). In this work, we adopt the analytic framework of approximate identities, bearing in mind that it is fundamentally equivalent to the probabilistic perspective of weakly converging measures. This choice is particularly convenient for the subsequent Fourier analysis, since Gaussian kernels admit an explicit Fourier transform. To this end, we shall briefly review the theory of kernels.

Following [40] and [41], we define the *convolution* between integrable functions $f, g : \mathbb{R}^K \rightarrow \mathbb{R}$ as

$$(f * g)(x) = \int_{\mathbb{R}^K} f(x - y) g(y) dy.$$

It is well defined and satisfies Young's inequality $\|f * g\|_{L^1} \leq \|f\|_{L^1} \|g\|_{L^1}$, since

$$\begin{aligned} \|f * g\|_{L^1} &= \int_{\mathbb{R}^K} |(f * g)(x)| dx = \int_{\mathbb{R}^K} \left| \int_{\mathbb{R}^K} f(x-y)g(y) dy \right| dx \\ &\leq \int_{\mathbb{R}^K} \int_{\mathbb{R}^K} |f(x-y)||g(y)| dy dx \stackrel{\dagger}{=} \int_{\mathbb{R}^K} |g(y)| \left(\int_{\mathbb{R}^K} |f(x-y)| dx \right) dy \\ &= \int_{\mathbb{R}^K} |g(y)| \|f\|_{L^1} dy = \|f\|_{L^1} \|g\|_{L^1}, \end{aligned}$$

where $\dagger$ follows by Fubini-Tonelli applied to the measurable, nonnegative function $(x, y) \mapsto |f(x-y)||g(y)|$. Furthermore, it is clear from the definition that the convolution is commutative. A *kernel* is an integrable function $\rho : \mathbb{R}^K \rightarrow \mathbb{R}$ such that

$$\int_{\mathbb{R}^K} \rho(x) dx = 1.$$

A family of kernels $(\rho_n)_{n \in \mathbb{N}}$ is an *approximate identity* if $\rho_n \geq 0$ for every $n \in \mathbb{N}$ and, for every $\delta > 0$,

$$\lim_{n \rightarrow \infty} \int_{\{\|x\| \geq \delta\}} |\rho_n(x)| dx = 0.$$

The elements of such a family are often referred to as *good kernels*. The next example shows the existence of good kernels:

Example 5.3 (Gaussian Kernel). The K -dimensional *Gaussian kernel*, or *heat kernel*, or *mollifier*, with standard deviation $1/n$ is defined on $\mathbb{R}^K$ as

$$\rho_n(x) := \left(\frac{n}{\sqrt{2\pi}} \right)^K \exp \left(-\frac{\|nx\|^2}{2} \right). \quad (5.7)$$

To see that ρ_n is a kernel for every $n \in \mathbb{N}$, observe that, by a polar change of variables,

$$\left(\int_{-\infty}^{\infty} e^{-\pi t^2} dt \right)^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\pi(t_1^2 + t_2^2)} dt_1 dt_2 = \int_0^{2\pi} \int_0^{\infty} e^{-\pi r^2} r dr d\theta = \left[-e^{-\pi r^2} \right]_0^{\infty} = 1.$$

Hence, by Fubini-Tonelli, since the integrand is nonnegative,

$$\begin{aligned} \int_{\mathbb{R}^K} \rho_n dx &= \int_{\mathbb{R}^K} \frac{n^K}{(2\pi)^{K/2}} e^{-\frac{\sum_{i=1}^K n^2 x_i^2}{2}} dx = \int_{\mathbb{R}^K} \left(\prod_{i=1}^K \frac{n}{\sqrt{2\pi}} e^{-\frac{n^2 x_i^2}{2}} \right) dx \\ &= \left(\int_{-\infty}^{\infty} \frac{n}{\sqrt{2\pi}} e^{-\frac{n^2 t^2}{2}} dt \right)^K \stackrel{u=nt/\sqrt{2\pi}}{=} \left(\int_{-\infty}^{+\infty} e^{-\pi u^2} du \right)^K = 1. \end{aligned}$$

Moreover, ρ_n is a good kernel: it is nonnegative and, for every $\delta > 0$,

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_{\|x\| \geq \delta} |\rho_n(x)| dx &= \lim_{n \rightarrow \infty} \int_{\|x\| \geq \delta} \frac{n^K}{(2\pi)^{K/2}} e^{-\frac{\|nx\|^2}{2}} dx \\ &\stackrel{y=nx/\sqrt{2\pi}}{=} \lim_{n \rightarrow \infty} \int_{\|y\| \geq \frac{n\delta}{\sqrt{2\pi}}} e^{-\pi\|y\|^2} dy \\ &= \int_{\bigcap_{n \in \mathbb{N}} \{\|y\| \geq \frac{n\delta}{\sqrt{2\pi}}\}} e^{-\pi\|y\|^2} dy \\ &= \int_{\emptyset} e^{-\pi\|y\|^2} dy = 0, \end{aligned}$$

since

$$\{\|y\| \geq \frac{(n+1)\delta}{\sqrt{2\pi}}\} \subseteq \{\|y\| \geq \frac{n\delta}{\sqrt{2\pi}}\}$$

for every $n \in \mathbb{N}$.

The importance of good kernels is highlighted by the following theorem (cf. [41, Thm. 4.1], [40, Thm. 8.15]):

Theorem 5.4 (Convergence of approximate identities). *If $(\rho_n)_{n \in \mathbb{N}}$ is an approximate identity on $\mathbb{R}^K$ and $f : \mathbb{R}^K \rightarrow \mathbb{R}$ is integrable, then*

$$\lim_{n \rightarrow \infty} (f * \rho_n)(x) = f(x) \quad (5.8)$$

whenever f is continuous at $x \in \mathbb{R}^K$.

The next lemmas are instrumental for the proof of Theorem 5.1.

Lemma 5.5. *The sequence $(\rho_n)_{n \in \mathbb{N}}$ defined by (5.5) is an approximate identity. Moreover, ρ_n is continuous, bounded, and even for every $n \in \mathbb{N}$.*

Proof. We already showed that $(\rho_n)_{n \in \mathbb{N}}$ is an approximate identity by Example 5.3. To see it is bounded, observe that

$$0 \leq \rho_n(t) \leq \rho_n(0) = \left(\frac{n}{\sqrt{2\pi}} \right)^K,$$

since $0 \leq e^{-\frac{\|nx\|^2}{2}} \leq 1$ for every $n \in \mathbb{N}$. ρ_n is continuous since $e^{-\frac{\|nx\|^2}{2}}$ is continuous for every $n \in \mathbb{N}$. Finally, $\rho_n(t) = \rho_n(-t)$ by inspecting the definition. $\square$

Lemma 5.6. *If $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is nonnegative and continuous, then the fiber integral*

$$A_\Phi(t) := \int_{\bar{z}^{-1}(t)} \Phi(G) d\mathcal{H}^d(G) = \int_{\bar{z}^{-1}(t)} f(G) \mathbf{1}_Q(G) d\mathcal{H}^d(G)$$

is continuous at $t = 0$.¹

Proof. Fix $\varepsilon > 0$. Let $\varphi_\varepsilon^- : \mathbb{R}^N \rightarrow \mathbb{R}$ and $\varphi_\varepsilon^+ : \mathbb{R}^N \rightarrow \mathbb{R}$ be defined by

$$\varphi_\varepsilon^-(G) = \min \left\{ 1, \frac{d_\infty(G, \mathcal{Q}^c)}{\varepsilon} \right\}, \quad \varphi_\varepsilon^+(G) = \max \left\{ 0, 1 - \frac{d_\infty(G, \mathcal{Q})}{\varepsilon} \right\},$$

where

$$d_\infty(G, A) = \begin{cases} \inf_{H \in A} \|G - H\|_\infty, & A \neq \emptyset, \\ +\infty, & A = \emptyset. \end{cases}$$

Since $G \mapsto d_\infty(G, A)$ is 1-Lipschitz with respect to $\|\cdot\|_\infty$, and since the functions

$$t \mapsto \min\{1, t/\varepsilon\}, \quad t \mapsto \max\{0, 1 - t/\varepsilon\}$$

are continuous, the functions φ_ε^- and φ_ε^+ are continuous. Moreover, it easy to verify that, by construction,

$$0 \leq \varphi_\varepsilon^-(G) \leq \mathbf{1}_{\mathcal{Q}}(G) \leq \varphi_\varepsilon^+(G) \leq 1 \quad \forall G \in \mathbb{R}^N. \quad (5.9)$$

Finally, setting $\psi_\varepsilon^\pm := f \cdot \varphi_\varepsilon^\pm$, it also follows that ψ_ε^- and ψ_ε^+ have compact support, since $\mathcal{Q}$ is bounded.

Define

$$A_{\Phi, \varepsilon}^-(t) = \int_{\bar{\varepsilon}^{-1}(t)} f(G) \varphi_\varepsilon^-(G) \, d\mathcal{H}^d(G),$$

and

$$A_{\Phi, \varepsilon}^+(t) = \int_{\bar{\varepsilon}^{-1}(t)} f(G) \varphi_\varepsilon^+(G) \, d\mathcal{H}^d(G).$$

We first prove that $A_{\Phi, \varepsilon}^-$ is continuous. Writing $\bar{\varepsilon}(G) = LG - b$, where $L : \mathbb{R}^N \rightarrow \mathbb{R}^K$ is linear and surjective and $b \in \mathbb{R}^K$, we see that $G \in \bar{\varepsilon}^{-1}(t)$ if and only if $LG = b + t$, while $G' \in \bar{\varepsilon}^{-1}(0)$ if and only if $LG' = b$. Hence, if $R : \mathbb{R}^K \rightarrow \mathbb{R}^N$ is such that $LR = \text{id}_{\mathbb{R}^K}$, then

$$G \in \bar{\varepsilon}^{-1}(t) \iff LG = b + t \iff L(G - Rt) = b \iff G - Rt \in \bar{\varepsilon}^{-1}(0).$$

Therefore, $G \in \bar{\varepsilon}^{-1}(t)$ if and only if $G \in \bar{\varepsilon}^{-1}(0) + Rt$, and thus

$$\bar{\varepsilon}^{-1}(t) = \bar{\varepsilon}^{-1}(0) + Rt.$$

¹The following proof is based on an argument sketched by Prof. Alessandro Pigati.

Setting $\psi_\varepsilon^- := f \cdot \varphi_\varepsilon^- \in C_c(\mathbb{R}^N)$, we compute, for every $t \in \mathbb{R}^K$,

$$\begin{aligned} \lim_{h \rightarrow 0} |A_{\Phi, \varepsilon}^-(t+h) - A_{\Phi, \varepsilon}^-(t)| &= \lim_{h \rightarrow 0} \left| \int_{\bar{\varepsilon}^{-1}(0)} [\psi_\varepsilon^-(G + R(t+h)) - \psi_\varepsilon^-(G + Rt)] d\mathcal{H}^d(G) \right| \\ &\stackrel{\dagger}{\leq} \lim_{h \rightarrow 0} \int_{\bar{\varepsilon}^{-1}(0)} |\psi_\varepsilon^-(G + R(t+h)) - \psi_\varepsilon^-(G + Rt)| d\mathcal{H}^d(G) \\ &\stackrel{\ddagger}{=} \int_{\bar{\varepsilon}^{-1}(0)} \lim_{h \rightarrow 0} |\psi_\varepsilon^-(G + R(t+h)) - \psi_\varepsilon^-(G + Rt)| d\mathcal{H}^d(G) \\ &= 0. \end{aligned}$$

Here, $\dagger$ is an application of the triangle inequality, while $\ddagger$ follows from the dominated convergence theorem, since the integrand converges pointwise to zero and, for h sufficiently small, is supported in a fixed compact subset of $\bar{\varepsilon}^{-1}(0)$ and uniformly bounded by $2\|\psi_\varepsilon^-\|_\infty$.

The same argument proves that $A_{\Phi, \varepsilon}^+$ is continuous. From $f \geq 0$, (5.9), and monotonicity of the integral,

$$A_{\Phi, \varepsilon}^-(t) \leq A_\Phi(t) \leq A_{\Phi, \varepsilon}^+(t) \quad \forall t \in \mathbb{R}^K. \quad (5.10)$$

We eventually compute the limit $\lim_{t \rightarrow 0} A_\Phi(t)$ by squeezing it using (5.10). By continuity of $A_{\Phi, \varepsilon}^\pm(t)$,

$$A_{\Phi, \varepsilon}^-(0) = \liminf_{t \rightarrow 0} A_{\Phi, \varepsilon}^-(t) \leq \liminf_{t \rightarrow 0} A_\Phi(t) \leq \limsup_{t \rightarrow 0} A_\Phi(t) \leq \limsup_{t \rightarrow 0} A_{\Phi, \varepsilon}^+(t) = A_{\Phi, \varepsilon}^+(0). \quad (5.11)$$

If we can show that $|A_{\Phi, \varepsilon}^-(0) - A_{\Phi, \varepsilon}^+(0)| \rightarrow 0$ as $\varepsilon \rightarrow 0$, the continuity of A_Φ at 0 would follow by combining (5.10) and (5.11). To this end, we observe that, for every $G \in \bar{\varepsilon}^{-1}(0)$,

$$\lim_{\varepsilon \rightarrow 0} \psi_\varepsilon^-(G) = f(G)\mathbf{1}_{\text{int}(\mathcal{Q})}(G) \quad \text{and} \quad \lim_{\varepsilon \rightarrow 0} \psi_\varepsilon^+(G) = f(G)\mathbf{1}_{\bar{\mathcal{Q}}}(G) = f(G)\mathbf{1}_{\mathcal{Q}}(G),$$

since $\mathcal{Q}$ is closed. Therefore, by the dominated convergence theorem - applied essentially as in the previous case -

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} A_{\Phi, \varepsilon}^-(0) &= \lim_{\varepsilon \rightarrow 0} \int_{\bar{\varepsilon}^{-1}(0)} f(G)\varphi_\varepsilon^-(G) d\mathcal{H}^d(G) = \int_{\bar{\varepsilon}^{-1}(0)} \lim_{\varepsilon \rightarrow 0} f(G)\varphi_\varepsilon^-(G) d\mathcal{H}^d(G) \\ &= \int_{\bar{\varepsilon}^{-1}(0)} f(G)\mathbf{1}_{\text{int}(\mathcal{Q})}(G) d\mathcal{H}^d(G) \end{aligned}$$

and

$$\begin{aligned} \lim_{\varepsilon \rightarrow 0} A_{\Phi, \varepsilon}^+(0) &= \lim_{\varepsilon \rightarrow 0} \int_{\bar{\varepsilon}^{-1}(0)} f(G)\varphi_\varepsilon^+(G) d\mathcal{H}^d(G) = \int_{\bar{\varepsilon}^{-1}(0)} \lim_{\varepsilon \rightarrow 0} f(G)\varphi_\varepsilon^+(G) d\mathcal{H}^d(G) \\ &= \int_{\bar{\varepsilon}^{-1}(0)} f(G)\mathbf{1}_{\bar{\mathcal{Q}}}(G) d\mathcal{H}^d(G) = \int_{\bar{\varepsilon}^{-1}(0)} f(G)\mathbf{1}_{\mathcal{Q}}(G) d\mathcal{H}^d(G). \end{aligned}$$

We see that the limits of $A_{\Phi,\varepsilon}^-(0)$ and $A_{\Phi,\varepsilon}^+(0)$ as $\varepsilon \rightarrow 0$ coincide up to the integral of Φ on the set $\bar{\varepsilon}^{-1}(0) \cap \partial(\mathcal{Q})$. This set is, however, negligible: having $\mathcal{H}^d(\bar{\varepsilon}^{-1}(0) \cap \partial\mathcal{Q}) \neq 0$ would force $\bar{\varepsilon}^{-1}(0)$, and hence $\mathcal{G}(\mu, \nu)$, to be contained in some face of $\mathcal{Q}$, that is, there exists $(i, j) \in [I] \times [J]$ such that $G_{ij} = 0$ for every $G \in \mathcal{G}(\mu, \nu)$, or $G_{ij} = 1$ for every $G \in \mathcal{G}(\mu, \nu)$. Luckily, this fact never occurs, by Lemma 1.2. Hence

$$\lim_{\varepsilon \rightarrow 0} A_{\Phi,\varepsilon}^-(0) = \lim_{\varepsilon \rightarrow 0} A_{\Phi,\varepsilon}^+(0) = A_{\Phi}(0).$$

Letting $\varepsilon \rightarrow 0$ in (5.11), we obtain

$$A_{\Phi}(0) \leq \liminf_{t \rightarrow 0} A_{\Phi}(t) \leq \limsup_{t \rightarrow 0} A_{\Phi}(t) \leq A_{\Phi}(0).$$

Therefore,

$$\lim_{t \rightarrow 0} A_{\Phi}(t) = A_{\Phi}(0),$$

as desired. □

Eventually, we provide a complete proof of Theorem 5.1.

Proof. Apply the coarea formula to the function $g := \Phi \cdot (\rho_n \circ \bar{\varepsilon})$ and to the Lipschitz map $\bar{\varepsilon} : \mathbb{R}^N \rightarrow \mathbb{R}^K$. Since $J_{\bar{\varepsilon}}$ is constant, we obtain

$$J_{\bar{\varepsilon}} \int_{\mathbb{R}^N} \Phi(G) \rho_n(\bar{\varepsilon}(G)) dG = \int_{\mathbb{R}^K} \left(\int_{\bar{\varepsilon}^{-1}(t)} \Phi(G) \rho_n(\bar{\varepsilon}(G)) d\mathcal{H}^d(G) \right) dt.$$

The hypotheses of the coarea formula are satisfied: $f \in C(\mathcal{Q})$, $\bar{\varepsilon}$ is Lipschitz, $\rho_n \in C(\mathbb{R}^K)$ by Lemma 5.5, and

$$g = f(\rho_n \circ \bar{\varepsilon}) \mathbf{1}_{\mathcal{Q}} \in L^1(\mathbb{R}^N),$$

because it is bounded and supported in the compact set $\mathcal{Q}$.

By the very definition of the fiber $\bar{\varepsilon}^{-1}(t)$, $\rho_n(\bar{\varepsilon}(G)) = \rho_n(t) = \rho_n(-t)$ for every $G \in \bar{\varepsilon}^{-1}(t)$, by Lemma 5.5. Hence

$$\begin{aligned} \int_{\mathbb{R}^N} \Phi(G) \rho_n(\bar{\varepsilon}(G)) dG &= \int_{\mathbb{R}^K} \left(\int_{\bar{\varepsilon}^{-1}(t)} \Phi(G) \rho_n(\bar{\varepsilon}(G)) d\mathcal{H}^d(G) \right) dt \\ &= \int_{\mathbb{R}^K} \rho_n(t) \left(\int_{\bar{\varepsilon}^{-1}(t)} \Phi(G) d\mathcal{H}^d(G) \right) dt \\ &= \int_{\mathbb{R}^K} \rho_n(0 - t) A_{\Phi}(t) dt \\ &= (\rho_n * A_{\Phi})(0). \end{aligned}$$

By the coarea formula applied to Φ , the function A_Φ is integrable on $\mathbb{R}^K$. Moreover, by Lemma 5.6, A_Φ is continuous at 0. Since $(\rho_n)_{n \in \mathbb{N}}$ is an approximate identity by Lemma 5.5, Theorem 5.4 yields

$$\lim_{n \rightarrow \infty} (\rho_n * A_\Phi)(0) = A_\Phi(0).$$

Therefore

$$\lim_{n \rightarrow \infty} J_{\bar{\varepsilon}} \int_{\mathbb{R}^N} \Phi(G) \rho_n(\bar{\varepsilon}(G)) dG = A_\Phi(0).$$

Using (5.6), we conclude that

$$\lim_{n \rightarrow \infty} J_{\bar{\varepsilon}} \int_{\mathcal{Q}} f(G) \rho_n(\bar{\varepsilon}(G)) dG = \int_{\mathcal{G}(\mu, \nu)} f(G) d\mathcal{H}_{\mu\nu}^d(G),$$

which is exactly (5.4). □

5.3 Fourier Representation of Kernel Mollifiers

In this section we continue the formalization of the physical heuristics discussed in Section 5.1 by introducing the Fourier representation of the Gaussian kernels.

Given ρ_n , f and $\bar{\varepsilon}$ as in Theorem 5.1, let us define

$$Z := \int_{\mathcal{G}(\mu, \nu)} f(G) d\mathcal{H}_{\mu\nu}^d(G) \quad \text{and} \quad Z^n := \int_{\mathcal{Q}} f(G) \rho_n(\bar{\varepsilon}(G)) dG.$$

Thus Z is the intrinsic integral over the transport polytope, whereas Z^n is its regularized approximation. Theorem 5.1 then reads as

$$Z = J_{\bar{\varepsilon}} \lim_{n \rightarrow \infty} Z^n.$$

For analytic convenience, we also write $f(G) = e^{-\varphi(G)}$ for some real-valued $\varphi \in C(\mathcal{Q})$. In the Boltzmann case relevant for [12], $\varphi(G) = \beta \langle C, G \rangle$. The main advantage of the regularized partition Z^n is that integration is performed over the hypercube $\mathcal{Q} = [0, 1]^N$, which, being a product space, can be factorized using Fubini-Tonelli's theorem. However, the integrand

$$f(G) \rho_n(\bar{\varepsilon}(G)) = \frac{n^K}{(2\pi)^{K/2}} e^{-\varphi(G)} e^{-\|n\bar{\varepsilon}(G)\|^2/2}$$

cannot be factorized for any $\varphi \in C(\mathcal{Q})$, due to the term

$$\|\bar{\varepsilon}(G)\|^2 = \sum_{i=1}^I \left(\sum_{j=1}^J G_{ij} - \mu_i \right)^2 + \sum_{j=1}^{J-1} \left(\sum_{i=1}^I G_{ij} - \nu_j \right)^2,$$

which contains mixed terms $G_{ij}G_{ik}$. Remarkably, the Fourier representation of $\rho_n(\bar{z}(G))$ linearizes the coupling by introducing variables $\xi \in \mathbb{R}^K$ dual to the constraint space.

By the standard Fourier transform formula for Gaussian functions (see, e.g., [40, Prop. 8.24], and Appendix D), with our convention and normalization constants, for a Gaussian kernel $\rho_n(t) \propto e^{-\frac{n^2}{2}\|t\|^2}$ it holds:

$$\rho_n(t) = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^K} e^{-i\langle \xi, t \rangle} e^{-\|\xi\|^2/(2n^2)} d\xi.$$

So

$$Z^n = \frac{1}{(2\pi)^K} \int_{\mathcal{Q}} e^{-\varphi(G)} \left(\int_{\mathbb{R}^K} e^{-i\langle \xi, \bar{z}(G) \rangle} e^{-\|\xi\|^2/(2n^2)} d\xi \right) dG.$$

The passage to the dual variables is justified by Fubini-Tonelli. Indeed, the integrand is absolutely integrable:

$$\begin{aligned} \int_{\mathcal{Q}} \int_{\mathbb{R}^K} |e^{-\varphi(G)}| \cdot |e^{-i\langle \xi, \bar{z}(G) \rangle}| \cdot |e^{-\|\xi\|^2/(2n^2)}| dG d\xi \\ = \left(\int_{\mathcal{Q}} e^{-\varphi(G)} dG \right) \cdot \left(\int_{\mathbb{R}^K} e^{-\|\xi\|^2/(2n^2)} d\xi \right) < \infty, \end{aligned}$$

where the first factor is finite because $e^{-\varphi(G)}$ is continuous on the compact domain $\mathcal{Q}$, and the second is a standard K -dimensional Gaussian integral (cf. [41, Section 2.4]). Thus:

$$Z^n = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^K} \left(\int_{\mathcal{Q}} e^{-\varphi(G)} e^{-i\langle \xi, \bar{z}(G) \rangle} dG \right) e^{-\|\xi\|^2/(2n^2)} d\xi.$$

To distinguish the I coordinates and the J coordinates, we set

$$\xi := (\theta_1, \dots, \theta_I, \eta_1, \dots, \eta_{J-1}).$$

As we see, the dual functional $e^{-i\langle \xi, \bar{z}(G) \rangle}$ can be factorized as

$$\begin{aligned} e^{-i\langle \xi, \bar{z}(G) \rangle} &= \exp \left(-i \sum_{i=1}^I \theta_i \left(\sum_{j=1}^J G_{ij} - \mu_i \right) - i \sum_{j=1}^{J-1} \eta_j \left(\sum_{i=1}^I G_{ij} - \nu_j \right) \right) \\ &= \exp \left(i \sum_{i=1}^I \theta_i \mu_i + i \sum_{j=1}^{J-1} \eta_j \nu_j \right) \left(\prod_{i=1}^I \prod_{j=1}^{J-1} e^{-i(\theta_i + \eta_j) G_{ij}} \right) \left(\prod_{i=1}^I e^{-i\theta_i G_{iJ}} \right). \end{aligned}$$

Equivalently, setting $\eta_J := 0$, this reads

$$e^{-i\langle \xi, \bar{z}(G) \rangle} = \exp \left(i \sum_{i=1}^I \theta_i \mu_i + i \sum_{j=1}^J \eta_j \nu_j \right) \prod_{i=1}^I \prod_{j=1}^J e^{-i(\theta_i + \eta_j) G_{ij}}. \quad (5.12)$$

We observe that the identification $\eta_J := 0$ is analogous to the invariance Kantorovich potentials described in Remark 1.8. This derivation gives a geometric interpretation of such invariance: the constraint map $\bar{\succ}$ takes values in a K -dimensional space because one marginal constraint is linearly dependent on the others. Consequently, the corresponding dual variables are defined only up to one redundant direction, exactly as the Kantorovich potentials are invariant under the translation

$$(u, v) \mapsto (u + c\mathbf{1}_I, v - c\mathbf{1}_J).$$

If $e^{-\varphi(G)}$ can also be factorized, the integral over $\mathcal{Q} = [0, 1]^{IJ}$ can be computed coordinate-wise. Hence, it is a reasonable modeling assumption that

$$\varphi(G) = \sum_{i=1}^I \sum_{j=1}^J \varphi_{ij}(G_{ij}).$$

This assumption is satisfied for the total *transport cost* of a plan G . Here, $\varphi(G) = \mathcal{U}(G) = \langle C, G \rangle = \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij}$, with $C \in \mathbb{R}_+^{IJ}$ representing the cost function matrix. In [12] and [34] a further multiplicative term $\beta > 0$ is introduced, so that φ takes the form

$$\varphi_\beta(G) = \beta \langle C, G \rangle.$$

Since φ_β depends on the parameter β , we relabel $Z \mapsto Z_\beta$ and $Z^n \mapsto Z_\beta^n$ accordingly, and we call Z_β the *partition function*, and Z_β^n its regularized version, to bridge with the physical interpretation.

Theorem 5.7 (Fourier representation of the regularized partition function).

For every $\beta > 0$ and $n > 0$, the Gaussian-regularized partition function satisfies

$$Z_\beta^n = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^I} \int_{\mathbb{R}^{J-1}} e^{-\frac{\|\theta\|^2 + \|\eta\|^2}{2n^2} + i\langle \theta, \mu \rangle + i\langle \eta, \nu \rangle} \prod_{i=1}^I \prod_{j=1}^J \frac{1 - e^{-(i\theta_i + i\eta_j + \beta C_{ij})}}{i\theta_i + i\eta_j + \beta C_{ij}} d\theta d\eta, \quad (5.13)$$

where $\eta_J := 0$.

Proof. We compute

$$\begin{aligned} \int_{\mathcal{Q}} e^{-\beta \langle C, G \rangle} e^{-i\langle \xi, \bar{\succ}(G) \rangle} dG &= \int_{[0,1]^{IJ}} e^{i\sum_{i=1}^I \theta_i \mu_i + i\sum_{j=1}^J \eta_j \nu_j} \prod_{i=1}^I \prod_{j=1}^J e^{-(i\theta_i + i\eta_j + \beta C_{ij}) G_{ij}} dG \\ &= \exp\left(i\langle \theta, \mu \rangle + i\langle \eta, \nu \rangle\right) \cdot \prod_{i=1}^I \prod_{j=1}^J \int_0^1 e^{-(i\theta_i + i\eta_j + \beta C_{ij}) G_{ij}} dG_{ij}. \end{aligned}$$

For every $(i, j) \in [I] \times [J]$ and $\beta > 0$,

$$\begin{aligned} \int_0^1 e^{-(i\theta_i + i\eta_j + \beta C_{ij})G_{ij}} dG_{ij} &= \frac{1}{i\theta_i + i\eta_j + \beta C_{ij}} \int_0^1 (i\theta_i + i\eta_j + \beta C_{ij}) e^{-(i\theta_i + i\eta_j + \beta C_{ij})G_{ij}} dG_{ij} \\ &= \frac{1}{i\theta_i + i\eta_j + \beta C_{ij}} \left[-e^{-(i\theta_i + i\eta_j + \beta C_{ij})G_{ij}} \right]_0^1 \\ &= \begin{cases} \frac{1 - e^{-(i\theta_i + i\eta_j + \beta C_{ij})}}{i\theta_i + i\eta_j + \beta C_{ij}} & \text{if } i\theta_i + i\eta_j + \beta C_{ij} \neq 0 \\ 1 & \text{otherwise} \end{cases}, \end{aligned}$$

The apparent singularity at $i\theta_i + i\eta_j + \beta C_{ij} = 0$ is removable, since

$$z \mapsto \frac{1 - e^{-z}}{z}$$

extends holomorphically to $z = 0$ with value 1. Hence

$$\int_{\mathcal{Q}} e^{-\beta \langle C, G \rangle} e^{-i \langle \xi, \bar{z}(G) \rangle} dG = e^{i \langle \theta, \mu \rangle + i \langle \eta, \nu \rangle} \prod_{i=1}^I \prod_{j=1}^J \frac{1 - e^{-(i\theta_i + i\eta_j + \beta C_{ij})}}{i\theta_i + i\eta_j + \beta C_{ij}}.$$

Substituting into the representation formula for Z_β^n , we obtain (5.13). $\square$

Combining Theorem 5.7 with the coarea approximation theorem gives, for the corresponding intrinsic integral,

$$Z_\beta^n = \frac{J_{\bar{z}}}{(2\pi)^K} \lim_{n \rightarrow \infty} \int_{\mathbb{R}^I} \int_{\mathbb{R}^{J-1}} e^{-\frac{\|\theta\|^2 + \|\eta\|^2}{2n^2} + i \langle \theta, \mu \rangle + i \langle \eta, \nu \rangle} \prod_{i=1}^I \prod_{j=1}^J \frac{1 - e^{-(i\theta_i + i\eta_j + \beta C_{ij})}}{i\theta_i + i\eta_j + \beta C_{ij}} d\theta d\eta.$$

If the normalized measure is used (cf. Remark 4.12), an additional factor depending on $\mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu))$ appears. However, this distinction is relevant at a logarithmic level up to an additive constant, which, as we shall see, is irrelevant for the finite-temperature algorithm. In the proof of Theorem 5.7 it naturally appears the holomorphic extension of the function h defined in section 2.2, namely

$$h(z) := \int_0^1 e^{-zt} dt = \frac{1 - e^{-z}}{z}, \quad h(0) := 1.$$

Then (5.13) can be written compactly as

$$Z_\beta^n = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^I} \int_{\mathbb{R}^{J-1}} e^{-\frac{\|\theta\|^2 + \|\eta\|^2}{2n^2} + i \langle \theta, \mu \rangle + i \langle \eta, \nu \rangle} \prod_{i=1}^I \prod_{j=1}^J h(\beta C_{ij} + i\theta_i + i\eta_j) d\theta d\eta. \quad (5.14)$$

5.4 Obstruction to the Naive Contour Deformation

The previous section gives a rigorous regularized Fourier representation of the partition function as in (5.14), for fixed $n < +\infty$ and $\eta_J := 0$. The variables $\xi = (\theta, \eta) \in \mathbb{R}^I \times \mathbb{R}^{J-1}$ are connected with Kantorovich potentials. However, this connection would require a change of variables of the form

$$u = i\theta/\beta, \quad v = i\eta/\beta.$$

Under this substitution, the regularized partition function can be written as

$$Z_\beta^n = \frac{(-i\beta)^{I+J-1}}{(2\pi)^K} \int_{(i\mathbb{R})^J} \int_{(i\mathbb{R})^{J-1}} e^{\frac{\beta^2 \|\theta\|^2 + \beta^2 \|\eta\|^2}{2n^2} + \beta \langle u, \mu \rangle + \beta \langle v, \nu \rangle} \prod_{i=1}^I \prod_{j=1}^J h(\beta(C_{ij} + u_i + v_j)) du dv.$$

If we can take the complex logarithm of h with no branch issues, we recover the effective energy functional that has been defined in Section 2.2. However, in order to integrate on the real axis rather than on the imaginary axis, a contour deformation is necessary. The aim of this subsection is to show that a naive contour deformation approach cannot be justified, for a fixed $n < +\infty$.

We show the obstruction in a two-dimensional model, which already contains the relevant analytic phenomenon. Consider

$$I_n := \int_{\mathbb{R}^2} f(\theta, \eta) d\theta d\eta,$$

where

$$f(\theta, \eta) := e^{-\frac{\theta^2 + \eta^2}{2n^2} + i(\mu\theta + \nu\eta)} h(\beta C + i\theta + i\eta),$$

with $\mu, \nu > 0$, $\beta > 0$ and $C \geq 0$. f is entire in the complex variables θ, η . Moreover, observe that

$$\begin{aligned} |f(\theta, \eta)| &= \left| e^{-\frac{\theta^2 + \eta^2}{2n^2}} \right| \left| e^{i(\mu\theta + \nu\eta)} \right| \frac{|1 - e^{-(\beta C + i\theta + i\eta)}|}{|\beta C + i\theta + i\eta|} \\ &= e^{-\frac{\theta^2 + \eta^2}{2n^2}} \cdot 1 \cdot \sqrt{\frac{1 + e^{-2\beta C} - 2e^{-\beta C} \cos(\theta + \eta)}{\beta^2 C^2 + (\theta + \eta)^2}} \\ &\leq e^{-\frac{\theta^2 + \eta^2}{2n^2}} \cdot \sqrt{\frac{(\beta C)^2 + 2e^{-\beta C} \frac{(\theta + \eta)^2}{2}}{\beta^2 C^2 + (\theta + \eta)^2}} \\ &\leq e^{-\frac{\theta^2 + \eta^2}{2n^2}} \sqrt{\frac{\beta^2 C^2 + (\theta + \eta)^2}{\beta^2 C^2 + (\theta + \eta)^2}} = e^{-\frac{\theta^2 + \eta^2}{2n^2}}. \end{aligned}$$

Hence, I_n converges absolutely because of the Gaussian factor:

$$\int_{\mathbb{R}^2} |f(\theta, \eta)| d\theta d\eta \leq \int_{\mathbb{R}^2} e^{-\frac{\theta^2 + \eta^2}{2n^2}} d\theta d\eta = 2\pi n^2 < \infty,$$

where the last equality is standard (cf. [41, Section 2.4]). This implies that the integral I_n is well-defined as an integration on the real axis.

We now try to rotate the positive half-line in the θ -variable. For $R > 0$, define

$$I_n^+(R) := \int_{\mathbb{R}} \int_0^R f(\theta, \eta) d\theta d\eta.$$

Define the sector

$$S_R := \{re^{-i\alpha} : 0 \leq r \leq R, 0 \leq \alpha \leq \pi/2\}.$$

Its boundary consists of

$$R_S = [0, R], \quad \text{arc}_R := \{Re^{-i\alpha} : 0 \leq \alpha \leq \pi/2\}, \quad C_S = [-iR, 0].$$

Since f is holomorphic in the θ -variable, Cauchy's theorem gives, for each fixed η ,

$$\int_0^R f(\theta, \eta) d\theta + \int_{\text{arc}_R} f(\theta, \eta) d\theta + \int_{-iR}^0 f(\theta, \eta) d\theta = 0.$$

Parametrizing the last segment by $\theta = -it$ and rescaling gives the equivalent form

$$I_n^+(R) = - \int_{\mathbb{R}} \int_{\text{arc}_R} f(\theta, \eta) d\theta d\eta + i\beta \int_{\mathbb{R}} \int_0^{R/\beta} f(-i\beta a, \eta) da d\eta.$$

If the arc contribution vanished as $R \rightarrow +\infty$, and if the rotated integral converged, then the contour rotation would justify the formal change of variables $(\theta, \eta) \mapsto (u, v)$. This is precisely where the argument fails. Along the rotated contour,

$$f(-i\beta a, \eta) = e^{+\frac{\beta^2 a^2}{2n^2} - \beta\mu a} e^{-\frac{\eta^2}{2n^2} + i\nu\eta} h(\beta C + \beta a + i\eta).$$

This integral diverges exponentially, since

$$h(\beta C + \beta a + i\eta) = \frac{1 - e^{-(\beta C + \beta a + i\eta)}}{\beta C + \beta a + i\eta} \sim \frac{1}{\beta C + \beta a + i\eta},$$

whereas the Gaussian term grows quadratically exponentially.

5.5 Method of Steepest Descents

While the previous section shows that a simple contour integration does not properly justify the shift from Z_β^n to F_β , the aim of this section is to formulate a steepest-descent framework under which such a passage could, in principle, be justified. We do not claim to fully verify the hypotheses needed for this method, but we isolate the analytic structures that would be required for that. The main methodological reference is [42], whose main results are collected in Appendix E.

The steepest descent method, also known as saddle-point method in $\mathbb{C}$, is a generalization of Laplace's principle for real integrals, yielding asymptotic formulae for integrals of the form

$$I(\beta) = \int_{\Gamma} e^{\beta f(z)} g(z) dz, \quad z \in \mathbb{C}, \quad \beta \rightarrow +\infty. \quad (5.15)$$

Under certain conditions on Γ and f , if it exists an admissible saddle point $z^* \in \Gamma$ as in Definitions E.1, E.2, then, by Corollary E.8,

$$-\frac{1}{\beta} \log I_\beta = -f(z^*) + o(1). \quad (5.16)$$

The integral (5.13) belongs to the form in (5.15): starting from

$$Z_\beta^n = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^I} \int_{\mathbb{R}^{J-1}} e^{-\frac{\|\theta\|^2 + \|\eta\|^2}{2n^2} + i\langle\theta, \mu\rangle + i\langle\eta, \nu\rangle} \prod_{i=1}^I \prod_{j=1}^J h(\beta C_{ij} + i\theta_i + i\eta_j) d\theta d\eta, \quad (5.17)$$

performing the change of variables $(\theta, \eta) := \beta(p, q)$,

$$Z_\beta^n = \frac{\beta^K}{(2\pi)^K} \int_{\mathbb{R}^{I+J-1}} e^{-\beta^2 \frac{\|p\|^2 + \|q\|^2}{2n^2} + \beta i\langle p, \mu\rangle + \beta i\langle q, \nu\rangle} \prod_{i=1}^I \prod_{j=1}^J h(\beta(ip_i + iq_j + C_{ij})) dp dq \quad (5.18)$$

$$= \frac{\beta^K}{(2\pi)^K} \int_{\mathbb{R}^{I+J-1}} e^{\beta \left(-\frac{\|p\|^2 + \|q\|^2}{2n^2} + i\langle p, \mu\rangle + i\langle q, \nu\rangle \right)} \cdot e^{\sum_{i,j} \log h(\beta(ip_i + iq_j + C_{ij}))} dp dq \quad (5.19)$$

$$= \frac{\beta^K}{(2\pi)^K} \int_{\mathbb{R}^{I+J-1}} e^{\beta \Psi_{\beta,n}(p,q)} dp dq, \quad (5.20)$$

where

$$\Psi_{\beta,n}(p, q) = i\langle p, \mu\rangle + i\langle q, \nu\rangle - \beta \frac{\|p\|^2 + \|q\|^2}{2n^2} + \frac{1}{\beta} \sum_{i=1}^I \sum_{j=1}^J \log h(\beta(C_{ij} + ip_i + iq_j)) \quad (5.21)$$

and, as usual, $v_J = \eta_J = p_J = 0$. We observe that $\Psi_{\beta,n}(p, q)$ is related with the effective free energy F_β defined and studied in Section 2 by

$$-\Psi_{\beta,n}(-iu, -iv) = -\langle u, \mu \rangle - \langle v, \nu \rangle + \frac{1}{\beta} \sum_{i=1}^I \sum_{j=1}^J \log h(\beta(C_{ij} + u_i + v_j)) - \beta \frac{\|u\|^2 + \|v\|^2}{2n^2},$$

hence

$$-\Psi_{\beta,n}(-iu, -iv) = F_{\beta}(u, v) - \beta \frac{\|u\|^2 + \|v\|^2}{2n^2}. \quad (5.22)$$

Therefore, if we are able to find a saddle point $(p^{\beta,n}, q^{\beta,n})$ of $\Psi_{\beta,n}$ on the imaginary axes, we can hopefully relate it with (u^{β}, v^{β}) , i.e., the unique global maximizer of F_{β} established in Theorem 2.8: fixing $v_J := 0$,

$$(u^{\beta}, v^{\beta}) := \arg \max_{(u,v) \in \mathbb{R}^{I+J-1}} F_{\beta}(u, v)$$

For this reason, we start by establishing a candidate imaginary saddle point and record its elementary properties:

Lemma 5.8. *For every $n, \beta > 0$, there exists a unique point*

$$z_{\beta,n}^{\star} := (p^{\beta,n}, q^{\beta,n}) = (-iu^{\beta,n}, -iv^{\beta,n})$$

on the imaginary axes

$$\{(p, q) \in (i\mathbb{R})^I \times (i\mathbb{R})^J : q_J = 0\},$$

which is an admissible saddle point of $\Psi_{\beta,n}$. More precisely,

$$\nabla \Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n}) = 0,$$

and, writing

$$z = (z_1, \dots, z_K) := (p_1, \dots, p_I, q_1, \dots, q_{J-1}), \quad K := I + J - 1,$$

one has, for every $\ell = 1, \dots, K$,

$$H_{\ell}(p^{\beta,n}, q^{\beta,n}) \neq 0,$$

where

$$H_{\ell}(p^{\beta,n}, q^{\beta,n}) := \det \begin{pmatrix} \frac{\partial^2 \Psi_{\beta,n}}{\partial z_1^2} & \cdots & \frac{\partial^2 \Psi_{\beta,n}}{\partial z_1 \partial z_{\ell}} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 \Psi_{\beta,n}}{\partial z_{\ell} \partial z_1} & \cdots & \frac{\partial^2 \Psi_{\beta,n}}{\partial z_{\ell}^2} \end{pmatrix}_{z=(p^{\beta,n}, q^{\beta,n})}.$$

Furthermore, there exists a radius $R > 0$ such that, for every $\beta > 0$ sufficiently large and $n > 0$,

$$\|(p^{\beta,n}, q^{\beta,n})\|^2 := \sum_{i=1}^I |p_i^{\beta,n}|^2 + \sum_{j=1}^{J-1} |q_j^{\beta,n}|^2 < R^2,$$

Proof. Let

$$\mathcal{P} := \{(u, v) \in \mathbb{R}^{I+J} : v_J = 0\}$$

and define

$$\tilde{F}_{\beta,n}(u, v) := F_{\beta}(u, v) - \beta \frac{\|u\|^2 + \|v\|^2}{2n^2}, \quad (u, v) \in \mathcal{P}. \quad (5.23)$$

Since, by Lemma 2.6, F_{β} is strictly concave on $\mathcal{P}$, and the map

$$(u, v) \mapsto -\beta \frac{\|u\|^2 + \|v\|^2}{2n^2}$$

is strictly concave, it follows that $\tilde{F}_{\beta,n}$ is strictly concave on $\mathcal{P}$. To see that $\tilde{F}_{\beta,n}$ has a unique global maximizer, observe that

$$\tilde{F}_{\beta,n}(0, 0) = F_{\beta}(0, 0) \geq 0,$$

and, if $F_{\beta}(u, v) < 0$, then

$$\tilde{F}_{\beta,n}(u, v) = F_{\beta}(u, v) - \beta \frac{\|u\|^2 + \|v\|^2}{2n^2} < 0 \leq \tilde{F}_{\beta,n}(0, 0).$$

Therefore,

$$\{\tilde{F}_{\beta,n}(u, v) \geq 0\} \subseteq \{F_{\beta}(u, v) \geq 0\}.$$

By the proof of Theorem 2.7, $\{F_{\beta} \geq 0\}$ is compact in $\mathcal{P}$, and $\tilde{F}_{\beta,n}$ is continuous, so $\{\tilde{F}_{\beta,n}(u, v) \geq 0\}$ is also compact, and, in particular, $\tilde{F}_{\beta,n}$ attains its maximum on $\mathcal{P}$. Since $\tilde{F}_{\beta,n}$ is strictly concave, this maximizer is unique. We denote it by

$$(u^{\beta,n}, v^{\beta,n}) := \arg \max_{(u,v) \in \mathcal{P}} \tilde{F}_{\beta,n}(u, v).$$

Since $\tilde{F}_{\beta,n}(u, v)$ is continuously differentiable, it holds

$$\nabla \tilde{F}_{\beta,n}(u^{\beta,n}, v^{\beta,n}) = 0.$$

Define

$$(p^{\beta,n}, q^{\beta,n}) := (-iu^{\beta,n}, -iv^{\beta,n}) \in \{(p, q) \in (\mathbb{i}\mathbb{R})^I \times (\mathbb{i}\mathbb{R})^J : q_J = 0\}.$$

By Equation (5.22), it holds

$$\Psi_{\beta,n}(-iu, -iv) = -\tilde{F}_{\beta,n}(u, v).$$

Observe that since $h(y) > 0$ for every $y \in \mathbb{R}$, all factors $h(\beta(C_{ij} + u_i^{\beta,n} + v_j^{\beta,n}))$ are real and nonzero. Therefore, $\Psi_{\beta,n}$ is holomorphic near $(p^{\beta,n}, q^{\beta,n})$. Differentiating in a neighbourhood of $(p^{\beta,n}, q^{\beta,n})$,

$$\mathbb{i} \nabla \Psi_{\beta,n}(-iu, -iv) = \nabla \tilde{F}_{\beta,n}(u, v) \quad (5.24)$$

Hence

$$i\nabla\Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n}) = \nabla\tilde{F}_{\beta,n}(u^{\beta,n}, v^{\beta,n}) = 0.$$

Dividing by i , it follows that $(p^{\beta,n}, q^{\beta,n})$ is a saddle point of $\Psi_{\beta,n}$.

By uniqueness of $(u^{\beta,n}, v^{\beta,n})$ on $\mathcal{P}$, it follows uniqueness of the saddle point $(p^{\beta,n}, q^{\beta,n})$ on the imaginary axes.

We now prove admissibility. Using the canonical coordinate ordering, differentiating (5.24) and using (5.23),

$$\nabla^2\Psi_{\beta,n}(-iu, -iv) = \nabla^2\tilde{F}_{\beta,n}(u, v) = \nabla^2F_{\beta}(u, v) - \frac{\beta}{n^2}I.$$

Hence

$$\nabla^2\Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n}) = \nabla^2F_{\beta}(u^{\beta,n}, v^{\beta,n}) - \frac{\beta}{n^2}I.$$

By Lemma 2.6, the Hessian ∇^2F_{β} is negative definite on $\mathcal{P}$. Therefore

$$\nabla^2F_{\beta} - \frac{\beta}{n^2}I$$

is also negative definite. Hence

$$\nabla^2\Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n})$$

is a real symmetric negative definite matrix. Every principal submatrix of a negative definite symmetric matrix is negative definite, hence has nonzero determinant. Therefore, for every $\ell = 1, \dots, K$,

$$H_{\ell}(p^{\beta,n}, q^{\beta,n}) \neq 0.$$

Thus $(p^{\beta,n}, q^{\beta,n})$ is an admissible saddle point of $\Psi_{\beta,n}$.

It remains to prove the uniform boundedness statement. By Lemma 2.9, there exist $\beta_0 > 0$ and $R > 0$ such that

$$\|(u^{\beta}, v^{\beta})\| < R \quad \text{for every } \beta \geq \beta_0,$$

where (u^{β}, v^{β}) is the unique maximizer of the effective free energy F_{β} . We compare $(u^{\beta,n}, v^{\beta,n})$ with (u^{β}, v^{β}) . We clearly have, by definition,

$$\tilde{F}_{\beta,n}(u^{\beta,n}, v^{\beta,n}) \geq \tilde{F}_{\beta,n}(u^{\beta}, v^{\beta}).$$

Equivalently,

$$F_{\beta}(u^{\beta,n}, v^{\beta,n}) - \frac{\beta}{2n^2}\|(u^{\beta,n}, v^{\beta,n})\|^2 \geq F_{\beta}(u^{\beta}, v^{\beta}) - \frac{\beta}{2n^2}\|(u^{\beta}, v^{\beta})\|^2. \quad (5.25)$$

Since (u^β, v^β) maximizes F_β , we also have

$$F_\beta(u^{\beta,n}, v^{\beta,n}) \leq F_\beta(u^\beta, v^\beta). \quad (5.26)$$

Therefore,

$$-\frac{\beta}{2n^2} \|(u^{\beta,n}, v^{\beta,n})\|^2 \geq -\frac{\beta}{2n^2} \|(u^\beta, v^\beta)\|^2,$$

or,

$$\|(u^{\beta,n}, v^{\beta,n})\| \leq \|(u^\beta, v^\beta)\| < R.$$

Hence

$$\|(p^{\beta,n}, q^{\beta,n})\| = \|(u^{\beta,n}, v^{\beta,n})\| < R,$$

concluding the proof. $\square$

The previous lemma provides the key starting point for a multidimensional steepest-descent analysis, as it shows that $\Psi_{\beta,n}$ has a unique admissible saddle point on the imaginary axes and, in particular, that these saddle points remain uniformly confined in a compact subset of $\mathbb{C}^K$ for β sufficiently large, $n > 0$. This makes it natural to relate Z_β^n , the phase $\Psi_{\beta,n}$, and the effective free energy F_β , in the spirit of the heuristic argument in [12, Eq. 24]. The multidimensional steepest-descent framework of Pinna-Viola [42, Thm. 4.2] provides a theoretical framework for making this asymptotic argument rigorous. There are, however, two main points which require care in adapting this framework to our setting. First, one has to replace the original integration path over $\mathbb{R}^{I+J-1}$ by an equivalent complex integration cycle passing through $(p^{\beta,n}, q^{\beta,n})$ and satisfying suitable dominance conditions. Second, the relevant saddle point depends on the asymptotic parameter β : therefore, an integration cycle independent of β and passing through $(p^{\beta,n}, q^{\beta,n})$, as prescribed by Pinna-Viola, cannot be obtained. As a consequence, the hypotheses of Pinna-Viola's theorem, if adapted to our setting, should be made uniform in β . We do not provide such an extension in this work. Nevertheless, there are reasons that make such an adaptation plausible:

- First, we have just proven that the saddle points are uniformly confined in a compact region of $\mathbb{C}^K$. Which suggests that the local geometry near the saddle can be controlled uniformly in β .
- Second, we do not need a full generalization of Theorem 4.2 of Pinna-Viola. Indeed, we are only looking for a logarithmic asymptotic of Z_β^n , as in (5.16). Therefore, possible β -dependent changes in the statement of the theorem may produce error

terms and contributions that are below-exponential, hence being harmless for the validity of Corollary E.8, which provides a logarithmic asymptotics.

As for the existence of a path satisfying the dominance assumptions of Appendix E, the problem is genuinely nontrivial. A favourable point is that the integrand, in the form of (5.18), is holomorphic on $\mathbb{C}^K$, as it does not introduce the logarithm yet. Hence, we can define the contour deformation through this function, before introducing any logarithm. Meanwhile, the function $\Psi_{\beta,n}$ is needed only locally near the saddle. Luckily, $\Psi_{\beta,n}$ is well-defined in a neighbourhood of $(p^{\beta,n}, q^{\beta,n})$ after choosing a branch of the logarithm, as

$$\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n}) = \beta(C_{ij} + u_i^{\beta,n} + v_j^{\beta,n}) \in \mathbb{R},$$

and $h(x) > 0$ for $x \in \mathbb{R}$, so our integrand has no effective branch issues.

The Translated Cycle. In what follows, we investigate the most natural candidate cycle, namely the translated path

$$\Gamma_{\beta,n} := \{z \in \mathbb{C}^K : z - (p^{\beta,n}, q^{\beta,n}) \in \mathbb{R}^K\},$$

oriented in some coordinate order as in Appendix E. We show that this cycle is a natural candidate and that it yields the desired logarithmic asymptotics provided some technical assumptions from Pinna-Viola are satisfied.

This cycle passes through the admissible saddle point constructed in Lemma 5.8 and depends on β only through a uniformly bounded translation. To show that this contour deformation is valid, it suffices to show that, fixing $\beta, n > 0$, the lateral contributions vanish as $R \rightarrow +\infty$. We estimate the contribution to the integral of only one of the $2K$ lateral faces. Since there are finitely many of them and the same estimate applies to each face, showing that one of them has negligible contribution to the integral is sufficient.

Consider the face

$$z := (x_1 + tp_1^{\beta,n}, x_2 + tp_2^{\beta,n}, \dots, x_I + tp_I^{\beta,n}, x_{I+1} + tq_1^{\beta,n}, \dots, x_{I+J-1} + tq_{J-1}^{\beta,n}),$$

where $t \in [0, 1]$, $x_1 = R$, and

$$(x_2, \dots, x_{I+J-1}) \in [-R, R]^{K-1}.$$

We aim to bound the corresponding lateral contribution, namely

$$\frac{\beta^K}{(2\pi)^K} \int_0^1 \int_{[-R, R]^{K-1}} e^{-\frac{\beta^2}{2n^2} \|z\|^2 + i\beta \langle z, (\mu, \nu) \rangle} \prod_{i=1}^I \prod_{j=1}^J h(\beta(C_{ij} + iz_i + iz_{I+j})) dx dt, \quad z_{I+J:=0}$$

We observe that, along $z = x + t(p^{\beta,n}, q^{\beta,n})$, using $h(y) = \int_0^1 e^{-sy} ds$ from (2.6),

$$|h(\beta(C_{ij} + iz_i + iz_{I+j}))| \leq \int_0^1 |e^{-s\beta(C_{ij} + iz_i + iz_{I+j})}| ds \quad (5.27)$$

$$= \int_0^1 e^{-s\beta\Re(C_{ij} + itp_i^{\beta,n} + itq_j^{\beta,n})} ds \quad (5.28)$$

$$\leq \int_0^1 e^{s\beta(|p_i^{\beta,n}| + |q_j^{\beta,n}|)} ds, \quad (5.29)$$

which is independent of x , t , and R . Similarly,

$$|e^{i\beta\langle z, (\mu, \nu) \rangle}| = |e^{i\beta\langle x, (\mu, \nu) \rangle} e^{i\beta t \langle (p^{\beta,n}, q^{\beta,n}), (\mu, \nu) \rangle}| \leq e^{\beta |\langle (p^{\beta,n}, q^{\beta,n}), (\mu, \nu) \rangle|}$$

which is independent of x , t and R . Thus, it remains to show that

$$\int_0^1 \int_{[-R, R]^{K-1}} |e^{-\frac{\beta^2}{2n^2} \|z\|^2}| dx dt \xrightarrow{R \rightarrow +\infty} 0,$$

where $\|z\|^2 = \sum_{\ell=1}^K z_\ell^2$. Since $z = x + t(p^{\beta,n}, q^{\beta,n})$, with $x_1 = R$, and since $(p^{\beta,n}, q^{\beta,n})$ lies on the imaginary axes, we have

$$\begin{aligned} |e^{-\frac{\beta^2}{2n^2} \|z\|^2}| &= \exp \left(-\frac{\beta^2}{2n^2} \sum_{i=1}^I [x_i^2 - t^2 |p_i^{\beta,n}|^2] - \frac{\beta^2}{2n^2} \sum_{j=1}^{J-1} [x_{I+j}^2 - t^2 |q_j^{\beta,n}|^2] \right) \\ &\leq \exp \left(-\frac{\beta^2}{2n^2} \left[R^2 + \sum_{i=1}^I -|p_i^{\beta,n}|^2 + \sum_{j=1}^{J-1} -|q_j^{\beta,n}|^2 \right] \right) \\ &= \text{Const.} \times e^{-\frac{\beta^2}{2n^2} R^2}. \end{aligned}$$

Therefore the whole lateral contribution is bounded by

$$\text{Const.} \times (1 + R)^{K-1} e^{-\frac{\beta^2}{2n^2} R^2} \xrightarrow{R \rightarrow +\infty} 0.$$

Since this deformation is a simple translation, we can write the new integral through $\Gamma_{\beta,n}$ in a particularly simple way as

$$Z_\beta^n = \frac{\beta^K}{(2\pi)^K} \int_{\mathbb{R}^K} e^{\beta \Psi_{\beta,n}((p^{\beta,n}, q^{\beta,n}) + (p, q))} dp dq = \frac{\beta^K}{(2\pi)^K} e^{\beta \Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n})} I_{\beta,n}, \quad (5.30)$$

where

$$I_{\beta,n} := \int_{\mathbb{R}^K} e^{\beta (\Psi_{\beta,n}((p^{\beta,n}, q^{\beta,n}) + (p, q)) - \Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n}))} dp dq. \quad (5.31)$$

Equivalently,

$$I_{\beta,n} = \int_{\mathbb{R}^K} e^{-\beta^2 \frac{\|p\|^2 + \|q\|^2}{2n^2} + i\beta \langle p, \mu \rangle + i\beta \langle q, \nu \rangle} e^{-\beta^2 \frac{\langle p, p^{\beta,n} \rangle + \langle q, q^{\beta,n} \rangle}{n^2}} \\ \times \prod_{i=1}^I \prod_{j=1}^J \frac{h\left(\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n} + i(p_i + q_j))\right)}{h\left(\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n})\right)} dp dq, \quad q_J = q_J^{\beta,n} = 0. \quad (5.32)$$

The explicit representation (5.32) is actually preferred because it does not introduce a logarithm that would require a global branch choice. Accordingly, (5.31) should be understood as a shorthand for (5.32). Moreover, (5.30) is useful because it isolates the saddle contribution from the remaining factor $I_{\beta,n}$. The latter is described, in the explicit form (5.32), by a Gaussian factor multiplied by a ratio of h -factors. This ratio can be bounded, as the next lemma shows:

Lemma 5.9. *For every $\beta, n > 0$,*

$$|I_{\beta,n}| \leq (2\pi)^{K/2} \left(\frac{n}{\beta}\right)^K.$$

In particular, if $n(\beta) = \beta^{3/4}$, then

$$\log |I_{\beta,n(\beta)}| \leq o(\beta).$$

Proof. As in (5.27), using $h(y) = \int_0^1 e^{-sy} ds$ from (2.6), for every (i, j) we have

$$\left| h\left(\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n} + i(p_i + q_j))\right) \right| \leq \int_0^1 \left| e^{-s\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n} + i(p_i + q_j))} \right| ds \quad (5.33)$$

$$= \int_0^1 e^{-s\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n})} ds \quad (5.34)$$

$$= h\left(\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n})\right) > 0, \quad (5.35)$$

since $(p^{\beta,n}, q^{\beta,n})$ lie on the imaginary axes. Hence

$$\left| \frac{h\left(\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n} + i(p_i + q_j))\right)}{h\left(\beta(C_{ij} + ip_i^{\beta,n} + iq_j^{\beta,n})\right)} \right| \leq 1,$$

with a strictly positive denominator, since $h(x) > 0$ for every $x \in \mathbb{R}$. From (5.32), we get

$$\begin{aligned}
 |I_{\beta,n}| &\leq \int_{\mathbb{R}^K} \left| e^{-\beta^2 \frac{\|p\|^2 + \|q\|^2}{2n^2} + i\beta \langle p, \mu \rangle + i\beta \langle q, \nu \rangle} \right| \left| e^{-\beta^2 \frac{\langle p, p^{\beta,n} \rangle + \langle q, q^{\beta,n} \rangle}{n^2}} \right| dp dq \\
 &= \int_{\mathbb{R}^K} \left| e^{-\beta^2 \frac{\|p\|^2 + \|q\|^2}{2n^2}} \right| \left| e^{-\beta^2 \frac{\langle p, p^{\beta,n} \rangle + \langle q, q^{\beta,n} \rangle}{n^2}} \right| dp dq \\
 &= \int_{\mathbb{R}^K} e^{-\beta^2 \frac{\|p\|^2 + \|q\|^2}{2n^2}} dp dq \\
 &= (2\pi)^{K/2} \left(\frac{n}{\beta} \right)^K.
 \end{aligned}$$

where the last cancellation is due to the fact that (p, q) are real and $(p^{\beta,n}, q^{\beta,n})$ lie on the imaginary axes, so the second term has modulus 1. The gaussian integral has been computed explicitly (see Example 5.3 for a derivation).

For $n(\beta) = \beta^{3/4}$ we obtain

$$|I_{\beta,n(\beta)}| \leq (2\pi)^{K/2} \beta^{-K/4},$$

so

$$\log |I_{\beta,n(\beta)}| \leq \frac{K}{2} \log(2\pi) - \frac{K}{4} \log \beta = o(\beta).$$

□

If we could prove a lower bound of the form

$$\log |I_{\beta,n(\beta)}| \geq -o(\beta),$$

then, together with the upper bound of Lemma 5.9, we would obtain

$$\log |I_{\beta,n(\beta)}| = o(\beta).$$

Consequently,

$$\begin{aligned}
 -\frac{1}{\beta} \log Z_{\beta}^{n(\beta)} &= -\frac{1}{\beta} \log |Z_{\beta}^{n(\beta)}| = -\frac{1}{\beta} \log \left[\frac{\beta^K}{(2\pi)^K} e^{\beta \Psi_{\beta,n(\beta)}(p^{\beta,n(\beta)}, q^{\beta,n(\beta)})} |I_{\beta,n(\beta)}| \right] \\
 &= -\frac{1}{\beta} [K \log \beta - K \log(2\pi) + \beta \Psi_{\beta,n(\beta)}(p^{\beta,n(\beta)}, q^{\beta,n(\beta)}) + \log |I_{\beta,n(\beta)}|] \\
 &= -\Psi_{\beta,n(\beta)}(p^{\beta,n(\beta)}, q^{\beta,n(\beta)}) + o(1) \\
 &\stackrel{(5.22)}{=} F_{\beta}(u^{\beta,n(\beta)}, v^{\beta,n(\beta)}) - \beta \frac{\|u^{\beta,n(\beta)}\|^2 + \|v^{\beta,n(\beta)}\|^2}{2[n(\beta)]^2} + o(1)
 \end{aligned}$$

If $n(\beta) \gg \sqrt{\beta}$, since saddle points are bounded by Lemma 5.8, then

$$-\frac{1}{\beta} \log Z_{\beta}^{n(\beta)} = F_{\beta}(u^{\beta,n(\beta)}, v^{\beta,n(\beta)}) + o(1).$$

We now recall Equations (5.25) and (5.26) found in Lemma 5.8: it holds

$$F_\beta(u^{\beta,n}, v^{\beta,n}) - \frac{\beta}{2n^2} \|(u^{\beta,n}, v^{\beta,n})\|^2 \geq F_\beta(u^\beta, v^\beta) - \frac{\beta}{2n^2} \|(u^\beta, v^\beta)\|^2.$$

and

$$F_\beta(u^{\beta,n}, v^{\beta,n}) \leq F_\beta(u^\beta, v^\beta).$$

Therefore,

$$0 \leq F_\beta(u^\beta, v^\beta) - F_\beta(u^{\beta,n(\beta)}, v^{\beta,n(\beta)}) \leq \frac{\beta}{2n^2} \|(u^\beta, v^\beta)\|^2 \stackrel{5.8}{\leq} \frac{\beta}{2n(\beta)^2} R^2 \xrightarrow{\beta \rightarrow +\infty} 0.$$

Hence

$$-\frac{1}{\beta} \log Z_\beta^{n(\beta)} = F_\beta(u^{\beta,n(\beta)}, v^{\beta,n(\beta)}) + o(1) = F_\beta(u^\beta, v^\beta) + o(1).$$

Recalling that

$$Z_\beta = J_{\bar{\mathbf{z}}} \lim_{n \rightarrow +\infty} Z_\beta^n,$$

where $J_{\bar{\mathbf{z}}}$ is a Jacobian factor, if the convergence is compatible with the joint limit $(\beta, n(\beta)) \rightarrow +\infty$ up to a negligible error, then

$$-\frac{1}{\beta} \log Z_\beta = F_\beta(u^\beta, v^\beta) + o(1), \tag{SPA}$$

which is the rigorous form of the saddle-point approximation corresponding to [12, Eq. 24]. This is the conditional asymptotic relation that will be used in Section 6.4 to justify the statistical physics derivation of the finite-temperature optimal transport algorithm.

Observe that Lemma 5.9 works as long as $n(\beta) = o(\beta)$. Since we just required $n(\beta) \gg \sqrt{\beta}$, it is a natural to set $n(\beta) := \beta^{3/4}$.

An upper bound for $|I_{\beta,n}|$ has been easily established thanks to the inequalities (5.27) and (5.33), that bound from above $|h(z)|$, and to the triangle inequality. Unfortunately, there does not seem to be a similar way to determine a lower bound. Indeed, $|h(z)|$ can be arbitrarily small near the complex zeros of h .

For this reason, we rely on the saddle-point framework, which directly yields the right asymptotic growth for $I_{\beta,n}$ under suitable hypotheses. This means that, strictly speaking, Lemma 5.9 may not be necessary. However, it provides an interesting upper bound, and supports the idea that the translated path

$$\Gamma_{\beta,n} := \{z \in \mathbb{C}^K : z - (p^{\beta,n}, q^{\beta,n}) \in \mathbb{R}^K\}$$

is a reasonable candidate cycle. Furthermore, it also shows absolute convergence of the integral, which is an assumption for Pinna-Viola's theorem. After the translation, the

cycle passes through the saddle, which is admissible by Lemma 5.8, and located at the origin.

What is not proved is that the translated cycle (or another cycle that satisfies analytic properties similar to those discussed above) can be realized, or replaced without changing the integral up to subexponential errors, by an admissible iterated path $\Gamma_{\lambda,\beta}$ satisfying the Third, Fourth and Fifth Conditions of Appendix E, uniformly in β at logarithmic scale. If those conditions were realized, then Theorem E.6 yields the asymptotic

$$I_{\beta,n(\beta)} = \left(\frac{2\pi}{\beta}\right)^{K/2} \frac{1}{\sqrt{\det(-H_{\beta,n(\beta)})}} e^{o(\beta)},$$

where

$$H_{\beta,n} := \nabla^2 \Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n}).$$

Under these assumptions,

$$\log |I_{\beta,n(\beta)}| = -\frac{K}{2} \log \beta - \frac{1}{2} \log |\det(-H_{\beta,n(\beta)})| + o(\beta) = -\frac{1}{2} \log |\det(-H_{\beta,n(\beta)})| + o(\beta).$$

Thus $\log |I_{\beta,n(\beta)}| = o(\beta)$ provided

$$\log |\det H_{\beta,n(\beta)}| = o(\beta).$$

To justify (SPA), it remains to prove this final condition.

Lemma 5.10. *Let $n(\beta) = \beta^{3/4}$, and let*

$$H_{\beta,n} := \nabla^2 \Psi_{\beta,n}(p^{\beta,n}, q^{\beta,n})$$

be the Hessian of $\Psi_{\beta,n}$ computed on the saddle $(p^{\beta,n}, q^{\beta,n})$ with respect to the coordinates

$$z = (z_1, \dots, z_K) := (p_1, \dots, p_I, q_1, \dots, q_{J-1}), \quad K := I + J - 1.$$

Then

$$\log |\det H_{\beta,n(\beta)}| = o(\beta).$$

Proof. By the proof of Lemma 5.8, one has

$$H_{\beta,n} = \nabla^2 F_{\beta}(u^{\beta,n}, v^{\beta,n}) - \frac{\beta}{n^2} I_K.$$

Using the explicit Hessian formula in Lemma 2.6, the entries of $H_{\beta,n}$ are finite sums of terms $f_{\beta}''(x_{ij}^{\beta,n})$, minus a diagonal term

$$\frac{\beta}{n(\beta)^2} = \frac{\beta}{(\beta^{3/4})^2} = \frac{1}{\sqrt{\beta}}.$$

We will now show that the terms $f''_{\beta}(x_{ij}^{\beta,n})$ have a polynomial growth at most. By Lemma 2.3,

$$f''_{\beta}(x_{ij}^{\beta,n}) = \beta\varphi'(\beta x_{ij}^{\beta,n}).$$

and, using

$$\varphi(y) = \frac{1}{y} - \frac{1}{e^y - 1},$$

we get, for $y \neq 0$,

$$\varphi'(y) = -\frac{1}{y^2} + \frac{e^y}{(e^y - 1)^2}.$$

Therefore, if $|\beta x_{ij}^{\beta,n}| \rightarrow +\infty$ ($x_{ij}^{\beta,n} \neq 0$),

$$f''_{\beta}(x_{ij}^{\beta,n}) = \beta \left(-\frac{1}{(\beta x_{ij}^{\beta,n})^2} + \frac{e^{\beta x_{ij}^{\beta,n}}}{(e^{\beta x_{ij}^{\beta,n}} - 1)^2} \right) \Rightarrow f''_{\beta}(x_{ij}^{\beta,n}) \sim -\frac{1}{\beta(x_{ij}^{\beta,n})^2}$$

If instead $|\beta x_{ij}^{\beta,n}| \rightarrow c$ for some finite $c \in \mathbb{R}$, then

$$f''_{\beta}(x_{ij}^{\beta,n}) = \beta\varphi'(\beta x_{ij}^{\beta,n}) \sim \beta\varphi'(c).$$

Since

$$\varphi'(-y) = -\frac{1}{(-y)^2} + \frac{e^{-y}}{(e^{-y} - 1)^2} = -\frac{1}{y^2} + \frac{e^y}{(1 - e^y)^2} = -\frac{1}{y^2} + \frac{e^y}{(e^y - 1)^2} = \varphi'(y),$$

and φ' is continuous by Lemma 2.2, then φ' is unbounded if and only if $\varphi'(0) \in \{\pm\infty\}$ or $\lim_{y \rightarrow +\infty} \varphi'(y) \in \{\pm\infty\}$. We already know that $\varphi'(y) \sim 1/y^2$ as $y \rightarrow +\infty$. To show that $\varphi'(0)$ is bounded, observe that, by Lemma 2.2,

$$\varphi'(y) = \left(\frac{\int_0^1 t e^{-ty} dt}{\int_0^1 e^{-ty} dt} \right)^2 - \frac{\int_0^1 t^2 e^{-ty} dt}{\int_0^1 e^{-ty} dt}.$$

hence

$$\varphi'(0) = \left(\frac{\int_0^1 t e^{-t \cdot 0} dt}{\int_0^1 e^{-t \cdot 0} dt} \right)^2 - \frac{\int_0^1 t^2 e^{-t \cdot 0} dt}{\int_0^1 e^{-t \cdot 0} dt} = \frac{1}{4} - \frac{1}{3} = -\frac{1}{12}.$$

Since the entries of $H_{\beta,n}$ are sums of terms with growth at most β , we conclude that the entries of the Hessian are bounded in absolute value, by $C\beta$, for some constant $C > 0$. Hence, the determinant of the Hessian is bounded by

$$|\det H_{\beta,n(\beta)}| \leq K! \cdot (C\beta)^K.$$

On the other hand, a lower bound can be determined as follows: by Lemma 2.6, the Hessian $\nabla^2 F_\beta$ is symmetric and negative definite on $\mathcal{P}$. Hence $H_{\beta,n}$ is negative definite, and every eigenvalue λ_m of $H_{\beta,n}$ satisfies

$$\lambda_m \leq -\frac{\beta}{n^2} \implies |\lambda_m| \geq \frac{\beta}{n(\beta)^2} = \frac{\beta}{(\beta^{3/4})^2} = \frac{1}{\sqrt{\beta}}.$$

Hence

$$|\det H_{\beta,n(\beta)}| = \prod_{m=1}^K |\lambda_m| \geq \beta^{-K/2}.$$

Combining the lower and upper bounds, we obtain

$$\beta^{-K/2} \leq |\det H_{\beta,n(\beta)}| \leq K!(C\beta)^K.$$

Taking logarithms gives

$$\log |\det H_{\beta,n(\beta)}| = o(\beta).$$

□

6 Statistical-Physics Framework on the Transport Polytope

In this chapter we present the statistical-physics framework introduced in [12]. It follows the standard statistical-physics viewpoint on optimization: a cost function is interpreted as an energy, a Boltzmann weight $e^{-\beta E}$ defines a Gibbs ensemble, the partition function normalizes the ensemble, and the low-temperature limit $\beta \rightarrow \infty$ extracts minimizers of the energy [13]. In the discrete optimal-transport setting, this means defining a Gibbs system on the transport polytope $\mathcal{G}(\mu, \nu)$, with the transport cost $\mathcal{U}$ as energy and the inverse temperature β controlling the concentration of the Gibbs measure around optimal transport plans. This formulation has two main advantages. First, it develops a competitive algorithm for solving the discrete Kantorovich problem by leveraging powerful asymptotic tools such as Laplace-type limits and zero-temperature arguments, as well as the analytical machinery developed in Chapter 5. Second, the statistical-physics viewpoint provides a structural interpretation of the problem, and yields some properties of the finite-temperature objective that are not immediate in the original constrained formulation, which will be analyzed in Chapter 7.

6.1 The Exact Finite-Temperature System

Definition 6.1 (Exact Gibbs measure and partition function). Let $(\mathcal{S}, \mathcal{F})$ be a measurable space, $\mu, \nu \in \mathcal{P}_f(\mathcal{S})$ two probability measures, $C : \mathcal{S} \times \mathcal{S} \rightarrow [0, \infty)$ a cost function and $\beta > 0$ an inverse-temperature parameter. We define the Gibbs measure at inverse temperature β as the probability measure

$$\mathbb{P}_{\mu, \nu}^{\beta} \in \mathcal{P}(\mathcal{G}(\mu, \nu))$$

that is absolutely continuous with respect to $\overline{\mathcal{H}}_{\mu\nu}^d$ with density

$$P_{\mu, \nu}^{\beta}(G) = \frac{1}{Z_{\beta}(\mu, \nu)} e^{-\beta \mathcal{U}(G)} \cdot \mathbf{1}_{\mathcal{G}(\mu, \nu)}(G),$$

where the *partition function* $Z_{\beta}(\mu, \nu)$ is a normalizing constant defined by

$$Z_{\beta}(\mu, \nu) := \int_{\mathcal{G}(\mu, \nu)} \exp(-\beta \mathcal{U}(G)) d\overline{\mathcal{H}}_{\mu\nu}^d,$$

with $\overline{\mathcal{H}}_{\mu\nu}^d$ d -dimensional normalized Hausdorff measure restricted on $\mathcal{G}(\mu, \nu)$ constructed in Section 4.2.

As pointed out in Remark 4.12, the choice of normalizing the Hausdorff measure on the transport polytope differs from the derivation in Chapter 5, and reflects the statistical-physics interpretation of the Gibbs density as a Radon-Nikodym derivative taken with respect to a reference probability measure $\overline{\mathcal{H}}_{\mu\nu}^d$. This choice affects finite-temperature quantities such as entropy and free energy, but not the main argument. Throughout this section, we will remark how proofs differ whether assuming or not that the measure has been normalized. We return to normalization issues in Chapter 7.

Definition 6.2 (Entropy). The entropy of the system is defined by

$$\begin{aligned} S_\beta(\mu, \nu) &:= - \int_{\mathcal{G}(\mu, \nu)} P_{\mu, \nu}^\beta(G) \log P_{\mu, \nu}^\beta(G) d\overline{\mathcal{H}}_{\mu\nu}^d \\ &= - \frac{1}{Z_\beta(\mu, \nu)} \int_{\mathcal{G}(\mu, \nu)} e^{-\beta \mathcal{U}(G)} \log \left(\frac{e^{-\beta \mathcal{U}(G)}}{Z_\beta(\mu, \nu)} \right) d\overline{\mathcal{H}}_{\mu\nu}^d. \end{aligned}$$

Definition 6.3 (Exact Free energy). The free energy of the system is defined by

$$\mathcal{F}_\beta(\mu, \nu) := -\frac{1}{\beta} \log Z_\beta(\mu, \nu).$$

Definition 6.4 (Average energy). The average energy of the system with respect to the Gibbs measure is defined by

$$E_\beta(\mu, \nu) := \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) d\mathbb{P}_{\mu, \nu}^\beta = \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) \frac{e^{-\beta \mathcal{U}(G)}}{Z_\beta(\mu, \nu)} d\overline{\mathcal{H}}_{\mu\nu}^d(G).$$

Since $\mathcal{U}(G) \geq 0$ and $\overline{\mathcal{H}}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)) = 1$, then $0 < e^{-\beta \mathcal{U}(G)} \leq 1$ and $0 < Z_\beta \leq 1$. In particular, $Z_\beta(\mu, \nu)$ is finite and strictly positive, so Definitions 6.1, 6.2, 6.3, and 6.4 are well posed.

We now state the main result of this section:

Theorem 6.5. *For sufficiently large $\beta > 0$, the free energy $\mathcal{F}_\beta(\mu, \nu)$ and the average energy $E_\beta(\mu, \nu)$ are monotonically decreasing functions of β . Moreover,*

$$\lim_{\beta \rightarrow +\infty} E_\beta(\mu, \nu) = \lim_{\beta \rightarrow +\infty} \mathcal{F}_\beta(\mu, \nu) = \text{OT}_C(\mu, \nu).$$

Remark 6.6. Theorem 6.5 is trivial if $\mathcal{U}$ is constant. Specifically, if $\mathcal{U} = a$ $\overline{\mathcal{H}}_{\mu\nu}^d$ -a.e. for some constant $a \geq 0$, then

$$Z_\beta(\mu, \nu) = e^{-\beta a}, \quad \mathcal{F}_\beta(\mu, \nu) = a, \quad E_\beta(\mu, \nu) = a, \quad S_\beta(\mu, \nu) = 0, \quad \text{OT}_C(\mu, \nu) = a.$$

If one used the non-normalized Hausdorff measure instead, then

$$Z_\beta(\mu, \nu) = e^{-\beta a} \mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)) \quad \text{and} \quad E_\beta(\mu, \nu) = a$$

and therefore

$$\mathcal{F}_\beta(\mu, \nu) = a - \frac{1}{\beta} \log \mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)),$$

so, again, $\mathcal{F}_\beta$ and E_β trivially converge to $a = \text{OT}_C(\mu, \nu)$. Notably, we observe that both S_β and $\mathcal{F}_\beta$ may be positive or negative, depending on the volume of the transport polytope and the normalization chosen. This creates issues to the next results, which might not hold under specific conditions. Therefore, henceforth we will always assume that $\mathcal{U}$ is nonconstant.

The following is an exhaustive characterization of constant transport costs:

Lemma 6.7 (Constant transport costs). *The transport cost $\mathcal{U}$ is constant on $\mathcal{G}(\mu, \nu)$ if and only if there exist $a \in \mathbb{R}^I$ and $b \in \mathbb{R}^J$ such that, for every $(i, j) \in [I] \times [J]$,*

$$C_{ij} = a_i + b_j.$$

Proof. Suppose that there exist $a \in \mathbb{R}^I$ and $b \in \mathbb{R}^J$ such that $C_{ij} = a_i + b_j$ for every $(i, j) \in [I] \times [J]$. Then, for every $G \in \mathcal{G}(\mu, \nu)$,

$$\mathcal{U}(G) = \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij} = \sum_{i=1}^I \sum_{j=1}^J (a_i + b_j) G_{ij} = \sum_{i=1}^I a_i \mu_i + \sum_{j=1}^J b_j \nu_j,$$

which is independent of G . Hence $\mathcal{U}$ is constant on $\mathcal{G}(\mu, \nu)$.

Conversely, let $Ax = b$ be the linear programming from of requiring $\sum_{i=1}^I G_{ij} = \nu_j$ and $\sum_{j=1}^J G_{ij} = \mu_i$, for every $(i, j) \in [I] \times [J]$. This has been already established in the proof of Proposition 1.5. A linear functional $\mathcal{U}$ is constant on $\mathcal{G}(\mu, \nu) = \{x \in \mathbb{R}_+^{IJ} : Ax = b\}$ if and only if it is constant on $\{x \in \mathbb{R}^{IJ} : Ax = b\}$, as detailed in [22, Section 13]. Let G^* be defined as $G_{ij}^* = \mu_i \nu_j$. As shown in Lemma 1.2, $G^* \in \mathcal{G}(\mu, \nu)$, hence we can write

$$\text{aff}(\mathcal{G}(\mu, \nu)) = \{x \in \mathbb{R}^{IJ} : Ax = b\} = G^* + \text{Ker}(A).$$

Therefore, $\mathcal{U}(G) = \langle C, G \rangle$ is constant on $G^* + \text{Ker}(A)$ if and only if $\langle C, G \rangle = 0$ on $\text{Ker}(A)$, which implies

$$C \in (\text{Ker}(A))^\perp = \text{Im}(A^T).$$

Since $C \in \text{Im}(A^T)$, there exists $(a, b) \in \mathbb{R}^I \times \mathbb{R}^J$ such that $C = A^T(a, b)$. By the definition of the matrix A , for every $G \in \mathbb{R}^{I \times J}$,

$$\langle AG, (a, b) \rangle = \sum_i a_i \sum_j G_{ij} + \sum_j b_j \sum_i G_{ij} = \sum_{i,j} (a_i + b_j) G_{ij}.$$

Hence

$$(A^T(a, b))_{ij} = a_i + b_j,$$

and therefore, for every $(i, j) \in [I] \times [J]$,

$$C_{ij} = a_i + b_j$$

□

6.2 Thermodynamic Identities

The quantities introduced so far are linked through differentiation with respect to β . We first show these differentiations are valid.

Lemma 6.8 (Smoothness of the partition function). *The partition function $Z_\beta(\mu, \nu)$ is C^∞ on $(0, \infty)$ as a function of β .*

Proof. Since $\mathcal{G}(\mu, \nu)$ is compact and $\mathcal{U}$ is continuous, there exists a constant $M < \infty$ such that $|\mathcal{U}(G)| \leq M$ on $\mathcal{G}(\mu, \nu)$. Furthermore, the map $\beta \mapsto e^{-\beta \mathcal{U}(G)}$ is C^∞ for every fixed G . Since $\mathcal{U} \geq 0$, we can uniformly bound the partial derivatives of Z_β :

$$|\partial_\beta^k e^{-\beta \mathcal{U}(G)}| = |(-\mathcal{U}(G))^k e^{-\beta \mathcal{U}(G)}| \leq |\mathcal{U}(G)|^k \leq M^k.$$

Since every derivative is uniformly bounded and $\overline{\mathcal{H}}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)) < \infty$, it follows that $\beta \mapsto Z_\beta$ belongs to $C^\infty(0, \infty)$, and

$$\partial_\beta^k Z_\beta = \int_{\mathcal{G}(\mu, \nu)} (-\mathcal{U}(G))^k e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d.$$

□

We now derive the standard thermodynamic identities relating Z_β , $\mathcal{F}_\beta$, E_β , and S_β .

Lemma 6.9 (Thermodynamic identities). *For every $\beta > 0$, the following relations*

hold:

$$\begin{aligned}
 E_\beta(\mu, \nu) &= -\frac{\partial \log(Z_\beta(\mu, \nu))}{\partial \beta} & S_\beta(\mu, \nu) &= -\frac{\partial \mathcal{F}_\beta(\mu, \nu)}{\partial T} \\
 E_\beta(\mu, \nu) &= \frac{\partial(\beta \mathcal{F}_\beta(\mu, \nu))}{\partial \beta} & E_\beta &= \mathcal{F}_\beta + T S_\beta \\
 S_\beta(\mu, \nu) &= \beta^2 \frac{\partial \mathcal{F}_\beta(\mu, \nu)}{\partial \beta} & \frac{\partial E_\beta}{\partial \beta} &= -\text{Var}_{\mathbb{P}_{\mu, \nu}^\beta}(\mathcal{U}),
 \end{aligned}$$

where $T := 1/\beta$.

Proof. In this proof, differentiation is justified by Lemma 6.8 combined with the chain rule.

A) Direct computation shows that

$$\begin{aligned}
 -\frac{\partial \log(Z_\beta(\mu, \nu))}{\partial \beta} &= -\frac{1}{Z_\beta} \frac{\partial Z_\beta}{\partial \beta} = -\frac{1}{Z_\beta} \int_{\mathcal{G}(\mu, \nu)} \frac{\partial \exp(-\beta \mathcal{U}(G))}{\partial \beta} d\overline{\mathcal{H}}_{\mu\nu}^d \\
 &= \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) \frac{1}{Z_\beta} \exp(-\beta \mathcal{U}(G)) d\overline{\mathcal{H}}_{\mu\nu}^d = \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) d\mathbb{P}_{\mu, \nu}^\beta \\
 &= E_\beta(\mu, \nu).
 \end{aligned}$$

B) Since $\beta \mathcal{F}_\beta(\mu, \nu) = -\log Z_\beta(\mu, \nu)$, this relation follows from the previous.

C) Direct computation shows that

$$\begin{aligned}
 \beta^2 \frac{\partial \mathcal{F}_\beta(\mu, \nu)}{\partial \beta} &= -\beta^2 \frac{\partial \left(\frac{1}{\beta} \log Z_\beta \right)}{\partial \beta} = \log Z_\beta - \beta \frac{\partial(\log Z_\beta)}{\partial \beta} = \log Z_\beta - \frac{\beta}{Z_\beta} \frac{\partial Z_\beta}{\partial \beta} \\
 &= \log Z_\beta - \frac{\beta}{Z_\beta} \int_{\mathcal{G}(\mu, \nu)} \frac{\partial \exp(-\beta \mathcal{U}(G))}{\partial \beta} d\overline{\mathcal{H}}_{\mu\nu}^d \\
 &= \log Z_\beta + \frac{\beta}{Z_\beta} \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d \\
 &= \log Z_\beta \underbrace{\int_{\mathcal{G}(\mu, \nu)} \frac{e^{-\beta \mathcal{U}(G)}}{Z_\beta} d\overline{\mathcal{H}}_{\mu\nu}^d}_{=1} + \int_{\mathcal{G}(\mu, \nu)} \frac{e^{-\beta \mathcal{U}(G)}}{Z_\beta} \beta \mathcal{U}(G) d\overline{\mathcal{H}}_{\mu\nu}^d \\
 &= \int_{\mathcal{G}(\mu, \nu)} \frac{e^{-\beta \mathcal{U}(G)}}{Z_\beta} \left[\log Z_\beta + \beta \mathcal{U}(G) \right] d\overline{\mathcal{H}}_{\mu\nu}^d \\
 &= - \int_{\mathcal{G}(\mu, \nu)} \frac{e^{-\beta \mathcal{U}(G)}}{Z_\beta} \log \left(\frac{e^{-\beta \mathcal{U}(G)}}{Z_\beta(\mu, \nu)} \right) d\overline{\mathcal{H}}_{\mu\nu}^d = S_\beta.
 \end{aligned}$$

D) By the chain rule,

$$-\frac{\partial \mathcal{F}_\beta}{\partial T} = -\frac{\partial \mathcal{F}_\beta}{\partial \beta} \frac{\partial \beta}{\partial T} = -\frac{\partial \mathcal{F}_\beta}{\partial \beta} \frac{\partial(1/T)}{\partial T} = \frac{1}{T^2} \frac{\partial \mathcal{F}_\beta}{\partial \beta} = \beta^2 \frac{\partial \mathcal{F}_\beta(\mu, \nu)}{\partial \beta} = S_\beta(\mu, \nu).$$

E) By the Leibniz rule,

$$E_\beta = \frac{\partial(\beta \mathcal{F}_\beta)}{\partial \beta} = \mathcal{F}_\beta + \beta \frac{\partial \mathcal{F}_\beta}{\partial \beta} = \mathcal{F}_\beta + \frac{1}{\beta} S_\beta = \mathcal{F}_\beta + T S_\beta.$$

F) Direct computation shows that

$$\begin{aligned} \frac{\partial E_\beta}{\partial \beta} &= \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) \frac{\partial}{\partial \beta} \left(\frac{1}{Z_\beta} e^{-\beta \mathcal{U}(G)} \right) d\overline{\mathcal{H}}_{\mu\nu}^d \\ &= \int_{\mathcal{G}(\mu, \nu)} \left[-\mathcal{U}^2(G) \frac{1}{Z_\beta} - \frac{\mathcal{U}(G)}{Z_\beta^2} \frac{\partial Z_\beta}{\partial \beta} \right] e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d \\ &= -\frac{1}{Z_\beta^2} \frac{\partial Z_\beta}{\partial \beta} \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d - \frac{1}{Z_\beta} \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G)^2 e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d \\ &= \frac{1}{Z_\beta^2} \left(\int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d \right)^2 - \frac{1}{Z_\beta} \int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G)^2 e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d \\ &= - \left[\int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G)^2 d\mathbb{P}_{\mu, \nu}^\beta - \left(\int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) d\mathbb{P}_{\mu, \nu}^\beta \right)^2 \right] \\ &= -(\mathbb{E}_{\mathbb{P}_{\mu, \nu}^\beta}(\mathcal{U}^2) - (\mathbb{E}_{\mathbb{P}_{\mu, \nu}^\beta}(\mathcal{U}))^2) = -\text{Var}_{\mathbb{P}_{\mu, \nu}^\beta}(\mathcal{U}) \end{aligned}$$

□

6.3 Zero-Temperature Limit of the Exact Free Energy

The proof of Theorem 6.5 relies on the fact that, for a sufficiently large β , the entropy S_β is negative. The next lemmas are instrumental for showing this fact.

Lemma 6.10. *If $\mathcal{U}$ is nonconstant $\overline{\mathcal{H}}_{\mu\nu}^d$ -a.e., then*

$$\overline{\mathcal{H}}_{\mu\nu}^d(\mathcal{U}^{-1}(t)) = 0 \quad \forall t \in \mathbb{R}.$$

Proof. Let $A = x_0 + V$ be the affine hull of $\mathcal{G}(\mu, \nu)$, and consider $\mathcal{U}_A(G) : A \rightarrow \mathbb{R}$ as the transport cost computed on A . By Theorem 4.4, $\dim_{\text{aff}}(A) = (I-1)(J-1)$. Since $\mathcal{U}_A$ is linear between vector spaces (up to a translation) of dimension $d = (I-1)(J-1)$ and 1, respectively, and it is not constant, then $\dim_{\text{aff}}(\mathcal{U}_A^{-1}(t)) = d-1$, by rank-nullity theorem, for every $t \in \mathbb{R}$, unless empty. As a consequence, $\dim_{\text{aff}}(\mathcal{U}_A^{-1}(t) \cap \mathcal{G}(\mu, \nu)) \leq d-1$. Since the affine dimension coincides with the Hausdorff dimension on an affine hull, then, for every $t \in \mathbb{R}$,

$$\overline{\mathcal{H}}_{\mu\nu}^d(\{G \in \mathcal{G}(\mu, \nu) : \mathcal{U}(G) = t\}) = \overline{\mathcal{H}}_{\mu\nu}^d(\mathcal{U}_A^{-1}(t) \cap \mathcal{G}(\mu, \nu)) = 0$$

□

Lemma 6.11. *Assume that $\mathcal{U}$ is not identically zero on $\mathcal{G}(\mu, \nu)$. Then*

$$\lim_{\beta \rightarrow \infty} Z_\beta = 0.$$

Proof. Suppose that $\mathcal{U}(G) = a$ $\overline{\mathcal{H}}_{\mu\nu}^d$ -a.e. for some constant a ; then $a > 0$ since $C_{ij} \geq 0$ and $G_{ij} \geq 0$ for every $(i, j) \in [I] \times [J]$. So $Z_\beta = e^{-\beta a} \overline{\mathcal{H}}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)) = e^{-\beta a}$, which implies $\lim_{\beta \rightarrow \infty} Z_\beta = 0$.

Now we assume $\mathcal{U}$ is not constant $\overline{\mathcal{H}}_{\mu\nu}^d$ -a.e. on $\mathcal{G}(\mu, \nu)$. Direct computation shows that:

$$\begin{aligned} \lim_{\beta \rightarrow \infty} \int_{\mathcal{G}(\mu, \nu)} e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d &= \int_{\mathcal{G}(\mu, \nu)} \left(\lim_{\beta \rightarrow \infty} e^{-\beta \mathcal{U}(G)} \right) d\overline{\mathcal{H}}_{\mu\nu}^d = \int_{G: \mathcal{U}(G)=0} d\overline{\mathcal{H}}_{\mu\nu}^d \\ &= \overline{\mathcal{H}}_{\mu\nu}^d(\{G \in \mathcal{G}(\mu, \nu) : \mathcal{U}(G) = 0\}), \end{aligned}$$

where the exchange of limit and integral is valid by the dominated convergence theorem (as $e^{-\beta \mathcal{U}(G)} \leq 1$). By Lemma 6.10, the last quantity is zero, concluding the proof. $\square$

The following technical result gives the asymptotic growth of the partition function. It follows trivially from Theorem C.2:

Lemma 6.12. *Let*

$$M := \arg \min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G), \quad m := \min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G),$$

and let

$$r := \dim_{\mathcal{H}}(M),$$

that is, the Hausdorff dimension of the set of minimizers of $\mathcal{U}$ over $\mathcal{G}(\mu, \nu)$. Assume that $\mathcal{U}$ is not constant on $\mathcal{G}(\mu, \nu)$. Then there exists a constant $\gamma \in (0, +\infty)$, depending on μ, ν and C , such that

$$Z_\beta(\mu, \nu) = \gamma e^{-\beta m} \beta^{-(d-r)} (1 + o(1))$$

and

$$\int_{\mathcal{G}(\mu, \nu)} (\mathcal{U}(G) - m) e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d(G) = \gamma(d-r) e^{-\beta m} \beta^{-(d-r+1)} (1 + o(1)).$$

The following lemma shows that S_β is eventually negative. It is necessary for making the proof of Theorem 6.5 independent from the normalization of the reference measure. If we select the normalized measure $\overline{\mathcal{H}}_{\mu\nu}^d$, then S_β is trivially negative, so there is no need of showing this lemma.

Lemma 6.13. *If $\mathcal{U}$ is nonconstant $\overline{\mathcal{H}}_{\mu\nu}^d$ -a.e., then*

$$\lim_{\beta \rightarrow \infty} S_\beta(\mu, \nu) = -\infty.$$

In particular, there exists $\beta_0 > 0$ such that $S_\beta < 0$ for all $\beta > \beta_0$.

Proof. By Relation **(E)** of Lemma 6.9,

$$E_\beta = \mathcal{F}_\beta + TS_\beta \implies S_\beta = \beta E_\beta - \beta \mathcal{F}_\beta = \beta E_\beta + \log Z_\beta$$

By Lemma 6.12,

$$Z_\beta(\mu, \nu) = \gamma e^{-\beta m} \beta^{-(d-r)} (1 + o(1))$$

Hence

$$\log Z_\beta(\mu, \nu) = \log \gamma + (r - d) \log \beta - \beta m + o(1).$$

Moreover, again by Lemma 6.12,

$$\int_{\mathcal{G}(\mu, \nu)} (\mathcal{U}(G) - m) e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d(G) = \gamma(d - r) e^{-\beta m} \beta^{-(d-r+1)} (1 + o(1)).$$

Dividing by $Z_\beta(\mu, \nu)$,

$$\begin{aligned} E_\beta(\mu, \nu) - m &= \frac{e^{-\beta m} (d - r) \gamma \beta^{-(d-r+1)} (1 + o(1))}{Z_\beta(\mu, \nu)} \\ &\stackrel{6.12}{=} \frac{e^{-\beta m} (d - r) \gamma \beta^{-(d-r+1)} (1 + o(1))}{e^{-\beta m} \gamma \beta^{-d+r} (1 + o(1))} \\ &= \frac{d - r}{\beta} + o(1/\beta). \end{aligned}$$

Hence

$$\begin{aligned} S_\beta(\mu, \nu) &= \beta E_\beta + \log Z_\beta = \beta \left(m + \frac{d - r}{\beta} + o(1/\beta) \right) + \log \gamma + (r - d) \log \beta - \beta m + o(1) \\ &= \beta m + d - r + \log \gamma + (r - d) \log \beta - \beta m + o(1) \\ &= \log \gamma + (r - d) \log \beta + d - r + o(1). \end{aligned}$$

Now observe that, by Lemma 6.10, $r < d$. Therefore,

$$\lim_{\beta \rightarrow \infty} S_\beta(\mu, \nu) = \lim_{\beta \rightarrow \infty} (r - d) \log \beta + [\log \gamma + d - r] = -\infty.$$

□

We now prove Theorem 6.5.

Proof. To show that $\mathcal{F}_\beta(\mu, \nu)$ is decreasing, recall that, by Lemma 6.13, S_β is eventually negative. Combining this observation with relation **C** of Lemma 6.9 gives

$$\frac{\partial \mathcal{F}_\beta(\mu, \nu)}{\partial \beta} = \frac{S_\beta}{\beta^2} \leq 0$$

for β large enough. Similarly, relation **F** of Lemma 6.9, yields

$$\frac{\partial E_\beta}{\partial \beta} = -\text{Var}_{\mathbb{P}_{\mu, \nu}}(\mathcal{U}(G)).$$

Since the variance is a non-negative quantity, the average energy E_β is monotonically decreasing in β . Since $E_\beta \geq 0$ by the monotonicity of the Lebesgue integral, we conclude that E_β admits a limit as $\beta \rightarrow \infty$, being bounded from below and monotonically decreasing.

Let us study now the zero-temperature limit of $\mathcal{F}_\beta$. It can be written as follows:

$$\lim_{\beta \rightarrow \infty} \mathcal{F}_\beta = \lim_{\beta \rightarrow \infty} \frac{\log Z_\beta}{-\beta} = \lim_{\beta \rightarrow \infty} \frac{\log \left[\int_{\mathcal{G}(\mu, \nu)} \exp(-\beta \mathcal{U}(G)) d\overline{\mathcal{H}}_{\mu\nu}^d \right]}{-\beta}.$$

By Lemma 6.11, the numerator diverges, producing an indeterminate form ∞/∞ . Hence, we can apply L'Hôpital's rule:

$$\begin{aligned} \lim_{\beta \rightarrow \infty} \frac{\log \left[\int_{\mathcal{G}(\mu, \nu)} \exp(-\beta \mathcal{U}(G)) d\overline{\mathcal{H}}_{\mu\nu}^d \right]}{-\beta} &= \lim_{\beta \rightarrow \infty} \frac{\frac{\partial}{\partial \beta} \left[\int_{\mathcal{G}(\mu, \nu)} \exp(-\beta \mathcal{U}(G)) d\overline{\mathcal{H}}_{\mu\nu}^d \right]}{-\int_{\mathcal{G}(\mu, \nu)} e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d} \\ &= \lim_{\beta \rightarrow \infty} \frac{\int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d}{\int_{\mathcal{G}(\mu, \nu)} e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d} \\ &= \lim_{\beta \rightarrow \infty} \frac{\int_{\mathcal{G}(\mu, \nu)} \mathcal{U}(G) e^{-\beta \mathcal{U}(G)} / Z_\beta d\overline{\mathcal{H}}_{\mu\nu}^d}{\int_{\mathcal{G}(\mu, \nu)} e^{-\beta \mathcal{U}(G)} / Z_\beta d\overline{\mathcal{H}}_{\mu\nu}^d} \\ &= \lim_{\beta \rightarrow \infty} \frac{E_\beta}{1} = \lim_{\beta \rightarrow \infty} E_\beta, \end{aligned}$$

concluding that E_β and $\mathcal{F}_\beta$ attain the same limit at zero temperature.

The differentiation under the integral sign used in the application of L'Hôpital's rule is justified by Lemma 6.8.

Finally, to compute $\lim_{\beta \rightarrow \infty} \mathcal{F}_\beta = \lim_{\beta \rightarrow \infty} E_\beta$, we observe that $\|e^{-\mathcal{U}}\|_\infty \leq 1$ and

$$\|\exp(-\mathcal{U})\|_1 = \int_{\mathcal{G}(\mu, \nu)} e^{-\mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d \leq \int_{\mathcal{G}(\mu, \nu)} d\overline{\mathcal{H}}_{\mu\nu}^d = 1.$$

Therefore, applying Lemma B.4 we deduce

$$\lim_{\beta \rightarrow \infty} \mathcal{F}_\beta = \lim_{\beta \rightarrow \infty} -\frac{1}{\beta} \log Z_\beta(\mu, \nu) = \operatorname{ess\,inf}_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G)$$

Since $\mathcal{U}$ is continuous and $\overline{\mathcal{H}}_{\mu\nu}^d$ has full support on $\mathcal{G}(\mu, \nu)$, the essential infimum coincides with the minimum. Hence

$$\operatorname{ess\,inf}_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G) = \min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G) = \operatorname{OT}_C(\mu, \nu).$$

Thus also $\lim_{\beta \rightarrow \infty} E_\beta = \lim_{\beta \rightarrow \infty} \mathcal{F}_\beta = \operatorname{OT}_C(\mu, \nu)$.

Of course, if we use the unnormalized measure $\mathcal{H}_{\mu\nu}^d$, the result still follows since $\mathcal{G}(\mu, \nu)$ is bounded, hence of finite volume. $\square$

6.4 Mean-Field Equations and Statistical Physics Interpretation

Theorem 6.5 motivates the statistical-physics interpretation. It states that the Kantorovich optimal transport cost $\operatorname{OT}_C(\mu, \nu)$ can be recovered as the zero-temperature limit of the free energy:

$$\operatorname{OT}_C(\mu, \nu) = \lim_{\beta \rightarrow \infty} \mathcal{F}_\beta(\mu, \nu) = \lim_{\beta \rightarrow \infty} -\frac{1}{\beta} \log Z_\beta(\mu, \nu).$$

Unfortunately, computing $\mathcal{F}_\beta$ directly is as hard as computing Z_β , since the latter involves a high-dimensional integral over $\mathcal{G}(\mu, \nu)$ which cannot be evaluated in closed form.

This is precisely why, in Chapter 5, we derived the following alternative representation of the partition function Z_β in terms of real Fourier coefficients $(\theta, \eta) \in \mathbb{R}^{I+J-1}$:

$$Z_\beta = \frac{J_{\mathbb{R}^I}}{(2\pi)^K} \lim_{n \rightarrow \infty} \int_{\mathbb{R}^I} \int_{\mathbb{R}^J} e^{-\frac{\|\theta\|^2 + \|\eta\|^2}{2n^2} - i\langle \theta, \mu \rangle + i\langle \eta, \nu \rangle} \prod_{i=1}^I \prod_{j=1}^J h(\beta C_{ij} + i\theta_i + i\eta_j) \, d\theta \, d\eta.$$

This led to the steepest-descent framework analyzed in Section 5.5, with the scope of justifying, under suitable assumptions, (SPA), namely

$$-\frac{1}{\beta} \log Z_\beta = F_\beta(u^\beta, v^\beta) + o(1),$$

where

$$F_\beta(u, v) = -\langle u, \mu \rangle - \langle v, \nu \rangle - \frac{1}{\beta} \sum_{i=1}^I \sum_{j=1}^J \log h(\beta(C_{ij} + u_i + v_j))$$

is the effective free energy already defined in Chapter 2, with the convention $v_J := 0$, and

$$(u^\beta, v^\beta) := \arg \max_{(u, v) \in \mathbb{R}^{I+J-1}} F_\beta(u, v).$$

By Theorem 6.5, the same limit converges to $\text{OT}_C(\mu, \nu)$. Hence, we recover the main identity of [12]:

$$\boxed{\text{OT}_C(\mu, \nu) = \lim_{\beta \rightarrow \infty} \mathcal{F}_\beta(\mu, \nu) = \lim_{\beta \rightarrow \infty} F_\beta(u^\beta, v^\beta).} \quad (6.1)$$

The fact that the optimal transport cost can be recovered as an optimization problem for F_β leads to the finite-temperature algorithm discussed in Chapter 2. We now summarize the logic that connects OT_C with the nonlinear system (2.1), which characterizes the temperature-annealing algorithm:

- i) By Theorem 6.5, the free energy $\mathcal{F}_\beta(\mu, \nu)$ monotonically converges to the optimal transport cost $\text{OT}_C(\mu, \nu)$;
- ii) By the discussion in Chapter 5, the free energy $\mathcal{F}_\beta$ can be approximated, in the large- β limit, by the maximum of an effective free energy F_β , giving a valid low-temperature approximation;
- iii) The effective free energy is a strictly concave function of the Kantorovich potentials. Hence, by Theorem 2.7, it admits a unique global maximizer, and its maximization yields the system (2.1), as shown in Theorem 2.8;
- iv) This system (2.1) gives, at inverse temperature β , admissible couplings $G^\beta \in \mathcal{G}(\mu, \nu)$;
- v) These couplings G^β converge, up to a subsequence, to a coupling G^∞ which, by Theorem 2.1, yields a transport cost

$$\mathcal{U}^\infty = \text{OT}_C(\mu, \nu).$$

This implies that, for the correctness of the algorithm, Theorems 2.1 and 2.8 are enough, and no analytic justification of the steepest-descent method is needed. This observation is crucial: it means that, sometimes, the most effective way of proving a theorem is to accept formal mean-field manipulations of physics and then prove the validity of the variational problem a posteriori through a different path. Indeed, the statistical-physics heuristics, even if not mathematically solid, suggest the correct form of the effective free energy functional F_β and, consequently, of the nonlinear function φ , which is the basis of our algorithm.

We remark that, even if the steepest-descent method is hard to fully justify mathematically, it can be understood as a low-temperature approximation scheme relating the exact constrained partition function to an effective soft-constrained partition function.

This line of reasoning allows us to give an even more convincing statistical-physics interpretation of the problem. It goes as follows.

We start from a system of IJ particles

$$G_{11}, G_{12}, \dots, G_{1J}, \dots, G_{IJ}.$$

In the initial formulation, which we call the hard-margin formulation, the configuration $G = (G_{ij})$ lies in the transport polytope $\mathcal{G}(\mu, \nu)$. This system has internal energy $\mathcal{U}(G) = \langle C, G \rangle$. The Lagrangian, or Hamiltonian, is the total energy of the system. In this case,

$$\mathcal{L}(G) = \mathcal{U}(G).$$

To this configuration, we associate a Gibbs measure at inverse temperature β , given by

$$d\mathbb{P}_\beta(G) = \frac{1}{Z_\beta(\mu, \nu)} e^{-\beta \mathcal{U}(G)} d\mathcal{H}_{\mu\nu}(G), \quad G \in \mathcal{G}(\mu, \nu),$$

where $d\mathcal{H}_{\mu\nu}$ denotes the natural Hausdorff measure on the affine hull of $\mathcal{G}(\mu, \nu)$. In this ensemble, the marginal constraints hold configuration by configuration.

The effective-field viewpoint consists in replacing this hard-constrained ensemble by a soft-constrained ensemble on the cube $[0, 1]^{IJ}$. The constraints are no longer imposed on each configuration. Instead, we introduce potential fields

$$(u, v) \in \mathbb{R}^I \times \mathbb{R}^{J-1}.$$

They define a corresponding potential energy, which penalizes or favors configurations according to their row and column masses. The corresponding Lagrangian is the sum of the internal energy $\mathcal{U}(G)$ and the potential energy:

$$\mathcal{L}(G, u, v) = \langle C, G \rangle + \sum_{i=1}^I u_i \left(\sum_{j=1}^J G_{ij} - \mu_i \right) + \sum_{j=1}^{J-1} v_j \left(\sum_{i=1}^I G_{ij} - \nu_j \right).$$

Equivalently, setting $v_J := 0$,

$$\mathcal{L}(G; u, v) = -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i=1}^I \sum_{j=1}^J (C_{ij} + u_i + v_j) G_{ij}.$$

Notice that, on the exact transport polytope, the constraint terms vanish and $\mathcal{L}(G; u, v) = \mathcal{U}(G)$. However, in the soft-constrained ensemble we allow $G \in [0, 1]^{IJ}$, and therefore the fields u_i, v_j affect the probability of configurations outside the exact constraint set.

For fixed fields u, v , we define the Gibbs measure

$$d\mathbb{P}_\beta^{u,v}(G) = \frac{1}{Z_\beta(u, v)} e^{-\beta \mathcal{L}(G; u, v)} dG, \quad G \in [0, 1]^{IJ},$$

where dG is the product Lebesgue measure on the hypercube. This is indeed a change of reference measure: the exact ensemble is supported on $\mathcal{G}(\mu, \nu)$ and uses $d\mathcal{H}_{\mu\nu}$, whereas the effective ensemble is defined on the full hypercube and uses dG . This change of measure is mathematically critical and requires the coarea formula and the corresponding Gaussian regularization.

Expanding the Lagrangian gives

$$e^{-\beta \mathcal{L}(G; u, v)} = e^{\beta \langle u, \mu \rangle + \beta \langle v, \nu \rangle} \prod_{i=1}^I \prod_{j=1}^J e^{-\beta (C_{ij} + u_i + v_j) G_{ij}}.$$

The prefactor is independent of G , and therefore cancels in the normalized law of G . The remaining factor is instead a product over the particles G_{ij} . Thus, in the effective ensemble, the total Gibbs measure decouples into one-particle measures. This means that, under the field (u, v) , or more precisely under the effective field

$$x_{ij} = C_{ij} + u_i + v_j,$$

for $(i, j) \in [I] \times [J]$, the particles G_{ij} become independent: their position is described by an independent random variable $g \in [0, 1]$ with local Gibbs measure

$$d\rho_{ij}^{u,v}(g) = \frac{e^{-\beta (C_{ij} + u_i + v_j)g}}{\int_0^1 e^{-\beta (C_{ij} + u_i + v_j)t} dt} dg.$$

The quantity $C_{ij} + u_i + v_j$ is the effective field seen by the particle G_{ij} . Its average position under this local measure is the first moment of $\rho_{ij}^{u,v}$:

$$G_{ij}^{\text{MF}} := \mathbb{E}_{\rho_{ij}^{u,v}}[g] = \frac{\int_0^1 g e^{-\beta (C_{ij} + u_i + v_j)g} dg}{\int_0^1 e^{-\beta (C_{ij} + u_i + v_j)g} dg}.$$

The mean-field equations are obtained by requiring that the original marginal constraints, instead of holding configuration by configuration, hold on average under the effective local measures. Namely, we require

$$\sum_{j=1}^J G_{ij}^{\text{MF}} := \sum_{j=1}^J \mathbb{E}_{\rho_{ij}^{u,v}}[g] = \mu_i, \quad i = 1, \dots, I,$$

and

$$\sum_{i=1}^I G_{ij}^{\text{MF}} := \sum_{i=1}^I \mathbb{E}_{\rho_{ij}^{u,v}}[g] = \nu_j, \quad j = 1, \dots, J-1.$$

By Lemma 2.2, $\varphi(y)$ is the expectation of $t \in [0, 1]$ under the probability density proportional to e^{-ty} . Hence this becomes

$$\sum_{j=1}^J \varphi(\beta(C_{ij} + u_i + v_j)) = \mu_i, \quad i = 1, \dots, I, \quad (6.2)$$

and

$$\sum_{i=1}^I \varphi(\beta(C_{ij} + u_i + v_j)) = \nu_j, \quad j = 1, \dots, J-1. \quad (6.3)$$

This is precisely the system (2.1). Together, these equations determine the fields that guarantee that mean occupations satisfy the original marginal constraints obtained in the soft-constrained system. Hence, they determine, *de-facto*, a *mean-field coupling* $G_\beta \in \mathcal{G}(\mu, \nu)$. By setting $\beta \rightarrow \infty$, the entropic contribution to the finite-temperature free energy becomes negligible with respect to the internal energy. Hence the finite-temperature variational problem formally collapses to the zero-temperature Kantorovich problem

$$\min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G).$$

We observe, however, that this argument does not rely on the notion of *effective free energy* $F_\beta(u, v)$, which is central in the mean-field derivation. Furthermore, it does not perform any saddle-point approximation: it imposes the mean-field equations directly, as self-consistency conditions on the one-particle Gibbs measures. One might therefore expect a different physical derivation of the same mean-field equations, closer in spirit to the steepest-descent method used by Koehl et al. [12]. In our view, no such physical derivation appears to be available, as we now explain.

We first observe that the partition function of the soft-margin system at fixed real fields (u, v) can be computed explicitly:

$$\begin{aligned} Z_\beta(u, v) &= \int_{[0,1]^{IJ}} e^{-\beta \mathcal{L}(G; u, v)} \, dG = e^{\beta \langle u, \mu \rangle + \beta \langle v, \nu \rangle} \int_{[0,1]^{IJ}} \prod_{i,j} e^{-\beta(C_{ij} + u_i + v_j)G_{ij}} \, dG \\ &= e^{\beta \langle u, \mu \rangle + \beta \langle v, \nu \rangle} \prod_{i,j} h(\beta(C_{ij} + u_i + v_j)). \end{aligned}$$

Taking the logarithm and dividing by $-\beta$, we obtain the free energy of the soft-margin system, which we call the *effective free energy*:

$$F_\beta(u, v) := -\frac{1}{\beta} \log Z_\beta(u, v) = -\langle u, \mu \rangle - \langle v, \nu \rangle + \sum_{i,j} f_\beta(x_{ij}), \quad f_\beta(x) := -\frac{1}{\beta} \log h(\beta x),$$

with $x_{ij} = C_{ij} + u_i + v_j$. A direct computation shows that the critical-point condition $\nabla F_\beta = 0$ reproduces *exactly* the mean-field system (2.1) obtained above. It is natural to ask what it means, physically, to maximize the effective free energy in order to obtain feasible optimal couplings for the hard-margin problem, and why this maximization yields $\text{OT}_C(\mu, \nu)$ at zero temperature. This remains unclear.

The route taken in the literature is to relate the hard-margin (canonical) partition function $Z_\beta(\mu, \nu)$ to the soft-margin (grand-canonical) one $Z_\beta(u, v)$ by representing each marginal constraint through its Fourier representation, which amounts to an identity of the type

$$Z_\beta(\mu, \nu) \text{ “} = \text{” } \int_{\mathbb{R}^{I+J-1}} Z_\beta(i\theta, i\eta) d\theta d\eta,$$

in which the dual variables are now imaginary fields. The structure of this identity is widespread in physics, from statistical mechanics to quantum mechanics [43, 44, 45, 46]. In our problem, however, the right-hand side is not well defined unless regularized as in Chapter 5. There, applying the steepest-descent method on an imaginary saddle shows that, under the analytic hypotheses isolated in Section 5.5, the regularized right-hand side is governed by the saddle at the maximizers (u^β, v^β) of F_β , where it is approximately equal to $e^{-\beta F_\beta(u^\beta, v^\beta)}$. This yields the approximation $\mathcal{F}_\beta(\mu, \nu) \approx F_\beta(u^\beta, v^\beta)$ and the corresponding justification of the method.

7 Metric Viewpoints

In this section we discuss the metric properties of the quantities introduced in the previous chapters, under the additional assumption that the cost matrix is induced by a metric on the underlying space. More precisely, we compare the classical optimal transport cost

$$\text{OT}_C(\mu, \nu) := \min_{G \in \mathcal{G}(\mu, \nu)} \mathcal{U}(G), \quad \mathcal{U}(G) := \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij},$$

with the finite-temperature quantities

$$Z_\beta(\mu, \nu), \quad \mathcal{F}_\beta(\mu, \nu), \quad \text{and} \quad \mathcal{U}_\beta(\mu, \nu).$$

Here $\mathcal{U}_\beta$ denotes the algorithmic finite-temperature transport cost defined in (2.3), namely

$$\mathcal{U}_\beta(\mu, \nu) := \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij}^\beta, \quad G_{ij}^\beta = \varphi(\beta x_{ij}^\beta),$$

The motivation for this discussion comes from Koehl et al. [12], where it is claimed that the statistical-physics formulation gives finite-temperature analogues of the metric properties of optimal transport. If correct, this would be a structural feature distinguishing their approach from the standard entropic regularization of Cuturi [21].

We first clarify the role of the normalization of the Hausdorff measure on the transport polytope. We then recall the classical metric properties of OT_C . Finally, we examine which of the corresponding claims remain valid for $\mathcal{F}_\beta$ and $\mathcal{U}_\beta$ at finite temperature.

7.1 On the Normalization of the Hausdorff Measure

The finite-temperature quantities depend on whether the reference measure on the transport polytope is normalized. As explained in Remark 4.12, we distinguish the intrinsic Hausdorff measure $\mathcal{H}_{\mu\nu}^d$ on $\mathcal{G}(\mu, \nu)$ with its normalized version $\overline{\mathcal{H}}_{\mu\nu}^d$, with total mass equal to 1.

Koehl et al. [12] do not explicitly normalize the Hausdorff measure. This breaks the interpretation of $P_{\mu,\nu}^\beta$ as a Radon-Nikodym derivative of $\mathbb{P}_{\mu,\nu}^\beta$ with respect to the reference probability measure $\overline{\mathcal{H}}_{\mu\nu}^d$. In particular, some probabilistic statements about entropy and free energy depend on the volume $\mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu))$,

In Appendix A of [12], the authors suggest that this volume is bounded by 1, because $\mathcal{G}(\mu, \nu) \subseteq [0, 1]^{I \times J}$ and the volume of the unit hypercube is 1. This argument is not valid as stated, as we cannot compare volumes living in different dimensions.

To show that their argument is false, it would suffice to exhibit a transport polytope satisfying

$$\mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)) > 1.$$

In this thesis, we do not address this problem, which remains open. Instead, we strengthened the argument by providing an alternative approach that bypasses the need for the bound $\mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)) \leq 1$ (cf. Remark 7.3). Specifically, we ensure the validity of the main results by proving that the entropy S_β is eventually negative (Theorem 6.13). Of course, we note that the inequality holds trivially if one chooses to work with the normalized measure $\overline{\mathcal{H}}_{\mu\nu}^d$.

In what follows we work with the normalized reference measure $\overline{\mathcal{H}}_{\mu\nu}^d$. We explicitly indicate how the statements change if the non-normalized measure $\mathcal{H}_{\mu\nu}^d$ is used instead.

Lemma 7.1. *For every $\beta > 0$,*

$$0 < Z_\beta(\mu, \nu) \leq 1,$$

with equality if and only if $C \equiv 0$.

Proof. Since $C_{ij} \geq 0$ and $G_{ij} \geq 0$, one has $\mathcal{U}(G) \geq 0$, hence

$$0 < e^{-\beta \mathcal{U}(G)} \leq 1.$$

Therefore, by monotonicity of the integral,

$$0 < Z_\beta(\mu, \nu) = \int_{\mathcal{G}(\mu, \nu)} e^{-\beta \mathcal{U}(G)} d\overline{\mathcal{H}}_{\mu\nu}^d(G) \leq \int_{\mathcal{G}(\mu, \nu)} d\overline{\mathcal{H}}_{\mu\nu}^d(G) = 1.$$

Equality holds if and only if $\mathcal{U}(G) \equiv 0$ $\overline{\mathcal{H}}_{\mu\nu}^d$ -a.e.

By Lemma 6.7, this implies that

$$C_{ij} = a_i + b_j \quad \forall (i, j) \in [I] \times [J].$$

Hence

$$\begin{aligned} \mathcal{U}(G) &= \sum_{i,j} C_{ij} G_{ij} = \sum_{i=1}^I \sum_{j=1}^J (a_i + b_j) G_{ij} = \sum_{i,j} a_i G_{ij} + \sum_{i,j} b_j G_{ij} \\ &= \sum_{i=1}^I a_i \mu_i + \sum_{j=1}^J b_j \nu_j = \sum_{i=1}^I a_i \mu_i \sum_{j=1}^J \nu_j + \sum_{j=1}^J b_j \nu_j \sum_{i=1}^I \mu_i \\ &= \sum_{i,j} (a_i + b_j) \mu_i \nu_j. \end{aligned}$$

Since $\mu_i, \nu_j > 0$ for every $(i, j) \in [I] \times [J]$, this implies that $a_i + b_j = 0$ for every $(i, j) \in [I] \times [J]$, hence $C \equiv 0$. $\square$

If the non-normalized Hausdorff measure is used instead, the same argument gives only

$$0 < Z_\beta(\mu, \nu) \leq \mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)).$$

Thus the sign of

$$\mathcal{F}_\beta(\mu, \nu) = -\frac{1}{\beta} \log Z_\beta(\mu, \nu)$$

may depend on the intrinsic volume of the transport polytope.

Lemma 7.2. *It holds*

$$S_\beta(\mu, \nu) \leq 0,$$

with equality if and only if $\mathcal{U}$ is constant $\overline{\mathcal{H}}_{\mu\nu}^d$ -a.e.

Proof. We recall the positive definiteness of the KL divergence between two probability measures $\mathbb{P}$ and $\mathbb{Q}$ defined on some measurable space $(X, \mathcal{F})$. If $\mathbb{P}$ and $\mathbb{Q}$ are absolutely continuous measures with respect to a reference measure on X , with densities p and q , their KL divergence is defined by

$$KL(\mathbb{P} \parallel \mathbb{Q}) = \int_X \log(p/q) d\mathbb{P} = \mathbb{E}_\mathbb{P} \left[\log \frac{p(X)}{q(X)} \right].$$

This quantity is nonnegative, due to the Jensen's inequality applied to the convex function $x \mapsto -\log x$:

$$\begin{aligned} KL(\mathbb{P} \parallel \mathbb{Q}) &= \int_X \log \frac{p(x)}{q(x)} p(x) dx = \mathbb{E}_\mathbb{P} \left[\log \frac{p(X)}{q(X)} \right] = \mathbb{E}_\mathbb{P} \left[-\log \frac{q(X)}{p(X)} \right] \\ &\geq -\log \left(\mathbb{E}_\mathbb{P} \left[\frac{q(X)}{p(X)} \right] \right) = -\log \left(\int_X (q/p) d\mathbb{P} \right) = -\log \int_X d\mathbb{Q} = -\log 1 = 0. \end{aligned}$$

Moreover, equality holds if and only if $\mathbb{P} = \mathbb{Q}$.

We apply this fact to the probability measures $\mathbb{P}_{\mu, \nu}^\beta$ and $\mathbb{U}$ on $\mathcal{G}(\mu, \nu)$, where $\mathbb{U}$ is uniform. Hence, its density is 1. This yields

$$\begin{aligned} 0 \leq KL(\mathbb{P}_{\mu, \nu}^\beta \parallel \mathbb{U}) &= \int_{\mathcal{G}(\mu, \nu)} P_{\mu, \nu}^\beta(G) (\log P_{\mu, \nu}^\beta(G) - \log 1) d\overline{\mathcal{H}}_{\mu\nu}^d(G) \\ &= \int_{\mathcal{G}(\mu, \nu)} P_{\mu, \nu}^\beta(G) \log P_{\mu, \nu}^\beta(G) d\overline{\mathcal{H}}_{\mu\nu}^d(G) = -S_\beta(\mu, \nu). \end{aligned}$$

Therefore,

$$S_\beta(\mu, \nu) \leq 0,$$

with equality if and only if $\mathbb{P}_{\mu,\nu}^\beta(G)$ is uniform, that is, $\mathcal{U}$ is constant $\overline{\mathcal{H}}_{\mu\nu}^d$ -almost everywhere on $\mathcal{G}(\mu, \nu)$. $\square$

Without normalization, the same computation gives

$$S_\beta(\mu, \nu) \leq \log \mathcal{H}_{\mu\nu}^d(\mathcal{G}(\mu, \nu)).$$

Thus the sign of the entropy is not controlled unless one has an independent estimate on the intrinsic volume of the transport polytope.

Remark 7.3. Under normalization, Lemma 7.2 simplifies the proof of Theorem 6.5. Indeed, from the thermodynamic identity

$$E_\beta = \mathcal{F}_\beta + \frac{1}{\beta} S_\beta,$$

one immediately obtains

$$E_\beta \leq \mathcal{F}_\beta.$$

However, in the proof of Theorem 6.5 we wanted an argument independent of the volume of $\mathcal{G}(\mu, \nu)$. For this reason, the proof was formulated using Lemma 6.13, which controls the entropy contribution without relying on the claim that the volume of $\mathcal{G}(\mu, \nu)$ is bounded by 1.

7.2 Metric Properties of the Wasserstein Distance

Before discussing finite-temperature quantities, we recall the standard metric property of optimal transport under a metric cost (see, e.g., [18, Chapter 6], [19, Lemma 5.4]).

We follow the discrete proof of [20, Prop 2.2.] because it is consistent with our setting. However, we recall the canonical description of measures as measures of distributed mass on a measurable space because it gives structure to our framework.

Let

$$X = \{x_1, \dots, x_I\}, \quad Y = \{y_1, \dots, y_J\}, \quad Z = \{z_1, \dots, z_L\}$$

be finite subsets of a metric space $(\mathcal{S}, d)$. Let

$$\mu = \sum_{i=1}^I \mu_i \delta_{x_i}, \quad \nu = \sum_{j=1}^J \nu_j \delta_{y_j}, \quad \rho = \sum_{\ell=1}^L \rho_\ell \delta_{z_\ell}$$

be finitely supported probability measures. As usual, we identify them with the corresponding vectors $\mu \in \mathbb{R}_+^I$, $\nu \in \mathbb{R}_+^J$, $\rho \in \mathbb{R}_+^L$, with strictly positive entries.

Furthermore, define *metric* cost matrices

$$C_{ij}^{XY} := d(x_i, y_j), \quad C_{j\ell}^{YZ} := d(y_j, z_\ell), \quad C_{i\ell}^{XZ} := d(x_i, z_\ell).$$

In particular, the metric structure gives

$$C_{ij}^{XY} \geq 0, \quad C_{ij}^{XY} = 0 \iff x_i = y_j,$$

and similarly for the other cost matrices. Moreover,

$$C_{ij}^{XY} = C_{ji}^{YX}$$

Finally, for every $i \in [I]$, $j \in [J]$, and $\ell \in [L]$, the triangle inequality gives

$$C_{i\ell}^{XZ} \leq C_{ij}^{XY} + C_{j\ell}^{YZ}.$$

Following [20], we start with the following lemma:

Lemma 7.4 (Gluing). *Let μ, ν, ρ be defined as above. Let*

$$G_{12} \in \mathcal{G}(\mu, \nu), \quad G_{23} \in \mathcal{G}(\nu, \rho).$$

Define $G_{13} \in \mathbb{R}_+^{I \times L}$ by

$$(G_{13})_{i\ell} := \sum_{j=1}^J \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j}.$$

Then

$$G_{13} \in \mathcal{G}(\mu, \rho).$$

Proof. Since $G_{12} \geq 0$, $G_{23} \geq 0$, and $\nu_j > 0$ for every j , we have

$$(G_{13})_{i\ell} \geq 0 \quad \forall (i, \ell) \in [I] \times [L].$$

Moreover, for every $i \in [I]$,

$$\begin{aligned} \sum_{\ell=1}^L (G_{13})_{i\ell} &= \sum_{\ell=1}^L \sum_{j=1}^J \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j} = \sum_{j=1}^J \frac{(G_{12})_{ij}}{\nu_j} \sum_{\ell=1}^L (G_{23})_{j\ell} \\ &= \sum_{j=1}^J \frac{(G_{12})_{ij}}{\nu_j} \nu_j = \sum_{j=1}^J (G_{12})_{ij} = \mu_i. \end{aligned}$$

Similarly, for every $\ell \in [L]$,

$$\begin{aligned} \sum_{i=1}^I (G_{13})_{i\ell} &= \sum_{i=1}^I \sum_{j=1}^J \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j} = \sum_{j=1}^J \frac{(G_{23})_{j\ell}}{\nu_j} \sum_{i=1}^I (G_{12})_{ij} \\ &= \sum_{j=1}^J \frac{(G_{23})_{j\ell}}{\nu_j} \nu_j = \sum_{j=1}^J (G_{23})_{j\ell} = \rho_\ell. \end{aligned}$$

□

Lemma 7.5. *Given $G_{12} \in \mathcal{G}(\mu, \nu)$, $G_{23} \in \mathcal{G}(\nu, \rho)$ and $G_{13} \in \mathcal{G}(\mu, \rho)$ as in Lemma 7.4. Then*

$$\mathcal{U}(G_{13}) \leq \mathcal{U}(G_{12}) + \mathcal{U}(G_{23}),$$

where transport costs are defined from cost matrices C^{XZ} , C^{XY} and C^{YZ} .

Proof. By definition of G_{13} ,

$$\begin{aligned} \sum_{i,\ell} C_{i\ell}^{XZ}(G_{13})_{i\ell} &= \sum_{i=1}^I \sum_{\ell=1}^L C_{i\ell}^{XZ} \sum_{j=1}^J \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j} \\ &= \sum_{i=1}^I \sum_{j=1}^J \sum_{\ell=1}^L C_{i\ell}^{XZ} \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j}. \end{aligned}$$

Since d is a metric,

$$C_{i\ell}^{XZ} = d(x_i, z_\ell) \leq d(x_i, y_j) + d(y_j, z_\ell) = C_{ij}^{XY} + C_{j\ell}^{YZ}.$$

Therefore

$$\begin{aligned} \sum_{i,\ell} C_{i\ell}^{XZ}(G_{13})_{i\ell} &= \sum_{i,j,\ell} C_{i\ell}^{XZ} \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j} \\ &\leq \sum_{i,j,\ell} C_{ij}^{XY} \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j} + \sum_{i,j,\ell} C_{j\ell}^{YZ} \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j} \\ &= \sum_{i,j} C_{ij}^{XY} (G_{12})_{ij} \frac{\sum_{\ell=1}^L (G_{23})_{j\ell}}{\nu_j} + \sum_{j,\ell} C_{j\ell}^{YZ} (G_{23})_{j\ell} \frac{\sum_{i=1}^I (G_{12})_{ij}}{\nu_j} \end{aligned}$$

Since G_{23} on Z and G_{12} on X marginalize to ν_j , we conclude

$$\sum_{i,\ell} C_{i\ell}^{XZ}(G_{13})_{i\ell} \leq \sum_{i,j} C_{ij}^{XY} (G_{12})_{ij} + \sum_{j,\ell} C_{j\ell}^{YZ} (G_{23})_{j\ell}.$$

□

Theorem 7.6 (Metric induced by discrete optimal transport). *Let C be induced by a metric d on $\mathcal{S}$. Then*

$$\text{OT}_C : \mathcal{P}_f(\mathcal{S}) \times \mathcal{P}_f(\mathcal{S}) \rightarrow [0, +\infty)$$

is a metric.

Proof. We prove the three metric properties.

Positive definiteness. Since $C_{ij} \geq 0$ and $G_{ij} \geq 0$, we have $\mathcal{U}(G) \geq 0$ for every $G \in \mathcal{G}(\mu, \nu)$. In particular,

$$\text{OT}_C(\mu, \nu) \geq 0.$$

If $\mu = \nu$, write

$$\mu = \sum_{i=1}^I \mu_i \delta_{x_i}.$$

The diagonal coupling

$$G_{ij}^\Delta := \begin{cases} \mu_i, & i = j, \\ 0, & i \neq j, \end{cases}$$

belongs to $\mathcal{G}(\mu, \mu)$, and satisfies

$$U(G^\Delta) = \sum_{i=1}^I C_{ii} \mu_i = 0.$$

Therefore

$$\text{OT}_C(\mu, \mu) = 0.$$

Conversely, assume

$$\text{OT}_C(\mu, \nu) = 0.$$

Let $G \in \mathcal{G}(\mu, \nu)$ be an optimal coupling. Then

$$\sum_{i,j} C_{ij} G_{ij} = 0.$$

Since all terms are nonnegative, we have $C_{ij} = 0$ whenever $G_{ij} > 0$. Since C is metric, this implies $x_i = y_j$ whenever $G_{ij} > 0$. Let $z \in \mathcal{S}$. Then

$$\mu(\{z\}) = \sum_{i:x_i=z} \sum_j G_{ij}.$$

But $G_{ij} = 0$ unless $x_i = y_j$. Therefore

$$\mu(\{z\}) = \sum_{i:x_i=z} \sum_{j:y_j=z} G_{ij} = \sum_{j:y_j=z} \sum_i G_{ij} = \nu(\{z\}).$$

Since this holds for every $z \in \mathcal{S}$, we conclude that

$$\mu = \nu.$$

Symmetry. For every $G \in \mathcal{G}(\mu, \nu)$, the transposed $G^T \in \mathbb{R}^{J \times I}$ belongs to $\mathcal{G}(\nu, \mu)$. Since d is symmetric, we get

$$\mathcal{U}(G^T) = \sum_{j=1}^J \sum_{i=1}^I d(y_j, x_i) (G^T)_{ji} = \sum_{i=1}^I \sum_{j=1}^J d(x_i, y_j) G_{ij} = \mathcal{U}(G)$$

Let $G^* \in \mathcal{G}(\mu, \nu)$ be optimal. Then

$$\text{OT}_C(\mu, \nu) \leq \mathcal{U}((G^*)^T) = \mathcal{U}(G^*) = \text{OT}_C(\mu, \nu).$$

Conversely, if $H^* \in \mathcal{G}(\nu, \mu)$ is optimal,

$$\text{OT}_C(\nu, \mu) \leq \mathcal{U}((H^*)^T) = \mathcal{U}(H^*) = \text{OT}_C(\mu, \nu),$$

Concluding the proof.

Triangle inequality. Let $G_{12}^* \in \mathcal{G}(\mu, \nu)$ and $G_{23}^* \in \mathcal{G}(\nu, \rho)$ be optimal couplings. By Lemma 7.4, define the glued coupling

$$(G_{13})_{i\ell} := \sum_{j=1}^J \frac{(G_{12}^*)_{ij} (G_{23}^*)_{j\ell}}{\nu_j}.$$

Then $G_{13} \in \mathcal{G}(\mu, \rho)$. Therefore, using Lemma 7.5,

$$\begin{aligned} \text{OT}_C(\mu, \rho) &\leq \sum_{i,\ell} C_{i\ell}^{XZ} (G_{13})_{i\ell} \\ &\leq \sum_{i,j} C_{ij}^{XY} (G_{12}^*)_{ij} + \sum_{j,\ell} C_{j\ell}^{YZ} (G_{23}^*)_{j\ell} \\ &= \text{OT}_C(\mu, \nu) + \text{OT}_C(\nu, \rho). \end{aligned}$$

This proves the triangle inequality. $\square$

7.3 Metric Properties of the Free Energy

We now examine the free energy functional $\mathcal{F}_\beta(\mu, \nu)$. Koehl et al. [12] claim that, when C is induced by a metric, the free energy satisfies the metric properties of symmetry and triangle-inequality, even at finite-temperature. This would be a remarkable difference with respect to entropic optimal transport, where the regularized cost does not define a metric.

We analyze the three metric properties separately. Symmetry holds. Positivity is satisfied under normalization, but the identity of indiscernibles fails. Finally, the proof for the triangle inequality provided by [12] is invalid, and the property itself is false, as demonstrated by an explicit counterexample.

7.3.1 Symmetry of the Free Energy

Under the assumption that C is induced by a metric, the symmetry of the free energy follows by direct computation as in [12]:

$$\begin{aligned} Z_\beta(\nu, \mu) &= \int_{\mathcal{G}(\nu, \mu)} \exp\left(-\beta \sum_{j=1}^J \sum_{i=1}^I d(y_j, x_i) H_{ji}\right) d\overline{\mathcal{H}}_{\nu\mu}^d(H) \\ &\stackrel{\dagger}{=} \int_{\mathcal{G}(\mu, \nu)} \exp\left(-\beta \sum_{i=1}^I \sum_{j=1}^J d(x_i, y_j) G_{ij}\right) d\overline{\mathcal{H}}_{\mu\nu}^d(G) \\ &= Z_\beta(\mu, \nu). \end{aligned}$$

Step $\dagger$ is justified by observing that $d(x_i, y_j) = d(y_j, x_i)$, and that the map

$$T : \mathcal{G}(\mu, \nu) \rightarrow \mathcal{G}(\nu, \mu), \quad T(G) = G^T,$$

is an isometry with respect to the Frobenius metric. Hence, it preserves the corresponding Hausdorff measures,

$$T_\# \overline{\mathcal{H}}_{\mu\nu}^d = \overline{\mathcal{H}}_{\nu\mu}^d,$$

both in the normalized and in the non-normalized convention, as observed in (4.1). Therefore

$$\mathcal{F}_\beta(\mu, \nu) = \mathcal{F}_\beta(\nu, \mu).$$

7.3.2 Positivity

Under the normalized convention, Lemma 7.1 gives

$$0 < Z_\beta(\mu, \nu) \leq 1.$$

Therefore

$$\mathcal{F}_\beta(\mu, \nu) = -\frac{1}{\beta} \log Z_\beta(\mu, \nu) \geq 0,$$

with equality if and only if $C \equiv 0$. If μ and ν have supports $\{x_1, \dots, x_I\}$ and $\{y_1, \dots, y_J\}$, respectively, then $C(x_1, y_j) = 0$ if and only if $x_1 = y_j$, for every $j \in [J]$. Similarly, $C(x_i, y_1) = 0$ if and only if $x_i = y_1$ for every $i \in [I]$. We conclude that $\mathcal{F}_\beta(\mu, \nu) = 0$ if and only if $\mu = \nu = \delta_x$ for some $x \in \mathcal{S}$. This implies that only one direction is verified. We can see it also by looking at this counterexample:

Example 7.7. Let

$$\mathcal{S} = \{x_1, x_2\}, \quad \mu = \frac{1}{2}\delta_{x_1} + \frac{1}{2}\delta_{x_2},$$

and let C be the discrete metric:

$$C_{11} = C_{22} = 0, \quad C_{12} = C_{21} = 1.$$

The transport polytope is given by

$$\mathcal{G}(\mu, \mu) = \left\{ \begin{pmatrix} t & \frac{1}{2} - t \\ \frac{1}{2} - t & t \end{pmatrix} \middle| t \in [0, 1/2] \right\}.$$

Its volume is $|J| \cdot \mathcal{L}^1([0, 1/2]) = \frac{1}{2}|J|$, where $|J|$ is the Jacobian factor, which we don't need to compute since it is simplified during the normalization. The cost is

$$\mathcal{U}(G(t)) = C_{12}G_{12} + C_{21}G_{21} = \left(\frac{1}{2} - t\right) + \left(\frac{1}{2} - t\right) = 1 - 2t.$$

Then

$$Z_\beta(\mu, \mu) = \frac{1}{2 \cdot \overline{\mathcal{H}}_{\mu\nu}^d(\mathcal{G}(\mu, \mu))} \int_0^{1/2} e^{-\beta(1-2t)} dt = 2 \frac{1 - e^{-2\beta \cdot \frac{1}{2}}}{2\beta} = \frac{1 - e^{-\beta}}{\beta} < 1.$$

If the Hausdorff measure is not normalized, even non-negativity may fail, because

$$Z_\beta(\mu, \nu) \leq \overline{\mathcal{H}}_{\mu\nu}^d(\mathcal{G}(\mu, \nu))$$

does not imply, in general, $Z_\beta(\mu, \nu) \leq 1$.

7.3.3 Triangle Inequality

We now discuss the triangle inequality. Following [12], the triangle inequality

$$\mathcal{F}_\beta(\mu, \rho) \leq \mathcal{F}_\beta(\mu, \nu) + \mathcal{F}_\beta(\nu, \rho).$$

is equivalent to

$$\log Z_\beta(\mu, \nu) + \log Z_\beta(\nu, \rho) \leq \log Z_\beta(\mu, \rho)$$

and, eventually,

$$Z_\beta(\mu, \nu)Z_\beta(\nu, \rho) \leq Z_\beta(\mu, \rho).$$

Gluing Map Argument. The authors suggest to consider a gluing map $\Phi : \mathcal{G}(\mu, \nu) \times \mathcal{G}(\nu, \rho) \rightarrow \mathcal{G}(\mu, \rho)$ such that

$$\Phi(G_{12}, G_{23}) = G_{13},$$

where the glued plan G_{13} is defined by

$$(G_{13})_{i\ell} = \sum_{j=1}^J \frac{(G_{12})_{ij}(G_{23})_{j\ell}}{\nu_j},$$

and belongs to $\mathcal{G}(\mu, \rho)$ by Lemma 7.4. Moreover, by Lemma 7.5,

$$\mathcal{U}_{\mu\rho}(G_{13}) \leq \mathcal{U}_{\mu\nu}(G_{12}) + \mathcal{U}_{\nu\rho}(G_{23}),$$

where

$$\mathcal{U}_{\mu\nu}(G_{12}) := \sum_{i=1}^I \sum_{j=1}^J C_{ij}^{XY}(G_{12})_{ij},$$

and analogously for $\mathcal{U}_{\nu\rho}$ and $\mathcal{U}_{\mu\rho}$. Passing to the exponents on both sides, we deduce, for every $\beta > 0$,

$$e^{-\beta \mathcal{U}_{\mu\rho}(\Phi(G_{12}, G_{23}))} \geq e^{-\beta \mathcal{U}_{\mu\nu}(G_{12})} e^{-\beta \mathcal{U}_{\nu\rho}(G_{23})}.$$

Integrating with respect to the product reference measure $\overline{\mathcal{H}}_{\mu\nu}^{d_{\mu\nu}} \otimes \overline{\mathcal{H}}_{\nu\rho}^{d_{\nu\rho}}$ yields

$$\begin{aligned} & \int_{\mathcal{G}(\mu, \nu) \times \mathcal{G}(\nu, \rho)} e^{-\beta \mathcal{U}_{\mu\rho}(\Phi(G_{12}, G_{23}))} d(\overline{\mathcal{H}}_{\mu\nu}^{d_{\mu\nu}} \otimes \overline{\mathcal{H}}_{\nu\rho}^{d_{\nu\rho}}) \\ & \geq \int_{\mathcal{G}(\mu, \nu) \times \mathcal{G}(\nu, \rho)} e^{-\beta \mathcal{U}_{\mu\nu}(G_{12})} e^{-\beta \mathcal{U}_{\nu\rho}(G_{23})} d(\overline{\mathcal{H}}_{\mu\nu}^{d_{\mu\nu}} \otimes \overline{\mathcal{H}}_{\nu\rho}^{d_{\nu\rho}}) \\ & = \left(\int_{\mathcal{G}(\mu, \nu)} e^{-\beta \mathcal{U}_{\mu\nu}(G_{12})} d\overline{\mathcal{H}}_{\mu\nu}^{d_{\mu\nu}} \right) \left(\int_{\mathcal{G}(\nu, \rho)} e^{-\beta \mathcal{U}_{\nu\rho}(G_{23})} d\overline{\mathcal{H}}_{\nu\rho}^{d_{\nu\rho}} \right) \\ & = Z_\beta(\mu, \nu) Z_\beta(\nu, \rho). \end{aligned}$$

Define the gluing pushforward measure on $\mathcal{G}(\mu, \rho)$ by

$$\overline{\mathcal{H}}_{\mu\rho}^g := \Phi_\#(\overline{\mathcal{H}}_{\mu\nu}^{d_{\mu\nu}} \otimes \overline{\mathcal{H}}_{\nu\rho}^{d_{\nu\rho}}).$$

Then

$$Z_\beta(\mu, \nu) Z_\beta(\nu, \rho) \leq \int_{\mathcal{G}(\mu, \rho)} e^{-\beta \mathcal{U}_{\mu\rho}(G)} d\overline{\mathcal{H}}_{\mu\rho}^g(G).$$

A natural hope would be to establish a dominance relation of the form

$$\overline{\mathcal{H}}_{\mu\rho}^g(E) \leq \overline{\mathcal{H}}_{\mu\rho}^{d_{\mu\rho}}(E),$$

for all Borel sets $E \subseteq \mathcal{G}(\mu, \rho)$.

However, the following counterexample shows that this dominance is not valid in general:

Example 7.8 (Failure of measure domination). Let

$$\mu = \frac{1}{2}\delta_{x_1} + \frac{1}{2}\delta_{x_2}, \quad \nu = b\delta_{y_1} + (1-b)\delta_{y_2}, \quad \rho = \frac{1}{2}\delta_{z_1} + \frac{1}{2}\delta_{z_2}, \quad b \in [1/2, 1).$$

The transport polytopes $\mathcal{G}(\mu, \nu)$, $\mathcal{G}(\nu, \rho)$ and $\mathcal{G}(\mu, \rho)$ are one-dimensional. They can be parametrized as

$$G_{12}(t) = \begin{pmatrix} t & \frac{1}{2} - t \\ b - t & t + \frac{1}{2} - b \end{pmatrix}, \quad t \in \left[b - \frac{1}{2}, \frac{1}{2} \right],$$

$$G_{23}(s) = \begin{pmatrix} s & b - s \\ \frac{1}{2} - s & \frac{1}{2} - b + s \end{pmatrix}, \quad s \in \left[b - \frac{1}{2}, \frac{1}{2} \right],$$

and

$$G_{13}(u) = \begin{pmatrix} u & \frac{1}{2} - u \\ \frac{1}{2} - u & u \end{pmatrix}, \quad u \in \left[0, \frac{1}{2} \right].$$

The gluing map $\Phi(G_{12}, G_{23})$ yields a transport plan G_{13} whose top-left entry g given by

$$u(t, s) = \frac{(G_{12})_{11}(G_{23})_{11}}{\nu_1} + \frac{(G_{12})_{12}(G_{23})_{21}}{\nu_2} = \frac{ts}{b} + \frac{(1/2 - t)(1/2 - s)}{1 - b}.$$

To find a counterexample, it would suffice to find $b \in [1/2, 1)$ such that

$$\overline{\mathcal{H}}_{\mu\rho}^1(\text{Im}(\Phi)) \leq \overline{\mathcal{H}}_{\mu\rho}^g(\text{Im}(\Phi)).$$

Let us study the image of the gluing map. We already know that the gluing map has image in $\mathcal{G}(\mu, \rho)$. Hence, it suffices to analyse the image of $u(t, s)$ in $\mathbb{R}$. Because $u(t, s)$ is continuous on the square $[b - 1/2, 1/2]^2$, its image is the interval

$$\text{Im}(u) = [\min u(t, s), \max u(t, s)].$$

Taking partial derivatives we find that critical points must satisfy

$$\begin{cases} \frac{s}{b} - \frac{1/2 - s}{1 - b} = 0 \\ \frac{t}{b} - \frac{1/2 - t}{1 - b} = 0 \end{cases} \implies (t, s) = \left(\frac{b}{2}, \frac{b}{2} \right).$$

If $(t, s) = (b/2, b/2)$, then $u(t, s) = 1/4$. By the Fermat's theorem, optimal points are either $(b/2, b/2)$ or on the boundary $\partial[b - 1/2, 1/2]^2$. On the boundary, u is affine in each variable separately, so the extrema are attained at the vertices. Direct computation gives

$$u\left(b - \frac{1}{2}, b - \frac{1}{2}\right) = u\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{4b},$$

and

$$u\left(b - \frac{1}{2}, \frac{1}{2}\right) = u\left(\frac{1}{2}, b - \frac{1}{2}\right) = \frac{2b - 1}{4b}.$$

One verifies easily that these two values are indeed minimizer and maximizer of u over $[b - 1/2, 1/2]^2$ by comparing them with $u(b/2, b/2) = 1/4$. Thus

$$\overline{\mathcal{H}}_{\mu\rho}^1(\text{Im}(\Phi)) = \frac{\text{Length}(\text{Im}(u))}{\text{Length}(\mathcal{G}(\mu, \rho))} = \frac{\mathcal{L}\left(\left[\frac{2b-1}{4b}, \frac{1}{4b}\right]\right)}{\mathcal{L}\left([0, \frac{1}{2}]\right)} = \frac{\frac{1-b}{2b}}{\frac{1}{2}} = \frac{1-b}{b}.$$

On the other hand, by definition of the pushforward measure,

$$\overline{\mathcal{H}}_{\mu\rho}^g(\text{Im}(\Phi)) = \Phi_{\#}(\overline{\mathcal{H}}_{\mu\nu}^{d_{\mu\nu}} \otimes \overline{\mathcal{H}}_{\nu\rho}^{d_{\nu\rho}})(\text{Im}(\Phi)) = 1.$$

Hence measure domination would imply

$$1 \leq \frac{1-b}{b}.$$

This inequality holds only when $b \leq 1/2$. Taking, for instance, $b = 2/3$, yields a counterexample.

A counterexample for the unnormalized measure would follow from the same strategy, studying a different Borel set. However, we propose an elegant counterexample due to Prof. Alessandro Pigati:

Example 7.9 (Failure of measure domination). Let

$$\mu = \frac{1}{2}\delta_{x_1} + \frac{1}{2}\delta_{x_2}, \quad \nu = \delta_y, \quad \rho = \frac{1}{2}\delta_{z_1} + \frac{1}{2}\delta_{z_2}.$$

Then $\mathcal{G}(\mu, \nu)$ and $\mathcal{G}(\nu, \rho)$ are singletons:

$$G_{12} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \quad G_{23} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Hence the gluing map is constant and its image is the single plan

$$G_0 = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

On the other hand,

$$\mathcal{G}(\mu, \rho) = \left\{ \begin{pmatrix} t & \frac{1}{2} - t \\ \frac{1}{2} - t & t \end{pmatrix} : t \in \left[0, \frac{1}{2}\right] \right\}$$

is one-dimensional. Therefore, $\mathcal{H}_{\mu\rho}^1$ is proportional to the Lebesgue measure on $[0, 1/2]$. Meanwhile, the gluing pushforward measure is

$$\mathcal{H}_{\mu\rho}^d = \delta_{G_0}.$$

Thus

$$1 = \mathcal{H}_{\mu\rho}^d(\{G_0\}) \leq \mathcal{H}_{\mu\rho}^1(\{G_0\}) = 0$$

is false.

Example 7.9 does not by itself disprove the triangle inequality for $\mathcal{F}_\beta$. To find a counterexample, we start from the following asymptotic argument:

Proposition 7.10. *Assume that μ, ν, ρ are measures for which the zero-temperature triangle inequality holds tightly:*

$$\text{OT}_C(\mu, \rho) = \text{OT}_C(\mu, \nu) + \text{OT}_C(\nu, \rho). \quad (7.1)$$

Let $d_{\mu\nu}, d_{\nu\rho}, d_{\mu\rho}$ be the Hausdorff dimensions of $\mathcal{G}(\mu, \nu), \mathcal{G}(\nu, \rho), \mathcal{G}(\mu, \rho)$, respectively, and $r_{\mu\nu}, r_{\nu\rho}, r_{\mu\rho}$ be the Hausdorff dimensions of the set of minimizers of $\mathcal{U}_{\mu\nu}, \mathcal{U}_{\nu\rho}, \mathcal{U}_{\mu\rho}$, as in Lemma 6.12. If

$$d_{\mu\rho} - r_{\mu\rho} > d_{\mu\nu} - r_{\mu\nu} + d_{\nu\rho} - r_{\nu\rho}, \quad (7.2)$$

and $\mathcal{U}_{\mu\nu}, \mathcal{U}_{\nu\rho}$ and $\mathcal{U}_{\mu\rho}$ are nonconstant on $\mathcal{G}(\mu, \nu), \mathcal{G}(\nu, \rho)$ and $\mathcal{G}(\mu, \rho)$, respectively, then the finite-temperature free energy violates the triangle inequality for all sufficiently large β .

Proof. We want to prove that, for β large enough,

$$\mathcal{F}_\beta(\mu, \rho) - \mathcal{F}_\beta(\mu, \nu) - \mathcal{F}_\beta(\nu, \rho) > 0. \quad (7.3)$$

By Lemma 6.12, there exist positive constants $\gamma_{\mu\nu}, \gamma_{\nu\rho}, \gamma_{\mu\rho}$ such that

$$\log Z_\beta(\mu, \nu) = \log \gamma_{\mu\nu} + (r_{\mu\nu} - d_{\mu\nu}) \log \beta - \beta \text{OT}_C(\mu, \nu) + o(1),$$

and similarly for $Z_\beta(\nu, \rho)$ and $Z_\beta(\mu, \rho)$. Therefore

$$\mathcal{F}_\beta(\mu, \nu) = \text{OT}_C(\mu, \nu) + \frac{d_{\mu\nu} - r_{\mu\nu}}{\beta} \log \beta - \frac{\log \gamma_{\mu\nu}}{\beta} + o(\beta^{-1}),$$

and similarly for $\mathcal{F}_\beta(\mu, \nu)$ and $\mathcal{F}_\beta(\mu, \rho)$. Hence, the asymptotic expansion of (7.3) is

$$\text{OT}_C(\mu, \rho) - \text{OT}_C(\mu, \nu) - \text{OT}_C(\nu, \rho) + \frac{d_{\mu\rho} - r_{\mu\rho} - (d_{\mu\nu} - r_{\mu\nu}) - (d_{\nu\rho} - r_{\nu\rho})}{\beta} \log \beta + O(1/\beta).$$

By assumption, the linear term cancels, and we remain with

$$\frac{[d_{\mu\rho} - r_{\mu\rho} - (d_{\mu\nu} - r_{\mu\nu}) - (d_{\nu\rho} - r_{\nu\rho})] \log \beta + O(1)}{\beta}.$$

Although this quantity converges to zero, it is eventually positive, implying that

$$\mathcal{F}_\beta(\mu, \rho) > \mathcal{F}_\beta(\mu, \nu) + \mathcal{F}_\beta(\nu, \rho)$$

for all sufficiently large β . □

The same conclusion can be derived for unnormalized measures, since $\mathcal{F}_\beta$ only changes by a constant $O(\beta^{-1})$.

Due to this proposition, finding a counterexample simplifies in finding measures and metric cost matrices such that (7.1) and (7.2) hold.

An explicit counterexample for the triangle inequality of the free energy. To determine a counterexample, we take measures μ, ν, ρ in a particularly simple metric space. A natural choice is to work on $(\mathbb{R}, |\cdot|)$ and to consider measures

$$\mu = \frac{1}{m} \sum_{i=1}^m \delta_{x_i}, \quad \nu = \frac{1}{n} \sum_{j=1}^n \delta_{y_j}, \quad \rho = \frac{1}{r} \sum_{\ell=1}^r \delta_{z_\ell},$$

so that the free parameters of our model are the locations of the points

$$\{x_1, \dots, x_m, y_1, \dots, y_n, z_1, \dots, z_r\} \subseteq \mathbb{R}.$$

Assuming without loss of generality that points are ordered, i.e.

$$x_1 \leq x_2 \leq \dots \leq x_m, \quad y_1 \leq y_2 \leq \dots \leq y_n, \quad z_1 \leq z_2 \leq \dots \leq z_r,$$

[20, Remark 2.28, 2.30] gives us a straightforward way to compute OT_C : if $Q_\mu : (0, 1) \rightarrow \mathbb{R}$ is the generalized quantile function associated with μ , and similarly for the other measures, then, for the cost $C(x, y) = |x - y|$, one has [20, Eq. 2.36]:

$$\text{OT}_C(\mu, \nu) = \int_0^1 |Q_\mu(t) - Q_\nu(t)| dt.$$

To further simplify the computation, we set $r = m$, so that the quantile formula reduces to [20, Eq. 2.33]:

$$\text{OT}_C(\mu, \rho) = \frac{1}{m} \sum_{i=1}^m |x_i - z_i|.$$

This choice forces

$$d_{\mu\nu} = (m-1)(n-1), \quad d_{\nu\rho} = (m-1)(n-1), \quad d_{\mu\rho} = (m-1)^2.$$

$m = 3$ would be likely too small for satisfying (7.2). We then try $m = 4$, and correspondingly $n = 2$. Then

$$d_{\mu\rho} - d_{\mu\nu} - d_{\nu\rho} = 9 - 3 - 3 = 3.$$

To keep some symmetry, we set

$$x_1 \leq x_2 \leq y_1 \leq z_1 \leq z_2 \leq z_3 \leq z_4 \leq y_2 \leq x_3 \leq x_4. \quad (7.4)$$

Thus

$$\text{OT}_C(\mu, \rho) = \frac{1}{4} \left[(z_1 - x_1) + (z_2 - x_2) + (x_3 - z_3) + (x_4 - z_4) \right],$$

$$\text{OT}_C(\mu, \nu) = \frac{1}{4} \left[(y_1 - x_1) + (y_1 - x_2) + (x_3 - y_2) + (x_4 - y_2) \right],$$

and

$$\text{OT}_C(\nu, \rho) = \frac{1}{4} \left[(z_1 - y_1) + (z_2 - y_1) + (y_2 - z_3) + (y_2 - z_4) \right].$$

Imposing (7.1) we see that

$$\text{OT}_C(\mu, \rho) = \text{OT}_C(\mu, \nu) + \text{OT}_C(\nu, \rho),$$

regardless of the specific values, provided that the numbers strictly satisfy (7.4). To satisfy (7.2), it is natural to try setting

$$(x_1, x_2, y_1, z_1, z_2, z_3, z_4, y_2, x_3, x_4) = (-5, -4, -3, -2, -1, 1, 2, 3, 4, 5).$$

The corresponding cost matrices, induced by the metric $C(x, y) = |x - y|$, are

$$C^{XY} = \begin{pmatrix} 2 & 8 \\ 1 & 7 \\ 7 & 1 \\ 8 & 2 \end{pmatrix}, \quad C^{YZ} = \begin{pmatrix} 1 & 2 & 4 & 5 \\ 5 & 4 & 2 & 1 \end{pmatrix}, \quad C^{XZ} = \begin{pmatrix} 3 & 4 & 6 & 7 \\ 2 & 3 & 5 & 6 \\ 6 & 5 & 3 & 2 \\ 7 & 6 & 4 & 3 \end{pmatrix}.$$

We can immediately check the third condition, namely that the transport cost is not constant on $\mathcal{G}(\mu, \nu)$, $\mathcal{G}(\nu, \rho)$ and $\mathcal{G}(\mu, \rho)$, by means of Lemma 6.7, which states that a cost matrix C yields a constant transport cost on the polytope if and only if $C_{ij} = a_i + b_j$ for all i, j . We then assume that C^{XY} , C^{YZ} and C^{XZ} can be written in that form, and we reach a contradiction:

- For C^{XY} we consider the second 2×2 matrix: we have $1 = a_2 + b_1$, $7 = a_2 + b_2$, $7 = a_3 + b_1$, $1 = a_3 + b_2$. Summing diagonally, we see that $a_2 + a_3 + b_1 + b_2$ is equal to both 2 and 14, contradiction. This method is valid in general: We must always have $a_i + a_{i+1} + b_j + b_{j+1}$ constant.
- For C^{YZ} we consider the first 2×2 matrix, where $C_{11}^{YZ} + C_{22}^{YZ} = 1 + 4 = 5 \neq 7 = 2 + 5 = C_{12}^{YZ} + C_{21}^{YZ}$.
- For C^{XZ} we consider the 2×2 matrix

$$\begin{pmatrix} C_{21}^{XZ} & C_{22}^{XZ} \\ C_{31}^{XZ} & C_{32}^{XZ} \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 6 & 5 \end{pmatrix}.$$

We see that $2 + 5 \neq 3 + 6$, reaching a contradiction.

Therefore, the transport cost is not constant on $\mathcal{G}(\mu, \nu)$, $\mathcal{G}(\nu, \rho)$, and $\mathcal{G}(\mu, \rho)$.

A generic coupling $G^{XY} \in \mathcal{G}(\mu, \nu)$ can be written as

$$G^{XY} = \begin{pmatrix} a & \frac{1}{4} - a \\ b & \frac{1}{4} - b \\ c & \frac{1}{4} - c \\ \frac{1}{2} - (a + b + c) & a + b + c - \frac{1}{4} \end{pmatrix},$$

with a, b, c constrained so that each entry is nonnegative. The corresponding cost is

$$\begin{aligned} \langle C^{XY}, G^{XY} \rangle &= 2a + 8\left(\frac{1}{4} - a\right) + b + 7\left(\frac{1}{4} - b\right) \\ &\quad + 7c + \left(\frac{1}{4} - c\right) + 8\left(\frac{1}{2} - (a + b + c)\right) + 2\left(a + b + c - \frac{1}{4}\right) \\ &= \frac{15}{2} - 12a - 12b. \end{aligned}$$

The cost is minimized when $a + b$ is maximized, with the constraints $a, b \in [0, 1/4]$ and $1/4 \leq a + b + c \leq 1/2$. This forces $c = 0$, and correspondingly $a = b = 1/4$:

$$(a^*, b^*, c^*) = \left(\frac{1}{4}, \frac{1}{4}, 0\right).$$

Since (a^*, b^*, c^*) is a point rather than a subspace, we conclude that $r_{\mu\nu} = 0$.

A generic coupling $G^{YZ} \in \mathcal{G}(\nu, \rho)$ can be written as

$$G^{YZ} = \begin{pmatrix} a & b & c & \frac{1}{2} - a - b - c \\ \frac{1}{4} - a & \frac{1}{4} - b & \frac{1}{4} - c & a + b + c - \frac{1}{4} \end{pmatrix},$$

with a, b, c constrained so that each entry is nonnegative. The corresponding cost is

$$\begin{aligned}\langle C^{YZ}, G^{YZ} \rangle &= a + 2b + 4c + 5\left(\frac{1}{2} - a - b - c\right) + 5\left(\frac{1}{4} - a\right) + 4\left(\frac{1}{4} - b\right) \\ &\quad + 2\left(\frac{1}{4} - c\right) + \left(a + b + c - \frac{1}{4}\right) \\ &= -8a - 6b - 2c + 5\end{aligned}$$

Since $a, b, c \leq 1/4$ and $a + b + c \leq 1/2$, we set a to be as big as possible, $a = 1/4$, then $b = 1/4$ and then $c = 0$. To see that this is the unique solution, observe that the coefficients can be ordered as $-8 < -6 < -2$, so that, for a fixed sum $a + b + c$, the cost is minimized assigning mass first to a , then to b , then to c . Of course, we expect $a + b + c = 1/2$ at the minimum, since the coefficients $-8, -6, -2$ are all negative, and increasing the total sum $a + b + c$ always decreases the total cost. Hence

$$a + b + c = \frac{1}{2}, \quad 0 \leq a, b \leq \frac{1}{4} \implies (a^*, b^*, c^*) = \left(\frac{1}{4}, \frac{1}{4}, 0\right).$$

Again, the optimal plan is unique, and we conclude that $r_{\nu\rho} = 0$.

A generic coupling $G^{XZ} \in \mathcal{G}(\mu, \rho)$ can be written as

$$G^{XZ} = \begin{pmatrix} a & b & c & \frac{1}{4} - a - b - c \\ d & e & f & \frac{1}{4} - d - e - f \\ g & h & k & \frac{1}{4} - g - h - k \\ \frac{1}{4} - a - d - g & \frac{1}{4} - b - e - h & \frac{1}{4} - c - f - k & -\frac{1}{2} + \sum_{\text{letters}} \end{pmatrix}$$

with $a, b, c, d, e, f, g, h, k$ constrained so that each entry is nonnegative. The corresponding cost is

$$\begin{aligned}\langle C^{XZ}, G^{XZ} \rangle &= 3a + 4b + 6c + 7\left(\frac{1}{4} - a - b - c\right) \\ &\quad + 2d + 3e + 5f + 6\left(\frac{1}{4} - d - e - f\right) \\ &\quad + 6g + 5h + 3k + 2\left(\frac{1}{4} - g - h - k\right) \\ &\quad + 7\left(\frac{1}{4} - a - d - g\right) + 6\left(\frac{1}{4} - b - e - h\right) \\ &\quad + 4\left(\frac{1}{4} - c - f - k\right) + 3\left(a + b + c + d + e + f + g + h + k - \frac{1}{2}\right) \\ &= \frac{13}{2} - 8a - 6b - 2c - 8d - 6e - 2f.\end{aligned}$$

The cost is minimized when $8(a + d) + 6(b + e) + 2(c + f)$ is maximized. Setting $A := a + d$, $B := b + e$, $C := c + f$, the column constraints imply $0 \leq A, B, C \leq \frac{1}{4}$, and also

$A + B + C \leq \frac{1}{2}$. This is precisely the scenario discussed before: if the original letters $a, b, c, d, e, f, g, h, k$ give enough flexibility, we expect

$$(A^*, B^*, C^*) = \left(\frac{1}{4}, \frac{1}{4}, 0\right) \implies (a^*, d^*, b^*, e^*, c^*, f^*) = \left(a, \frac{1}{4} - a, b, \frac{1}{4} - b, 0, 0\right).$$

With this choice, column constraints force

$$(g^*, h^*) = (0, 0),$$

so that the optimal coupling G^{XZ} can be rewritten as

$$(G^{XZ})^* = \begin{pmatrix} a^* & b^* & 0 & \frac{1}{4} - a^* - b^* \\ \frac{1}{4} - a^* & \frac{1}{4} - b^* & 0 & a^* + b^* - \frac{1}{4} \\ 0 & 0 & k^* & \frac{1}{4} - k^* \\ 0 & 0 & \frac{1}{4} - k^* & k^* \end{pmatrix}.$$

Imposing nonnegativity, we deduce

$$0 \leq a^*, b^*, k^* \leq \frac{1}{4}, \quad \frac{1}{4} \leq a^* + b^* \leq \frac{1}{4},$$

so that $b^* = \frac{1}{4} - a^*$: setting $(a^*, k^*) = (\alpha, \beta)$,

$$(G^{XZ})^* = \begin{pmatrix} \alpha & \frac{1}{4} - \alpha & 0 & 0 \\ \frac{1}{4} - \alpha & \alpha & 0 & 0 \\ 0 & 0 & \beta & \frac{1}{4} - \beta \\ 0 & 0 & \frac{1}{4} - \beta & \beta \end{pmatrix}, \quad 0 \leq \alpha, \beta \leq \frac{1}{4}.$$

We observe that $(G^{XZ})^*$ is feasible for every $\alpha, \beta \in [0, 1/4]$. The corresponding cost is

$$\langle C^{XZ}, G^{XZ} \rangle = \frac{13}{2} - 8A^* - 6B^* - 2C^* = \frac{13}{2} - 8 \cdot \frac{1}{4} - 6 \cdot \frac{1}{4} - 2 \cdot 0 = 3,$$

independent from α and β . We conclude that the set of minimizers for the transport cost is a subspace of dimension 2, and $r_{\mu\rho} = 2$. Hence

$$d_{\mu\rho} - r_{\mu\rho} \stackrel{?}{>} d_{\mu\nu} - r_{\mu\nu} + d_{\nu\rho} - r_{\nu\rho} \iff 9 - 2 > 3 - 0 + 3 - 0.$$

All the hypotheses of Proposition 7.10 are satisfied. Hence, for all sufficiently large β ,

$$\mathcal{F}_\beta(\mu, \rho) > \mathcal{F}_\beta(\mu, \nu) + \mathcal{F}_\beta(\nu, \rho).$$

7.4 Metric Properties of the Finite-Temperature Internal Energy

We finally examine the finite-temperature internal energy

$$\mathcal{U}_\beta := \sum_{i=1}^I \sum_{j=1}^J C_{ij} G_{ij}^\beta, \quad G_{ij}^\beta = \varphi(\beta(C_{ij} + u_i^\beta + v_j^\beta)),$$

with (u_i^β, v_j^β) satisfying the system (2.1), with $v_J = 0$. As in the case of $\mathcal{F}_\beta$, it is claimed that such a function satisfies the metric properties of symmetry and triangle-inequality. Symmetry is valid, while the triangle inequality argument provided by [12] is invalid. For completeness, we address also positive-definiteness.

Positive-definiteness Since $C_{ij} \geq 0$ and $\varphi > 0$ by Lemma 2.2, we have

$$\mathcal{U}_\beta(\mu, \nu) = \sum_{i,j} C_{ij} \varphi(\beta(C_{ij} + u_i^\beta + v_j^\beta)) \geq 0,$$

with equality if and only if $C_{ij} = 0$ for every $(i, j) \in [I] \times [J]$. As we saw in Section 7.3.2, this implies $\mu = \nu = \delta_x$ for some $x \in \mathcal{S}$. The converse implication is in general false, as it is easy to check with the same counterexample as in 7.7: indeed, if

$$\mu = \frac{1}{2}\delta_{x_1} + \frac{1}{2}\delta_{x_2},$$

and C is induced by the discrete metric, then

$$\mathcal{U}_\beta(\mu, \mu) = C_{12}G_{12}^\beta + C_{21}G_{21}^\beta = G_{12}^\beta + G_{21}^\beta > 0.$$

Symmetry Under the assumption that C is induced by a metric, the symmetry of the finite-temperature internal energy follows by direct computation as in [12]. Let $(u^\beta, v^\beta) \in \mathbb{R}^I \times \mathbb{R}^J$ be the solution to the system (2.1), with $v_J = 0$. Let $(\tilde{u}^\beta, \tilde{v}^\beta) = (v^\beta, u^\beta) \in \mathbb{R}^J \times \mathbb{R}^I$. It is clear that

$$\sum_{i=1}^I \varphi(\beta(C_{ji} + \tilde{u}_j^\beta + \tilde{v}_i^\beta)) = \sum_{i=1}^I \varphi(\beta(C_{ij} + u_i^\beta + v_j^\beta)), \quad j \in [J],$$

and

$$\sum_{j=1}^J \varphi(\beta(C_{ji} + \tilde{u}_j^\beta + \tilde{v}_i^\beta)) = \sum_{j=1}^J \varphi(\beta(C_{ij} + u_i^\beta + v_j^\beta)), \quad i \in [I].$$

Therefore, $(\tilde{u}^\beta, \tilde{v}^\beta)$ satisfies the system (2.1) for the problem with marginals ν and μ . By uniqueness of the solution of (2.1),

$$\begin{aligned} \mathcal{U}_\beta(\nu, \mu) &= \sum_{i=1}^I \sum_{j=1}^J d(y_j, x_i) \varphi(d(y_j, x_i) + \tilde{u}_i^\beta + \tilde{v}_j^\beta) \\ &\stackrel{\dagger}{=} \sum_{i=1}^I \sum_{j=1}^J d(x_i, y_j) \varphi(d(x_i, y_j) + \tilde{u}_i^\beta + \tilde{v}_j^\beta) \\ &= \sum_{i=1}^I \sum_{j=1}^J d(x_i, y_j) \varphi(d(x_i, y_j) + u_i^\beta + v_j^\beta) = \mathcal{U}_\beta(\mu, \nu), \end{aligned}$$

where $\dagger$ follows from the symmetry property of the metric.

Triangle inequality We show that the gluing argument provided in [12] has an obstruction that cannot be adjusted easily. We do not claim that the triangle inequality for $\mathcal{U}_\beta$ is false. We only show that the argument proposed in [12] does not prove it

We want to prove

$$\mathcal{U}_\beta(\mu, \rho) \leq \mathcal{U}_\beta(\mu, \nu) + \mathcal{U}_\beta(\nu, \rho),$$

that is,

$$\sum_{i=1}^I \sum_{\ell=1}^L C_{i\ell}^{XZ}(G_{13}^\beta)_{i\ell} \leq \sum_{i=1}^I \sum_{j=1}^J C_{ij}^{XY}(G_{12}^\beta)_{ij} + \sum_{j=1}^J \sum_{\ell=1}^L C_{j\ell}^{YZ}(G_{23}^\beta)_{j\ell}, \quad (7.5)$$

where G_{12}^β , G_{23}^β and G_{13}^β are the solutions to the systems of the form (2.1) associated with $\mathcal{G}(\mu, \nu)$, $\mathcal{G}(\nu, \rho)$ and $\mathcal{G}(\mu, \rho)$. Define the glued transport plan $\tilde{G}_{13} := \Phi(G_{12}^\beta, G_{23}^\beta) \in \mathcal{G}(\mu, \rho)$ by

$$(\tilde{G}_{13})_{i\ell} := \sum_{j=1}^J \frac{(G_{12}^\beta)_{ij}(G_{23}^\beta)_{j\ell}}{\nu_j}.$$

By Lemma 7.4, this is indeed an admissible coupling between μ and ρ . For every fixed $j \in [J]$, the metric triangle inequality gives

$$C_{i\ell}^{XZ} \leq C_{ij}^{XY} + C_{j\ell}^{YZ} \quad \forall (i, \ell) \in [I] \times [L].$$

Therefore

$$\begin{aligned}
 \sum_{i=1}^I \sum_{\ell=1}^L C_{i\ell}^{XZ} (G_{13}^\beta)_{i\ell} &\leq \sum_{i=1}^I \sum_{\ell=1}^L (C_{ij}^{XY} + C_{j\ell}^{YZ}) (G_{13}^\beta)_{i\ell} \\
 &= \sum_{i=1}^I C_{ij}^{XY} \sum_{\ell=1}^L (G_{13}^\beta)_{i\ell} + \sum_{\ell=1}^L C_{j\ell}^{YZ} \sum_{i=1}^I (G_{13}^\beta)_{i\ell} \\
 &= \sum_{i=1}^I C_{ij}^{XY} \mu_i + \sum_{\ell=1}^L C_{j\ell}^{YZ} \rho_\ell \\
 &= \sum_{i=1}^I C_{ij}^{XY} \sum_{\ell=1}^L (\tilde{G}_{13})_{i\ell} + \sum_{\ell=1}^L C_{j\ell}^{YZ} \sum_{i=1}^I (\tilde{G}_{13})_{i\ell}
 \end{aligned}$$

where we used the fact that both G_{13}^β and $\tilde{G}_{13}$ are in $\mathcal{G}(\mu, \rho)$, hence they satisfy the marginal constraints. Therefore, for every fixed $j \in [J]$,

$$\sum_{i=1}^I \sum_{\ell=1}^L C_{i\ell}^{XZ} (G_{13}^\beta)_{i\ell} \leq \sum_{i=1}^I \sum_{\ell=1}^L (C_{ij}^{XY} + C_{j\ell}^{YZ}) (\tilde{G}_{13})_{i\ell} \quad (7.6)$$

The authors now observe that, for each fixed pair (i, ℓ) , the set of real numbers $\{C_{ij}^{XY} + C_{j\ell}^{YZ} : j \in [J]\}$ is finite. Hence, it admits a minimal element, with index $j_{k(i, \ell)}$. Then, for every $h \in [J]$,

$$C_{ij_{k(i, \ell)}}^{XY} + C_{j_{k(i, \ell)}\ell}^{YZ} \leq C_{ih}^{XY} + C_{h\ell}^{YZ}.$$

The proof breaks down at the level of quantifiers: it is claimed that that

$$\sum_{i=1}^I \sum_{\ell=1}^L C_{i\ell}^{XZ} (G_{13}^\beta)_{i\ell} \leq \sum_{i=1}^I \sum_{\ell=1}^L (C_{ij_{k(i, \ell)}}^{XY} + C_{j_{k(i, \ell)}\ell}^{YZ}) (\tilde{G}_{13})_{i\ell}. \quad (7.7)$$

However, this inequality does not follow from (7.6). This is because j depends on i and ℓ , while (7.6) is valid for a fixed j . If (7.7) were valid, then

$$\begin{aligned}
 \sum_{i=1}^I \sum_{\ell=1}^L (C_{ij_{k(i, \ell)}}^{XY} + C_{j_{k(i, \ell)}\ell}^{YZ}) (\tilde{G}_{13})_{i\ell} &= \sum_{i=1}^I \sum_{\ell=1}^L (C_{ij_{k(i, \ell)}}^{XY} + C_{j_{k(i, \ell)}\ell}^{YZ}) \sum_{h=1}^J \frac{(G_{12}^\beta)_{ih} (G_{23}^\beta)_{h\ell}}{\nu_h} \\
 &= \sum_{i=1}^I \sum_{\ell=1}^L \sum_{h=1}^J (C_{ij_{k(i, \ell)}}^{XY} + C_{j_{k(i, \ell)}\ell}^{YZ}) \frac{(G_{12}^\beta)_{ih} (G_{23}^\beta)_{h\ell}}{\nu_h} \\
 &\leq \sum_{i=1}^I \sum_{\ell=1}^L \sum_{h=1}^J (C_{ih}^{XY} + C_{h\ell}^{YZ}) \frac{(G_{12}^\beta)_{ih} (G_{23}^\beta)_{h\ell}}{\nu_h}.
 \end{aligned}$$

Splitting the last sum gives

$$\mathcal{U}_\beta(\mu, \rho) \leq \sum_{i=1}^I \sum_{h=1}^J C_{ih}^{XY} (G_{12}^\beta)_{ih} \frac{\sum_{\ell=1}^L (G_{23}^\beta)_{h\ell}}{\nu_h} + \sum_{h=1}^J \sum_{\ell=1}^L C_{h\ell}^{YZ} (G_{23}^\beta)_{h\ell} \frac{\sum_{i=1}^I (G_{12}^\beta)_{ih}}{\nu_h}.$$

Since

$$\sum_{\ell=1}^L (G_{23}^\beta)_{h\ell} = \nu_h, \quad \sum_{i=1}^I (G_{12}^\beta)_{ih} = \nu_h,$$

we would conclude

$$\mathcal{U}_\beta(\mu, \rho) \leq \sum_{i=1}^I \sum_{h=1}^J C_{ih}^{XY} (G_{12}^\beta)_{ih} + \sum_{h=1}^J \sum_{\ell=1}^L C_{h\ell}^{YZ} (G_{23}^\beta)_{h\ell} = \mathcal{U}_\beta(\mu, \nu) + \mathcal{U}_\beta(\nu, \rho),$$

which is exactly the desired triangle inequality. However, (7.7) remains false, hence the argument has an obstruction. We observe that, for $\mathcal{F}_\beta$ we can argue about its validity through the asymptotic provided by Lemma 6.12. Meanwhile, a control of $\mathcal{U}_\beta$ would require us to study the value of $u_i^\beta + v_j^\beta$ for every $\beta > 0$ and $(i, j) \in [I] \times [J]$.

7.5 Summary of the Mathematical Status of the Claims in [12]

We conclude this chapter by summarizing the mathematical status of the main claims in [12]. We isolate six main mathematical arguments in the paper.

i) In [12, Section III], the authors argue that

$$F_\beta(u^\beta, v^\beta) \rightarrow \text{OT}_C(\mu, \nu)$$

by first showing that

$$\mathcal{F}_\beta(\mu, \nu) \rightarrow \text{OT}_C(\mu, \nu),$$

and then claiming that $F_\beta(u^\beta, v^\beta)$ has the same asymptotics as $\beta \rightarrow +\infty$, because it is obtained from a saddle-point approximation. We do not regard this argument as rigorous in its present form. Indeed, Chapter 5 is entirely devoted to identifying the additional hypotheses under which such an asymptotic argument can be made rigorous. Moreover, the derivation based on Dirac deltas and their Fourier representation remains heuristic in the form presented in [12]. On the other hand, the conclusion itself is valid:

$$F_\beta(u^\beta, v^\beta) \rightarrow \text{OT}_C(\mu, \nu),$$

as proved independently in Theorem 2.11.

ii) In [12, Appendix A], the authors argue that $\mathcal{F}_\beta$ satisfies the metric properties of symmetry and triangle inequality. In Section 7.3, we showed that symmetry is valid. By contrast, the proof of the triangle inequality relies on a flawed gluing argument:

the gluing map induces a measure on the target transport polytope through push-forward, and this measure is, in general, different from the reference measure on it, nor can one obtain a domination estimate in general. We also constructed an explicit counterexample in 7.3.3, concluding that triangle inequality is not valid.

- iii) In [12, Appendix B], the authors argue that $\mathcal{F}_\beta(\mu, \nu)$ and $E_\beta(\mu, \nu)$ are monotonically decreasing functions of β , and that they converge to the optimal transport cost $\text{OT}_C(\mu, \nu)$. The convergence to the optimal transport cost is not proved there, nor is a Laplace-type asymptotic argument provided. Nevertheless, the result is valid, as shown in Theorem 6.5. The authors correctly prove that E_β is monotonically decreasing in β . Their argument for the monotonicity of $\mathcal{F}_\beta$, however, uses the claim that $S_\beta < 0$. This claim is justified in [12, Appendix A] by asserting that the volume of the transport polytope is smaller than 1. In Section 7.1, we showed that this volume argument is invalid, although we do not claim that the volume bound itself is false. To avoid this issue, we proved an independent entropy estimate, Lemma 6.13, which implies that $\mathcal{F}_\beta$ is monotonically decreasing for all sufficiently large β .
- iv) In [12, Appendix C], the authors prove that F_β is strictly concave after fixing $v_J := 0$. This proof is valid.
- v) In [12, Appendix D], the authors claim that the finite-temperature internal energy, $\mathcal{U}_\beta$ satisfies the triangle inequality. In Section 7.4, we showed that the proof breaks down at the level of quantifiers. We do not claim that the triangle inequality for $\mathcal{U}_\beta$ is false; we only conclude that it is not established by the proof in [12].
- vi) In [12, Appendix E], the authors argue that $F_\beta(u^\beta, v^\beta)$ and $\mathcal{U}_\beta$ are monotonically decreasing functions of β , and that both converge to $\text{OT}_C(\mu, \nu)$. The convergence of $F_\beta(u^\beta, v^\beta)$ is justified in [12, Section III] through the saddle-point approximation discussed above, and therefore inherits the same limitations discussed in Chapter 5. For analogous reasons, we do not regard their argument as rigorous in its present form for the convergence

$$\mathcal{U}_\beta \rightarrow \text{OT}_C(\mu, \nu),$$

which relies on interpreting $\mathcal{U}_\beta$ as a mean-field approximation of the exact internal energy. The convergence itself is valid and was proved analytically in Theorem 2.1. Their argument for the monotonicity of $F_\beta(u^\beta, v^\beta)$ is valid. Their argument for the

monotonicity of $\mathcal{U}_\beta$ contains a mistake, but this mistake does not invalidate the result. A correct proof was given in Theorem [2.12](#), and the issue is discussed in Remark [2.14](#).

A Convex Analysis and Duality

In this appendix we collect the convex-analytic facts used in Chapter 3. The convex conjugacy results are taken from Rockafellar [22] and Rockafellar–Wets [30]; the strong-duality statements are taken from Boyd–Vandenberghe [23].

Definition A.1 (Proper lower semicontinuous function). A function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is called proper if there exists $x \in \mathbb{R}^d$ such that $f(x) < +\infty$. It is lower semicontinuous (lsc) at $x_0 \in \mathbb{R}^d$ if

$$\liminf_{x \rightarrow x_0} f(x) \geq f(x_0),$$

or, equivalently, if for every $\alpha \in \mathbb{R}$, the set

$$\{x \in \mathbb{R}^d : f(x) > \alpha\}$$

is open.

We state without proof the following standard representation result, a classical corollary of the Hahn–Banach theorem, reformulated from [30, Thm. 8.13].

Theorem A.2. *A function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is proper, convex and lower semicontinuous if and only if it is the pointwise supremum of a nonempty family of affine functions. More precisely, this representation takes the explicit form*

$$f(x) = \sup_{\xi \in \mathbb{R}^d} \left\{ \langle \xi, x \rangle + \inf_{z \in \mathbb{R}^d} \{f(z) - \langle \xi, z \rangle\} \right\}.$$

Definition A.3 (Legendre transform). Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$. Its Fenchel–Legendre–Moreau transform, also known as convex conjugate, is the function $f^* : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{\pm\infty\}$ given by

$$f^*(\xi) := \sup_{x \in \mathbb{R}^d} \{\langle \xi, x \rangle - f(x)\}.$$

Conjugation is useful because it induces a bijection on the class of proper, convex and lower semicontinuous functions, as the next theorem, adapted from [30, Thm. 11.1], shows.

Theorem A.4 (Fenchel–Moreau theorem). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper, convex and lower semicontinuous function. Then*

$$f = f^{**}.$$

Next, we define a surrogate of the differential for convex functions.

Definition A.5 (Subdifferential). Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper and convex. The subdifferential of f at $x \in \mathbb{R}^d$ is

$$\partial f(x) := \{\xi \in \mathbb{R}^d : f(z) \geq f(x) + \langle \xi, z - x \rangle \text{ for every } z \in \mathbb{R}^d\}.$$

If no such ξ exists, we set $\partial f(x) = \emptyset$.

The following theorem gives the relationship between differential calculus and convex conjugacy. It allows one to compute Legendre transforms easily for differentiable functions. It follows from [30, Section 11.A and Sec. 11.C].

Theorem A.6 (Fenchel–Young equality and subgradient inversion). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, convex and lower semicontinuous. Then, for $x, \xi \in \mathbb{R}^d$,*

$$\xi \in \partial f(x) \iff f(x) + f^*(\xi) = \langle \xi, x \rangle \iff x \in \partial f^*(\xi).$$

Corollary A.7 (Differentiable conjugacy). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be C^1 , strictly convex, and suppose that $f' : \mathbb{R} \rightarrow J$ is a bijection onto an open interval $J \subset \mathbb{R}$. Then f^* is differentiable on J , and*

$$(f^*)'(s) = (f')^{-1}(s), \quad s \in J.$$

Proof. Fix $s \in J$, and let

$$y = (f')^{-1}(s).$$

We claim that

$$\partial f^*(s) = \{y\}.$$

Since f is differentiable and strictly convex,

$$\partial f(y) = \{f'(y)\} = \{s\}.$$

By Theorem A.6,

$$y \in \partial f^*(s).$$

Furthermore, by strict convexity, the maximizer in

$$f^*(s) = \sup_{t \in \mathbb{R}} \{st - f(t)\}$$

is unique, because the first-order condition is

$$s = f'(t).$$

Hence

$$\partial f^*(s) = \{y\}.$$

Since f^* is one-dimensional and convex, its subdifferential at s is a singleton and s is an interior point of J because J is open. It follows that f^* is differentiable at s . Hence

$$\partial f^*(s) = \{y\} \iff (f^*)'(s) = y = (f')^{-1}(s).$$

□

Convex duality and KKT conditions. Next, we review the main theorems that characterize Lagrangian duality. We recall that a convex problem over a convex domain X has the form

$$P = \min_{x \in X} f(x) \quad \text{subject to } f_n(x) \leq 0, \quad h_m(x) = 0, \quad n \in [N], \quad m \in [M],$$

where $f, f_1, \dots, f_N$ are convex functions and $h_1, \dots, h_M$ are affine functions. Its Lagrangian dual is the problem

$$D = \sup_{\substack{\lambda \geq 0 \\ \eta \in \mathbb{R}^M}} \inf_{x \in X} \left\{ f(x) + \sum_{n=1}^N \lambda_n f_n(x) + \sum_{m=1}^M \eta_m h_m(x) \right\}, \quad (\text{A.1})$$

where $\lambda = (\lambda_1, \dots, \lambda_N) \in \mathbb{R}_+^N$ and $\eta = (\eta_1, \dots, \eta_M) \in \mathbb{R}^M$. The vectors $(\lambda, \eta) \in \mathbb{R}_+^N \times \mathbb{R}^M$ are called *Lagrange multipliers*, and the function

$$\mathcal{L}(x; \lambda, \eta) := f(x) + \sum_{n=1}^N \lambda_n f_n(x) + \sum_{m=1}^M \eta_m h_m(x)$$

is called the Lagrangian for (A.1). As remarked in [23, Sec. 5.2.3], a sufficient condition for strong duality, i.e.,

$$D = P,$$

is Slater's condition: there exists a point $x^{\text{int}} \in \text{relint } X$ such that

$$f_n(x^{\text{int}}) < 0 \quad n \in [N], \quad h_m(x^{\text{int}}) = 0 \quad m \in [M]. \quad (\text{A.2})$$

Moreover, if Slater's condition holds and the primal optimal value P is finite, then the dual optimum is attained. The convex counterpart of complementary slackness is given by the KKT conditions [23, Sec. 5.5.3], which give necessary and sufficient conditions for a point to be an optimum for the constrained optimization problem. Specifically, if Slater's

condition holds, then x^* is optimal if and only if there exist multipliers $(\lambda^*, \eta^*) \in \mathbb{R}_+^N \times \mathbb{R}^M$ such that

$$\begin{aligned} f_n(x^*) &\leq 0, & n &= 1, \dots, N, \\ h_m(x^*) &= 0, & m &= 1, \dots, M, \\ \lambda_n^* f_n(x^*) &= 0, & n &= 1, \dots, N, \\ 0 &\in \partial_x \mathcal{L}(x^*; \lambda^*, \eta^*), \end{aligned} \tag{A.3}$$

where ∂_x denotes the partial subdifferential with respect to x .

B Laplace's Principle

In this appendix we establish a special case of Laplace's principle used to prove Theorem 6.5. We follow Rudin [47] and [48].

The rigorous proof of the Laplace's principle follows from interpreting it as a limit of L^p norms as $p \rightarrow \infty$. Therefore, we recall the minimal formalism from the L^p theory.

L^p spaces. Given a measure space (X, Σ, μ) and $p \in [1, \infty)$, a measurable function $\varphi : X \rightarrow \mathbb{R}$ is *p-integrable* if

$$\|\varphi\|_p := \left(\int_X |\varphi|^p d\mu \right)^{1/p}$$

is finite. We denote by $\mathcal{L}^p(X)$ the space of p -integrable functions. We can extend this notion to $p = \infty$:

The *essential supremum* of a measurable function $\varphi : X \rightarrow \mathbb{R}$ is

$$\operatorname{ess\,sup}_{x \in X} \varphi(x) := \inf \{ a \in \mathbb{R} : \mu(\{x \in X : \varphi(x) > a\}) = 0 \}.$$

Similarly, one defines

$$\operatorname{ess\,inf}_{x \in X} \varphi(x) := - \operatorname{ess\,sup}_{x \in X} (-\varphi(x)).$$

A measurable function $\varphi : X \rightarrow \mathbb{R}$ is *essentially bounded* if $\operatorname{ess\,sup} |\varphi| < +\infty$. In this case, we set

$$\|\varphi\|_\infty := \operatorname{ess\,sup} |\varphi|.$$

We denote by $\mathcal{L}^\infty(X)$ the space of essentially bounded functions on X .

In order to construct a vector space, let us consider the following equivalence relation on $\mathcal{L}^p(X)$:

$$\varphi \sim \psi \iff \varphi \stackrel{a.e.}{=} \psi.$$

We denote by $L^p(X)$ the quotient space $\mathcal{L}^p(X)/\sim$. It is easy to see that $\sim$ is an equivalence relation and $L^p(X)$, endowed with the operations

$$[f] + [g] := [f + g] \quad \text{and} \quad \lambda[f] := [\lambda f], \quad \forall f, g \in L^p(X), \lambda \in \mathbb{R}$$

is a vector space. To simplify notation, we write $f \in L^p(X)$ in place of $[f] \in L^p(X)$. For every $p \in [1, \infty]$, we refer to $L^p(X)$ as the L^p -space on X .

The following inequalities, will be useful for the next theorem:

Theorem B.1 (Jensen's inequality). *Let (X, Σ, μ) be a probability space (i.e., $\mu(X) = 1$), let $f \in L^1(X)$, and let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function. Then*

$$\varphi \left(\int_X f \, d\mu \right) \leq \int_X (\varphi \circ f) \, d\mu.$$

Proof. By Theorem A.2, any proper, convex and lower semicontinuous function is the pointwise supremum of affine functions. Since $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a finite convex function, this supremum is attained at every $x_0 \in \mathbb{R}$, guaranteeing the existence of constants $a, b \in \mathbb{R}$ satisfying $\varphi(x_0) = ax_0 + b$ and $\varphi(x) \geq ax + b$ for all $x \in \mathbb{R}$. Let $x_0 = \int_X f \, d\mu$. A direct application of this observation gives

$$\varphi \left(\int_X f \, d\mu \right) = \varphi(x_0) = ax_0 + b = a \int_X f \, d\mu + \int_X b \, d\mu = \int_X (af + b) \, d\mu \leq \int_X (\varphi \circ f) \, d\mu,$$

where the third equality holds because $\int_X b \, d\mu = b\mu(X) = b$. □

Theorem B.2 (Hölder's inequality). *Let (X, Σ, μ) be a measure space, and let $p, q \in (1, \infty)$ satisfy*

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then, for any $f \in L^p(X)$ and $g \in L^q(X)$, we have that $fg \in L^1(X)$ and

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

Moreover, if $f \in L^\infty(X)$ and $g \in L^1(X)$, then $fg \in L^1(X)$ and

$$\|fg\|_1 \leq \|f\|_\infty \|g\|_1.$$

Proof. The conclusion is trivial if $\|f\|_p = 0$ or $\|g\|_q = 0$. Assume next that $\|f\|_p > 0$ and $\|g\|_q > 0$. By the Jensen's inequality (or by convexity of the function $\exp(x)$), for every $x_0, x_1 \in \mathbb{R}$ and $t \in (0, 1)$:

$$te^{x_0} + (1-t)e^{x_1} \geq e^{tx_0 + (1-t)x_1}.$$

Applying this inequality to the pair $\left(p \log \frac{|f(x)|}{\|f\|_p}, q \log \frac{|g(x)|}{\|g\|_q}\right)$ with weights $1/p$ and $1/q$, for every $x \in X$ such that $f(x) \neq 0$ and $g(x) \neq 0$, we get:

$$\begin{aligned} \frac{|f(x)|}{\|f\|_p} \frac{|g(x)|}{\|g\|_q} &= \exp \left(\frac{1}{p} \log \frac{|f(x)|}{\|f\|_p} + \frac{1}{q} \log \frac{|g(x)|}{\|g\|_q} \right) \\ &\leq \frac{1}{p} e^{p \log \frac{|f(x)|}{\|f\|_p}} + \frac{1}{q} e^{q \log \frac{|g(x)|}{\|g\|_q}} = \frac{|f(x)|^p}{p \|f\|_p^p} + \frac{|g(x)|^q}{q \|g\|_q^q}. \end{aligned}$$

(Note that the resulting inequality trivially holds even when $f(x) = 0$ or $g(x) = 0$).

Integrating over X with respect to μ yields:

$$\frac{\int_X |fg| d\mu}{\|f\|_p \|g\|_q} \leq \frac{1}{p} \frac{\int_X |f|^p d\mu}{\|f\|_p^p} + \frac{1}{q} \frac{\int_X |g|^q d\mu}{\|g\|_q^q} = \frac{1}{p} + \frac{1}{q} = 1,$$

hence

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

As a byproduct, $fg \in L^1(X)$.

For the second part, let $f \in L^\infty(X)$ and $g \in L^1(X)$. By definition of the essential supremum, we have $|f(x)| \leq \|f\|_\infty$ for μ -almost every $x \in X$. Multiplying by $|g(x)|$, we obtain:

$$|f(x)g(x)| \leq \|f\|_\infty |g(x)| \quad \mu\text{-a.e.}$$

Integrating both sides over X :

$$\|fg\|_1 = \int_X |f(x)g(x)| d\mu \leq \int_X \|f\|_\infty |g(x)| d\mu = \|f\|_\infty \int_X |g(x)| d\mu = \|f\|_\infty \|g\|_1.$$

This confirms $fg \in L^1(X)$ and concludes the proof. $\square$

Proposition B.3. *Let $\varphi \in L^1(X) \cap L^\infty(X)$. Then $\varphi \in L^p(X)$ for every $p \in [1, \infty]$ and*

$$\lim_{p \rightarrow \infty} \|\varphi\|_p = \|\varphi\|_\infty.$$

Proof. For $p \geq 1$, we split the integrand as $|\varphi|^p = |\varphi|^{p-1}|\varphi|$. Applying the L^∞ - L^1 case of Hölder's inequality (Theorem B.2) with $f = |\varphi|^{p-1}$ and $g = |\varphi|$, we obtain

$$\|\varphi\|_p^p = \int_X |\varphi|^{p-1} |\varphi| d\mu \leq \| |\varphi|^{p-1} \|_\infty \|\varphi\|_1 = \|\varphi\|_\infty^{p-1} \|\varphi\|_1.$$

Taking the p -th root yields

$$\|\varphi\|_p \leq (\|\varphi\|_\infty^{p-1} \|\varphi\|_1)^{1/p} = \|\varphi\|_\infty^{1-1/p} \|\varphi\|_1^{1/p},$$

so that, as $p \rightarrow \infty$,

$$\limsup_{p \rightarrow \infty} \|\varphi\|_p \leq \|\varphi\|_\infty.$$

As for the converse inequality, let $0 < a < \|\varphi\|_\infty$ (for $\|\varphi\|_\infty = 0$ the conclusion is trivial) and set $E_a := \{x \in X : |\varphi(x)| > a\}$. Then $|\varphi|^p \geq a^p$ on E_a , so

$$\|\varphi\|_p^p = \int_X |\varphi|^p d\mu \geq \int_{E_a} |\varphi|^p d\mu \geq \int_{E_a} a^p d\mu = a^p \mu(E_a).$$

Hence

$$\|\varphi\|_p \geq a \sqrt[p]{\mu(E_a)}.$$

Since $0 < \mu(E_a) < \infty$ (being bounded by $\|\varphi\|_1/a$), we have $\lim_{p \rightarrow \infty} \sqrt[p]{\mu(E_a)} = 1$. Therefore,

$$\liminf_{p \rightarrow \infty} \|\varphi\|_p \geq a.$$

Since this holds for every $0 < a < \|\varphi\|_\infty$, we conclude that

$$\liminf_{p \rightarrow \infty} \|\varphi\|_p \geq \|\varphi\|_\infty,$$

concluding the proof. □

Lemma B.4 (Laplace's Principle). *Let $f : X \rightarrow \mathbb{R}$ be a measurable function over a measure space (X, Σ, μ) such that $e^{-f} \in L^1(X) \cap L^\infty(X)$. Then*

$$\lim_{\beta \rightarrow \infty} -\frac{1}{\beta} \log \int_X e^{-\beta f(x)} d\mu(x) = \operatorname{ess\,inf}_{x \in X} f(x).$$

Proof. Observe that, for every $\beta \geq 1$,

$$\left(\int_X e^{-\beta f(x)} d\mu(x) \right)^{1/\beta} = \|e^{-f}\|_\beta,$$

since $e^{-f} \in L^\beta(X)$ for $\beta \geq 1$ by Proposition B.3. For the same Proposition,

$$\lim_{\beta \rightarrow \infty} \left(\int_X e^{-\beta f(x)} d\mu(x) \right)^{1/\beta} = \|e^{-f}\|_\infty.$$

By continuity of the logarithm, this implies

$$\lim_{\beta \rightarrow \infty} \frac{1}{\beta} \log \int_X e^{-\beta f(x)} d\mu(x) = \log(\|e^{-f}\|_\infty).$$

Hence

$$\begin{aligned}
 \lim_{\beta \rightarrow \infty} -\frac{1}{\beta} \log \int_X e^{-\beta f(x)} d\mu(x) &= -\log \|e^{-f}\|_\infty \\
 &= -\log \left(\operatorname{ess\,sup}_{x \in X} e^{-f(x)} \right) \\
 &= -\log \left(e^{-\operatorname{ess\,inf}_{x \in X} f(x)} \right) \\
 &= \operatorname{ess\,inf}_{x \in X} f(x).
 \end{aligned}$$

□

C Polyhedral Laplace Asymptotics

In this appendix we strengthen the Laplace's Principle [B.4](#) in the setting of affine functions over polytopes.

We recall that a (compact) convex polytope $G \subseteq \mathbb{R}^d$ is a bounded polyhedron, and a polyhedron is the intersection of finitely many closed half-spaces [[23](#), Sec. 2.2.4].

Definition C.1 (Tangent cone). Let $G \subset \mathbb{R}^d$ be a convex polytope and let $x_0 \in G$. The tangent cone of G at x_0 is the cone with vertex at the origin defined by

$$T_G(x_0) := \{\lambda(x - x_0) : x \in G, \lambda \geq 0\}.$$

In particular, $T_G(x_0)$ is invariant under positive dilations.

Theorem C.2 (Polyhedral Laplace asymptotics). Let $G \subset \mathbb{R}^d$ be a compact d -dimensional convex polytope, and let $f : G \rightarrow \mathbb{R}$ be affine and nonconstant. Set

$$m := \min_G f, \quad F := \{x \in G : f(x) = m\}, \quad k := \dim F.$$

Then there exists $\gamma \in (0, +\infty)$ such that, as $\beta \rightarrow +\infty$,²

$$\int_G e^{-\beta f(x)} dx = \gamma e^{-\beta m} \beta^{-(d-k)} (1 + o(1)),$$

and

$$\int_G (f(x) - m) e^{-\beta f(x)} dx = \gamma(d-k) e^{-\beta m} \beta^{-(d-k+1)} (1 + o(1)).$$

²The following proof is due to Prof. Alessandro Pigati.

Proof. Since f is affine and nonconstant on G , its gradient is a nonzero constant. Let

$$\tilde{f}(x) := \frac{f(x) - m}{|\nabla f|}, \quad \text{and} \quad \tilde{\beta} := \beta |\nabla f|.$$

Then

$$\int_G e^{-\beta f(x)} dx = \int_G e^{-\beta(m + |\nabla f| \tilde{f}(x))} dx = e^{-\beta m} \int_G e^{-\tilde{\beta} \tilde{f}(x)} dx.$$

Moreover,

$$\int_G (f(x) - m) e^{-\beta f(x)} dx = |\nabla f| e^{-\beta m} \int_G \tilde{f}(x) e^{-\tilde{\beta} \tilde{f}(x)} dx.$$

Therefore, henceforth we will assume that $m = 0$ and $|\nabla f| = 1$, so that $F = G \cap f^{-1}(0)$. Since f is affine, the set of its minimizers F defines an exposed face of G by definition (see [22, Sec. 18]). Furthermore, since f is not constant on G , F cannot coincide with G , making it a proper face. By [22, Cor. 18.1.3], any proper face of a convex set is entirely contained in its relative boundary, which implies that $k < d$.

Let $P := \text{aff } F$. Choose a point $x_0 \in \text{relint } F$. After translating x_0 to the origin and applying an orthogonal change of coordinates, we may assume that $P = \mathbb{R}^k \times \{0\} \subset \mathbb{R}^k \times \mathbb{R}^{d-k}$.

Since f is constant on F , and hence its linear part vanishes along $\mathbb{R}^k \times \{0\}$, we may also assume that, in this system, $f(p) = p_{k+1}$.

Let K_0 be the tangent cone of G at x_0 , as in Definition C.1. Since $x_0 \in \text{relint } F$, the tangent cone can be written as $K_0 = \mathbb{R}^k \times K_1$, for some cone $K_1 \subseteq \mathbb{R}^{d-k}$.

Fix two compact sets F' and F'' in P such that

$$F' \subset \text{relint } F, \quad F \subset \text{int}_P F'',$$

where $\text{int}_P F''$ denotes the interior of F'' with respect to the topology of P .

Claim 1: there exists $\rho > 0$ such that, inside the cylinder $\mathbb{R}^k \times B_\rho^{d-k}(0)$, one has the inclusions

$$F' \times (K_1 \cap B_\rho^{d-k}(0)) \subset G \cap (\mathbb{R}^k \times B_\rho^{d-k}(0)) \subset F'' \times (K_1 \cap B_\rho^{d-k}(0)).$$

Intuitively, the claim is locally (for a sufficiently small $\rho > 0$) propagating on $K_1 \cap B_\rho^{d-k}(0)$ the inclusions

$$F' \subset \text{relint } F \subseteq F \subset \text{int}_P F'',$$

using the fact that $F \times K_1$ is a local approximation of G .

To prove the claim, we study the geometry of G . Write G as the intersection of finitely many closed half-spaces. The inequalities active on F are exactly those whose supporting hyperplanes contain F . The inequalities inactive on F are strict on compact subsets of $\text{relint } F$, such as F' : hence, slightly perturbing these points in the direction $K_1 \cap B_\rho^{d-k}(0)$ they remain locally valid. Hence the first inclusion.

For the second inclusion, first observe that every point $(x, y) \in G \cap (\mathbb{R}^k \times B_\rho^{d-k}(0))$ has $y \in K_1$. Indeed, K_1 is determined by the inequalities active on F , and these inequalities contain $P = \text{aff } F$ in their supporting hyperplanes. Hence, in the coordinates $\mathbb{R}^k \times \mathbb{R}^{d-k}$, their restriction depends only on the variable y . Since $(x, y) \in G$, all these active inequalities are satisfied, and therefore $y \in K_1$. Now, suppose by contradiction that there exists a sequence of points $(x_n, y_n)_{n \in \mathbb{N}} \subset G$, with $y_n \rightarrow 0$, such that $x_n \notin F''$. Since G is compact, up to extracting a subsequence we may assume that the sequence converges to $(x, 0) \in G$. Here, $x \in F$, because $(x, 0) \in G$ and $G \cap \{\mathbb{R}^k \times \{0\}\} = F$. Therefore, $x \in F \subset \text{int}_P F''$. It follows that $x_n \in F''$ for all sufficiently large n (here, it is crucial that F lies in the interior of F''), contradicting $x_n \notin F''$. After possibly reducing ρ , one has $x \in F''$ for every $(x, y) \in G \cap (\mathbb{R}^k \times B_\rho^{d-k}(0))$, proving the second inclusion.

Claim 2: It holds

$$\int_{G \setminus (\mathbb{R}^k \times B_\rho^{d-k}(0))} e^{-\beta f(p)} dp = o(\beta^{-(d-k)})$$

and

$$\int_{G \setminus (\mathbb{R}^k \times B_\rho^{d-k}(0))} f(p) e^{-\beta f(p)} dp = o(\beta^{-(d-k+1)}).$$

Intuitively, the set $G \cap (\mathbb{R}^k \times B_\rho^{d-k}(0))$ is a neighbourhood of the whole minimizing face F , and all the mass of the integrals concentrates there. In particular, setting $C_\rho := G \setminus (\mathbb{R}^k \times B_\rho^{d-k}(0))$, C_ρ is compact and disjoint from F , hence there exists $m_\rho > 0$ such that $f(x) \geq m_\rho$ for every $x \in G \setminus (\mathbb{R}^k \times B_\rho^{d-k}(0))$. Define, for $t \in \mathbb{R}$, the slices

$$A_\rho(t) := \mathcal{H}^{d-1}(C_\rho \cap f^{-1}(t)).$$

Applying the coarea formula to the function

$$\varphi_\beta(p) := e^{-\beta f(p)} \mathbf{1}_{C_\rho}(p),$$

gives

$$\int_{\mathbb{R}^d} \varphi_\beta \|\nabla f\| d\mathcal{H}^d = \int_{\mathbb{R}} \left(\int_{f^{-1}(t)} \varphi_\beta d\mathcal{H}^{d-1} \right) dt.$$

Since $\|\nabla f\| = 1$, then

$$\int_{C_\rho} e^{-\beta f(p)} dp = \int_{\mathbb{R}} e^{-\beta t} A_\rho(t) dt. \quad (\text{C.1})$$

Since C_ρ is compact and f is continuous, $A_\rho(t)$ has support in $[m_\rho, M_\rho]$. Hence, for $\beta > 0$,

$$\begin{aligned} \int_{C_\rho} e^{-\beta f(p)} dp &= \int_{\mathbb{R}} e^{-\beta t} A_\rho(t) dt = \int_{m_\rho}^{M_\rho} e^{-\beta t} A_\rho(t) dt \\ &\leq e^{-\beta m_\rho} \int_{m_\rho}^{M_\rho} A_\rho(t) dt = e^{-\beta m_\rho} \mathcal{H}^d(C_\rho). \end{aligned}$$

Likewise, using $\varphi_\beta(p) := f(p)e^{-\beta f(p)} \mathbf{1}_{C_\rho}(p)$, one obtains

$$\begin{aligned} \int_{C_\rho} f(p) e^{-\beta f(p)} dp &= \int_{\mathbb{R}} t e^{-\beta t} A_\rho(t) dt = \int_{m_\rho}^{M_\rho} t e^{-\beta t} A_\rho(t) dt \\ &\leq M_\rho e^{-\beta m_\rho} \int_{m_\rho}^{M_\rho} A_\rho(t) dt = M_\rho e^{-\beta m_\rho} \mathcal{H}^d(C_\rho). \end{aligned}$$

In either case, the decay is exponential, so

$$\int_{C_\rho} e^{-\beta f(p)} dp = o(\beta^{-(d-k)}), \quad \int_{C_\rho} f(p) e^{-\beta f(p)} dp = o(\beta^{-(d-k+1)}).$$

It remains to compute

$$\int_{G \cap (\mathbb{R}^k \times B_\rho^{d-k}(0))} e^{-\beta f(p)} dp \quad \text{and} \quad \int_{G \cap (\mathbb{R}^k \times B_\rho^{d-k}(0))} f(p) e^{-\beta f(p)} dp.$$

By monotonicity of the integral and claim 1,

$$\int_{F' \times (K_1 \cap B_\rho^{d-k}(0))} e^{-\beta f(p)} dp \leq \int_{G \cap (\mathbb{R}^k \times B_\rho^{d-k}(0))} e^{-\beta f(p)} dp \leq \int_{F'' \times (K_1 \cap B_\rho^{d-k}(0))} e^{-\beta f(p)} dp$$

and similarly for the other integral. Since the reasoning is the same for F' and F'' , we simply consider a compact set $E \subset P = \mathbb{R}^k \times \{0\}$, and we define

$$I_E(\beta) := \int_{E \times (K_1 \cap B_\rho^{d-k}(0))} e^{-\beta f(p)} dp \quad \text{and} \quad J_E(\beta) := \int_{E \times (K_1 \cap B_\rho^{d-k}(0))} f(p) e^{-\beta f(p)} dp.$$

Write points as $p = (x, y) \in \mathbb{R}^k \times \mathbb{R}^{d-k}$. We recall that for our choice of the coordinates, the function f is independent of the first k coordinates: $f(x, y) = f(y)$. Hence, by Fubini,

$$I_E(\beta) = \mathcal{H}^k(E) \int_{K_1 \cap B_\rho^{d-k}(0)} e^{-\beta f(y)} dy, \quad J_E(\beta) = \mathcal{H}^k(E) \int_{K_1 \cap B_\rho^{d-k}(0)} f(y) e^{-\beta f(y)} dy.$$

Claim 3. Level sets of f in $\{0\} \times K_1$ are compact.

For $t < 0$, the level set is empty. Moreover, $K_1 \cap f^{-1}(0) = \{0\}$: otherwise, a nonzero direction in $K_1 \cap f^{-1}(0)$ would produce minimizing points outside $\text{aff } F$, contradicting the definition of F . Thus, for $t = 0$, the level set is $\{0\}$, and it remains to consider $t > 0$. Since K_1 is closed and $f^{-1}(t)$ is closed by continuity of f , it follows that level sets are closed. It remains to show that they are bounded. In our coordinates, where $f(y) = y_1$, $y_1 > 0$ for all $y \in K_1 \setminus \{0\}$. Since K_1 is the tangent cone of a polytope, it is a polyhedral cone ([30, Thm. 6.46]). Consequently, by the Minkowski-Weyl theorem for convex cones ([30, Thm. 3.52]) it is the conic hull of a finite number of non-zero generators $v_1, \dots, v_m \in \mathbb{R}^{d-k}$. Thus, any $y \in K_1$ can be written as $y = \sum_{i=1}^m \lambda_i v_i$ for some coefficients $\lambda_i \geq 0$. Moreover, since $v_i \in K_1 \setminus \{0\}$, we necessarily have $v_{i,1} > 0$ for every $i = 1, \dots, m$.

For any $y \in K_1$, it holds

$$|y| = \left| \sum_{i=1}^m \lambda_i v_i \right| \leq \sum_{i=1}^m \lambda_i |v_i| \leq C \sum_{i=1}^m \lambda_i v_{i,1} = C y_1, \quad C := \max_{1 \leq j \leq m} \frac{|v_j|}{v_{j,1}}.$$

This means that

$$K_1 \subseteq \{|y| \leq C y_1\}.$$

Therefore, for every $t > 0$,

$$K_1 \cap \{y_1 = t\} \subseteq B_{Ct}(0).$$

Since this ball is bounded, it follows that level sets are bounded. Being also closed, they are compact.

Claim 4. Level sets of f in $\{0\} \times K_1$ have an area ct^{d-k-1} for every $t > 0$ and some constant $c > 0$

Let $S_t := (\{0\} \times K_1) \cap f^{-1}(t)$ be the t -level set of f on the cone. Since K_1 is a cone with its vertex at the origin, it is invariant under positive dilations. By definition, a point y belongs to S_1 if and only if $y \in K_1$ and $y_1 = 1$. Scaling by a factor $t > 0$, we immediately see that $ty \in K_1$ and its first coordinate becomes t , implying that $ty \in S_t$. In particular, $S_t = tS_1$. Thus, by the scaling property of the Hausdorff measure, [31, Thm 2.1.2], we obtain

$$\mathcal{H}^{d-k-1}(S_t) = \mathcal{H}^{d-k-1}(tS_1) = t^{d-k-1} \mathcal{H}^{d-k-1}(S_1) = ct^{d-k-1}.$$

We only need to prove that $c := \mathcal{H}^{d-k-1}(S_1)$ is strictly positive and finite. It is finite because $S_1 = K_1 \cap \{f = 1\}$ is a compact section of the polyhedral cone K_1 . It is greater than zero because G is a d -dimensional polytope and F is a proper face of dimension $k < d$; hence, the tangent cone K_1 has a non-empty interior in $\mathbb{R}^{d-k}$. Since $f > 0$ on

$K_1 \setminus \{0\}$, the section $S_1 = K_1 \cap \{f = 1\}$ is a compact $(d - k - 1)$ -dimensional polytope with non-empty relative interior in the hyperplane $\{f = 1\}$. Therefore $\mathcal{H}^{d-k-1}(S_1) > 0$.

Claim 5. There exists a positive constant c' such that, for every compact set $E \subset P = \mathbb{R}^k \times \{0\}$, we have

$$I_E(\beta) := \int_{E \times (K_1 \cap B_\rho^{d-k}(0))} e^{-\beta f(p)} dp = c' \mathcal{H}^k(E) \beta^{-(d-k)} + o(\beta^{-(d-k)}),$$

and

$$J_E(\beta) := \int_{E \times (K_1 \cap B_\rho^{d-k}(0))} f(p) e^{-\beta f(p)} dp = (d - k) c' \mathcal{H}^k(E) \beta^{-(d-k+1)} + o(\beta^{-(d-k+1)}).$$

We emulate the same strategy used for Claim 2, based on the coarea formula. We define, for $t > 0$ (since for $t < 0$ the level sets are empty, while $t = 0$ is irrelevant for the integral), the slices

$$A_\rho(t) := \mathcal{H}^{d-k-1}(K_1 \cap B_\rho^{d-k}(0) \cap f^{-1}(t)).$$

Applying the coarea formula to the function

$$\varphi_\beta(y) := e^{-\beta f(y)} \mathbf{1}_{K_1 \cap B_\rho^{d-k}(0)}(y),$$

we obtain, as in (C.1),

$$\int_{K_1 \cap B_\rho^{d-k}(0)} e^{-\beta f(y)} dy = \int_0^\infty e^{-\beta t} A_\rho(t) dt.$$

By Claim 3, the level set $S_t = K_1 \cap f^{-1}(t)$ satisfies $S_t \subseteq B_{Ct}(0)$. Therefore, if $Ct \leq \rho$, i.e., $t \leq \rho/C$, the intersection with the ball is redundant and we have the exact equality

$$A_\rho(t) = \mathcal{H}^{d-k-1}(S_t) = ct^{d-k-1}.$$

For $t > \rho/C$, we simply use the trivial bound

$$A_\rho(t) \leq \mathcal{H}^{d-k-1}(S_t) = ct^{d-k-1}.$$

Splitting the integral accordingly, we get

$$\int_{K_1 \cap B_\rho^{d-k}(0)} e^{-\beta f(y)} dy = \int_0^{\rho/C} e^{-\beta t} ct^{d-k-1} dt + \int_{\rho/C}^\infty e^{-\beta t} A_\rho(t) dt.$$

The first integral can be computed by the change of variables $\beta t := s$, so that

$$\int_0^{\rho/C} e^{-\beta t} t^{d-k-1} dt = \beta^{-(d-k)} \int_0^{\beta \rho/C} e^{-s} s^{d-k-1} ds.$$

As $\beta \rightarrow \infty$, the integral $\int_0^{\beta\rho/C} e^{-s} s^{d-k-1} ds$ converges to the Gamma function $\Gamma(d-k)$. Thus we can write

$$\int_{K_1 \cap B_\rho^{d-k}(0)} e^{-\beta f(y)} dy = c\beta^{-(d-k)}\Gamma(d-k) - c\beta^{-(d-k)} \int_{\beta\rho/C}^{\infty} e^{-s} s^{d-k-1} ds + \int_{\rho/C}^{\infty} e^{-\beta t} A_\rho(t) dt.$$

We only need to bound asymptotically the two error terms. We have

$$\begin{aligned} \int_{\beta\rho/C}^{\infty} e^{-s} s^{d-k-1} ds &= \int_{\beta\rho/C}^{\infty} e^{-s/2} (e^{-s/2} s^{d-k-1}) ds \\ &\leq \max_{s \in [0, \infty)} (e^{-s/2} s^{d-k-1}) \cdot \int_{\beta\rho/C}^{\infty} e^{-s/2} ds = O(e^{-\text{const.} \times \beta}). \end{aligned}$$

As for the second, we similarly have

$$\begin{aligned} \int_{\rho/C}^{\infty} e^{-\beta t} A_\rho(t) dt &\leq c \int_{\rho/C}^{\infty} e^{-\beta t} t^{d-k-1} dt \\ &= c\beta^{-(d-k)} \int_{\beta\rho/C}^{\infty} e^{-s} s^{d-k-1} ds \\ &= c\beta^{-(d-k)} O(e^{-\text{const.} \times \beta}). \end{aligned}$$

Both error terms are $\beta^{-(d-k)} O(e^{-\text{const.} \times \beta})$, so they are negligible compared with the leading term $c\beta^{-(d-k)}\Gamma(d-k)$.

Applying the coarea formula to the function

$$\varphi_\beta(y) := f(y) e^{-\beta f(y)} \mathbf{1}_{K_1 \cap B_\rho^{d-k}(0)}(y),$$

we similarly get

$$\int_{K_1 \cap B_\rho^{d-k}(0)} f(y) e^{-\beta f(y)} dy = c \int_0^{\rho/C} e^{-\beta t} t^{d-k} dt + \int_{\rho/C}^{\infty} t e^{-\beta t} A_\rho(t) dt.$$

The computations and the conclusions are precisely the same, with now the leading term growing as

$$c \int_0^{\rho/C} e^{-\beta t} t^{d-k} dt = c\beta^{-(d-k+1)}\Gamma(d-k+1) + o(\beta^{-(d-k+1)}).$$

Since $\Gamma(d-k+1) = (d-k)\Gamma(d-k)$, this leading coefficient is exactly $(d-k)$ times the previous one. Again, the error terms are negligible compared with this leading term.

Setting

$$c' := c\Gamma(d-k),$$

we conclude that

$$I_E(\beta) = c' \mathcal{H}^k(E) \beta^{-(d-k)} + o(\beta^{-(d-k)}),$$

and

$$J_E(\beta) = (d-k) c' \mathcal{H}^k(E) \beta^{-(d-k+1)} + o(\beta^{-(d-k+1)}).$$

Conclusion. By Claim 1, Claim 2, and Claim 5, for $E = F'$ and $E = F''$ we obtain

$$c' \mathcal{H}^k(F') \beta^{-(d-k)} + o(\beta^{-(d-k)}) \leq \int_G e^{-\beta f(x)} dx \leq c' \mathcal{H}^k(F'') \beta^{-(d-k)} + o(\beta^{-(d-k)}).$$

Since $F' \subset \text{relint } F$ and $F'' \supset F$ were arbitrary, we may choose them so that

$$\mathcal{H}^k(F') \uparrow \mathcal{H}^k(F), \quad \mathcal{H}^k(F'') \downarrow \mathcal{H}^k(F).$$

Hence, setting

$$\tilde{\gamma} := c' \mathcal{H}^k(F) > 0,$$

we obtain

$$\int_G e^{-\beta f(x)} dx = \tilde{\gamma} \beta^{-(d-k)} (1 + o(1)).$$

The same argument applied to the second integral gives

$$\int_G f(x) e^{-\beta f(x)} dx = (d-k) \tilde{\gamma} \beta^{-(d-k+1)} (1 + o(1)).$$

We have proved the result under the normalization $m = 0$ and $\|\nabla f\| = 1$. Returning to the original affine function and setting

$$\gamma := \tilde{\gamma} \|\nabla f\|^{-(d-k)},$$

we obtain

$$\int_G e^{-\beta f(x)} dx = \gamma e^{-\beta m} \beta^{-(d-k)} (1 + o(1)).$$

Similarly,

$$\int_G (f(x) - m) e^{-\beta f(x)} dx = (d-k) \gamma e^{-\beta m} \beta^{-(d-k+1)} (1 + o(1)).$$

□

D Fourier Transform

In this appendix, we collect the Fourier-analytic facts used in Section 5.3. The standard reference for this material is Folland [40].

For any $f \in L^1(\mathbb{R}^K)$, we define its Fourier transform as the function

$$\widehat{f} : \mathbb{R}^K \rightarrow \mathbb{C}, \quad \widehat{f}(\xi) := \int_{\mathbb{R}^K} e^{-i\langle \xi, t \rangle} f(t) dt.$$

This function is well-defined, bounded, and uniformly continuous. Specifically, a standard integral bound yields

$$\|\widehat{f}\|_\infty \leq \|f\|_1.$$

Under mild regularity assumptions, the Fourier transform can be inverted. The Fourier Inversion Theorem [40, Thm. 8.26] states that if both $f \in L^1(\mathbb{R}^K)$ and $\widehat{f} \in L^1(\mathbb{R}^K)$, then, for almost every $t \in \mathbb{R}^K$ (and everywhere if f is continuous),

$$f(t) = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^K} e^{i\langle \xi, t \rangle} \widehat{f}(\xi) d\xi.$$

Applying these tools to the sequence of Gaussian regularizers defined by

$$\rho_n(t) := \left(\frac{n}{\sqrt{2\pi}} \right)^K \exp \left(-\frac{n^2}{2} \|t\|^2 \right), \quad t \in \mathbb{R}^K, \quad n > 0,$$

we have that $\rho_n \in L^1(\mathbb{R}^K)$, and its Fourier transform is given by [40, Prop. 8.24]

$$\widehat{\rho}_n(\xi) = \exp \left(-\frac{\|\xi\|^2}{2n^2} \right).$$

Clearly, $\widehat{\rho}_n \in L^1(\mathbb{R}^K)$ as well. Consequently, the Fourier inversion formula yields, for every $t \in \mathbb{R}^K$,

$$\rho_n(t) = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^K} e^{i\langle \xi, t \rangle} e^{-\|\xi\|^2/(2n^2)} d\xi.$$

Equivalently, since the function $\xi \mapsto \widehat{\rho}_n(\xi)$ is even, substituting ξ with $-\xi$ in the integral shows that

$$\rho_n(t) = \frac{1}{(2\pi)^K} \int_{\mathbb{R}^K} e^{-i\langle \xi, t \rangle} e^{-\|\xi\|^2/(2n^2)} d\xi.$$

E Multidimensional Steepest Descent

In this appendix we collect the multidimensional steepest-descent results used in Section 5.5. We rephrase [42], slightly adapting the framework (cf. Remark E.7), and we do not provide any proof.

Let $f : \Delta \rightarrow \mathbb{C}$ be a function holomorphic in an open set $\Delta \subseteq \mathbb{C}^N$ such that, for each $j = 1, \dots, N$, $\partial f / \partial z_j$ does not vanish identically.

Definition E.1 (Saddle point). A *saddle point* of f is a point $z^* = (z_1^*, \dots, z_N^*) \in \Delta$ such that

$$\nabla f(z^*) = 0.$$

It is a *nondegenerate* saddle point if

$$H(z^*) \neq 0, \quad \text{where} \quad H(z) := \det(\nabla^2 f(z)).$$

Clearly, the notion of nondegenerate saddle point is independent of the ordering $z_1, \dots, z_N$ of the variables.

Definition E.2 (Admissible saddle point). A saddle point $z^* \in \Delta$ of f is *admissible* with respect to the ordering $z_1, \dots, z_N$ of the variables if, for every $j = 1, \dots, N$,

$$H_j(z^*) \neq 0,$$

where

$$H_j(z_1, \dots, z_N) := \det \begin{pmatrix} \frac{\partial^2 f}{\partial z_1^2} & \cdots & \frac{\partial^2 f}{\partial z_1 \partial z_j} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial z_j \partial z_1} & \cdots & \frac{\partial^2 f}{\partial z_j^2} \end{pmatrix}.$$

We also set, for convenience, $H_0(z) := 1$.

Remark E.3. Since $H_N(z) = H(z)$, an admissible saddle-point z^* with respect to an ordering of the variables is also nondegenerate. Conversely, if a saddle point z^* is nondegenerate, there exists a linear, homogeneous and invertible change of variables that makes it admissible. Hence, admissibility is not essentially restrictive on the saddle point.

Lemma E.4. Let $z^* = (z_1^*, \dots, z_N^*)$ be an admissible saddle point of f with respect to the ordering $z_1, \dots, z_N$. Then, for every $j = 1, \dots, N - 1$, there exist an open neighbourhood

$$V_{j+1} \subseteq \mathbb{C}^{N-j}$$

of $(z_{j+1}^*, \dots, z_N^*)$ and j holomorphic functions

$$Z_{1j}, \dots, Z_{jj} : V_{j+1} \rightarrow \mathbb{C}$$

such that, for every $i = 1, \dots, j$,

$$z_i^* = Z_{ij}(z_{j+1}^*, \dots, z_N^*),$$

and, for every $(z_{j+1}, \dots, z_N) \in V_{j+1}$,

$$0 = \frac{\partial f}{\partial z_i}(Z_{1j}(z_{j+1}, \dots, z_N), \dots, Z_{jj}(z_{j+1}, \dots, z_N), z_{j+1}, \dots, z_N).$$

For convenience, we define, for every $j = 1, \dots, N-1$, the reduced vectors

$$\tilde{z}_j := (z_{j+1}, \dots, z_N) \in V_{j+1}$$

and the reduced functions $\tilde{f}_j : V_{j+1} \rightarrow \mathbb{C}$ by

$$\tilde{f}_j(\tilde{z}_j) := f(Z_{1j}(\tilde{z}_j), \dots, Z_{jj}(\tilde{z}_j), \tilde{z}_j).$$

Equivalently,

$$\tilde{f}_j(z_{j+1}, \dots, z_N) := f(Z_{1j}(z_{j+1}, \dots, z_N), \dots, Z_{jj}(z_{j+1}, \dots, z_N), z_{j+1}, \dots, z_N).$$

Finally, we set $\tilde{f}_0 := f$.

A piecewise continuously differentiable path is a function

$$\lambda : J \rightarrow \mathbb{C},$$

where $J \subseteq \mathbb{R}$ is an interval, possibly unbounded, for which there exists a finite partition of every compact subinterval of J such that, on each element of the partition, λ is C^1 .

Definition E.5 (Admissible path). Let $\Delta \subseteq \mathbb{C}^N$ be an open set, and let $(\lambda_1, \dots, \lambda_N)$ be a collection of piecewise continuously differentiable paths such that:

- for every $j = 1, \dots, N-1$, the path λ_j depends on $(z_{j+1}, \dots, z_N)$, where $z_i \in \lambda_i$ for every $i = j+1, \dots, N$;
- $(z_1, \dots, z_N) \in \Delta$ for every $z_N \in \lambda_N$, $z_{N-1} \in \lambda_{N-1}(z_N)$, $\dots$, $z_1 \in \lambda_1(z_2, \dots, z_N)$.

The associated *admissible path* is the real N -dimensional integration set

$$\Gamma_\lambda = \{(z_1, \dots, z_N) \in \mathbb{C}^N : z_N \in \lambda_N, z_{N-1} \in \lambda_{N-1}(z_N), \dots, z_1 \in \lambda_1(z_2, \dots, z_N)\}.$$

Next, we provide five conditions required for [42, Thm 4.2].

First Condition. We assume that $N \geq 2$, $f, g : \Delta \rightarrow \mathbb{C}$ are holomorphic functions in an open set $\Delta \subseteq \mathbb{C}^N$, $z^* \in \Delta$ is an admissible saddle point of f with respect to the ordering $z_1, \dots, z_N$, $g(z^*) \neq 0$, and Γ_λ is an admissible path. For every $\beta > 0$, set

$$I_\beta := \int_{\Gamma_\lambda} e^{\beta f(z)} g(z) dz_1 \wedge \dots \wedge dz_N,$$

where the integral over Γ_λ is understood as the iterated integral induced by the ordered family of paths $(\lambda_1, \dots, \lambda_N)$.

Second Condition. We assume that I_β is absolutely convergent for every sufficiently large β .

Third Condition. Let $z_N^* \in \lambda_N$ be an interior point of λ_N . We assume that

$$\Re \tilde{f}_{N-1}(z_N) < \Re f(z^*)$$

for every $z_N \in \lambda_N \cap V_N$, $z_N \neq z_N^*$.

Fourth Condition. For every $j = 1, \dots, N-1$ and for every $\tilde{z}_j = (z_{j+1}, \dots, z_N) \in V_{j+1}$ such that

$$z_N \in \lambda_N, \quad z_{N-1} \in \lambda_{N-1}(z_N), \quad \dots, \quad z_{j+1} \in \lambda_{j+1}(z_{j+2}, \dots, z_N),$$

we assume that

$$Z_{jj}(\tilde{z}_j) \in \lambda_j(\tilde{z}_j)$$

is an interior point of $\lambda_j(\tilde{z}_j)$, and that

$$\Re \tilde{f}_{j-1}(z_j, \tilde{z}_j) < \Re \tilde{f}_j(\tilde{z}_j)$$

for every

$$z_j \in \lambda_j(\tilde{z}_j), \quad z_j \neq Z_{jj}(\tilde{z}_j), \quad (z_j, \tilde{z}_j) \in V_j.$$

Fifth Condition. For every neighbourhood σ of z^* in Γ_λ , there exists $\delta_\sigma > 0$ such that

$$\Re f(z) \leq \Re f(z^*) - \delta_\sigma$$

for every $z \in \Gamma_\lambda \setminus \sigma$.

Theorem E.6. *Under these five conditions, the following asymptotic formula holds as $\beta \rightarrow +\infty$:*

$$I_\beta := \int_{\Gamma_\lambda} e^{\beta f(z)} g(z) dz_1 \wedge \dots \wedge dz_N = \left(\frac{2\pi}{\beta} \right)^{N/2} \frac{C e^{\beta f(z^*)} g(z^*)}{\sqrt{H(z^*)}} (1 + o(1)).$$

Here $H(z^*) = \det(\nabla^2 f(z^*))$, and $C \in \mathbb{C}$ is a unit norm constant encoding orientation of the steepest descent.

Remark E.7. The statement below is the exponential-phase version of [42, Thm. 4.2]. In the original formulation the integrand is written as

$$f(z)^n g(z), \quad n \in \mathbb{N},$$

whereas here we write it as

$$e^{\beta f(z)} g(z), \quad \beta > 0.$$

As discussed by the authors in [42, Sec. 1.2], when f has the form of an exponential, the parameter β may be taken real. Accordingly, the dominance assumptions are written in terms of $\Re f$, rather than in terms of the modulus of the function raised to the large power (in our case, $|e^f| = e^{\Re f}$). The factor $C \in \mathbb{C}$ determined by the orientation of the local descent directions is irrelevant for the logarithmic asymptotics used in our thesis, hence we do not track it.

Corollary E.8. *After choosing, for β sufficiently large, a branch of the logarithm compatible with $f(z^*)$, under the above conditions, the following asymptotic formula holds as $\beta \rightarrow \infty$:*

$$-\frac{1}{\beta} \log I_\beta = -f(z^*) + o(1).$$

Proof. By Theorem E.6, taking the logarithms,

$$\log I_\beta = \beta f(z^*) - \frac{N}{2} \log \beta + \frac{N}{2} \log(2\pi) + \log g(z^*) - \frac{1}{2} \log H(z^*) + o(\beta).$$

Dividing by $-\beta$ yields the claim. □

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